GROUPS IN THE NEW MATHEMATICS

An elementary introduction to mathematical groups through familiar examples

IRVING ADLER
Author of THE NEW MATHEMATICS

With diagrams by Ruth Adler and Ellen Viereck

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GROUPS
IN THE NEW MATHEMATICS

By IRVING ADLER

With diagrams by RUTH ADLER
and ELLEN VIERECK

Once more Irving Adler leads the way with this treatment of a mathematical subject that will be getting steadily increasing attention in the future—as he did with *The New Mathematics* and *Magic House of Numbers.*

Just as the concept of sets plays a major part in "the new mathematics," so does that of groups.

Groups are important not only in algebra, geometry, and other branches of mathematics, but also in art and in science. They are useful in the study of symmetry. Symmetry is found very often in nature, and wherever symmetry is found, there is a group that is related to it. For this reason, group theory is used today in many of the sciences. It is used, for example, by crystallographers studying the symmetries of crystals of minerals. It is used by physicists and chemists studying the symmetries of particles and of fields of force. The importance of group theory was emphasized very dramatically recently when some physicists using group theory predicted the existence of a particle that had never been observed before, and described the properties that it should have. Later experiments proved that this particle really exists and has those properties.

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Groups in the New Mathematics

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IRVING ADLER

GROUPS in the NEW MATHEMATICS

An elementary introduction to mathematical groups through familiar examples

WITH DIAGRAMS BY RUTH ADLER AND ELLEN VIERECK

THE JOHN DAY COMPANY
NEW YORK

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Foreword

This book is about the mathematical structure known as a *group*.

The idea of a group was born in the eighteenth century through the study of certain groups that arise in algebra. Later it was found that there are also groups that are important in geometry and other branches of mathematics. Then it was found that groups play an important part in art and in science, too.

Groups are important because they are useful for the study of symmetry, as we shall see in Chapter 7. Symmetry is found very often in nature, and wherever symmetry is found, there is a group that is related to it. For this reason, group theory is used today in many of the sciences. It is used, for example, by crystallographèrs studying the symmetries of crystals of minerals. It is used by physicists and chemists studying the symmetries of particles and of fields of force. The importance of group theory was emphasized very dramatically recently when some physicists using group theory predicted the existence of a particle that had never been observed before, and described the properties that it should have. Later experiments proved that this particle really exists and has those properties.

From the point of view of the general reader, there is a serious gap in the available literature on groups. On the one hand, there are many textbooks on the theory of groups. But these are too abstract, too technical, and too difficult for the general reader. On the other hand, there are many books for the general reader

that mention groups, but do not give enough detail to give the reader a real understanding of what groups are. This book is intended to bridge that gap by developing a substantial part of the elements of group theory through simple, concrete examples, and by requiring of the reader that he know no more than sixth-grade arithmetic.

Because groups have so many uses, it is important that young people become acquainted with them early. This book shows how the idea of a group may be introduced through familiar examples that can be understood by school children above the sixth grade. It is likely that some day in the near future groups will be a part of the school mathematics curriculum as early as the seventh grade, and experiences from which the idea of a group may be developed will be included as early as the second grade. To prepare for that day, schoolteachers have to learn now what they may have to teach later about groups. This book, written for the general reader, will also help achieve that purpose for the teacher.

Groups in the New Mathematics

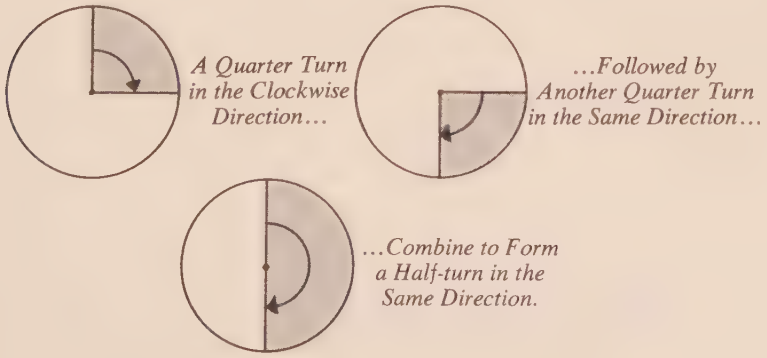
1

Things That Behave Like Numbers

Any two numbers can be "combined" by addition to form a third number called their *sum*. For example, if the numbers are *2* and *3*, they can be added to form the sum *5*.

Any two numbers can also be "combined" by multiplication to form a third number called their *product*. For example, if the numbers are $\frac{1}{2}$ and $\frac{1}{3}$, they can be multiplied to form the product $\frac{1}{6}$.

There are many sets of things that are not numbers but that resemble numbers in this respect. There is a rule of combination by which any two things in the set can be combined to form a third thing. A familiar example is the set of all possible *rotations* or turns of a hand of a clock around the center of the face of the clock. Two turns can be combined by letting one turn follow the other. For example, if a quarter turn in the clockwise direction



is followed by another quarter turn in the clockwise direction, the two turns combine to form a half turn in the same direction.

In some sets in which there is a rule of combination, the resemblance to numbers goes even further. Not only can any two of the things be combined to form a third thing, but the rule of combination has properties that are like the properties that addition or multiplication have in some important sets of numbers. Four of these properties, described in Chapter 2, occur in many useful sets. Any set with a rule of combination that has these properties is called a *group*.

The purpose of this book is to explain what a group is, to learn a few things about the nature of a group, and to get acquainted with some important groups. As we shall see, the members of a group may or may not be numbers, but they behave like numbers in many ways.

2

Multiplication of Measuring Numbers

Counting Numbers

When we count, we use the numbers *1*, *2*, *3*, *4*, *5*, *6*, and so on. These numbers are called the *counting numbers*. It is useful to picture the counting numbers in the following way as points on a number line: Imagine a horizontal line that extends endlessly to the right and to the left. Choose any point on the line to use as *starting point*. Choose any unit of length. Divide the half-line to the right of the starting point into *segments* or pieces that have this length. In this way we get a set of equally spaced points to the right of the starting point. We let the first of these equally spaced points to the right of the starting point represent the



The Counting Numbers as Points on the Number Line

number *1*; we let the second of these points represent *2*; we let the third of these points represent *3*; and so on.

Exercises:

1. What counting number is represented by the ninth of these equally spaced points to the right of the starting point?

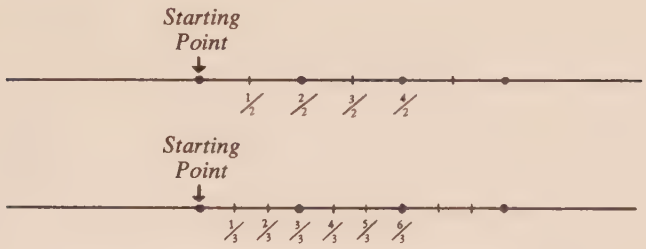
Measuring Numbers

When we make measurements, we use some other numbers besides the counting numbers. We use numbers that can be represented by fractions such as $\frac{1}{2}, \frac{2}{2}, \frac{3}{2}$, and so on; $\frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \frac{4}{3}$, and so on; $\frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{4}{4}, \frac{5}{4}$, and so on. In each of these fractions, the numerator and the denominator are counting numbers. Since the numbers represented by such fractions are used when we make measurements, we shall call them *measuring numbers*. Another name for them is *positive rational numbers*.

Every measuring number is represented by many different fractions. For example, the fractions $\frac{1}{2}, \frac{2}{4}, \frac{3}{6}$, and all other fractions in which the denominator is double the numerator, stand for the same measuring number.

The set of all measuring numbers includes the set of all counting numbers. So every counting number is also represented by many different fractions. For example, the fractions $\frac{1}{1}, \frac{2}{2}, \frac{3}{3}$, and so on, all stand for the counting number *1*. In fact, any fraction whose numerator and denominator are equal stands for the number *1*.

We can picture the measuring numbers as points on the number line, too. The points that represent the counting numbers divide the half-line to the right of the starting point into unit segments. Divide each of these segments into two equal parts. Then we have a series of equally spaced points that divide the half-line into segments whose length is one-half of a unit. Moving to the right from the starting point, label the first of these points $\frac{1}{2}$, label the second one $\frac{2}{2}$, label the third one $\frac{3}{2}$, and so on. Now start all over again, and divide each of the unit segments into three equal parts. Then we have a series of equally spaced points



Measuring Numbers as Points on the Number Line

that divide the half-line into segments whose length is one-third of a unit. Moving to the right from the starting point, label the first of these points $\frac{1}{3}$, label the second one $\frac{2}{3}$, label the third one $\frac{3}{3}$, and so on. In a similar way we locate points to be labeled with fractions whose denominator is *4*, fractions whose denominator is *5*, and so on. Each point that receives a label in this way stands for a measuring number.

If a point on the line has at least one of these labels attached to it, it has many of them. For example, the labels $\frac{1}{2}$, $\frac{2}{4}$, $\frac{3}{6}$, and so on, which all stand for the same measuring number, are all labels for the same point. The measuring number that a point stands for is the measuring number that is represented by any of the fractions that are labels for that point.

Exercises:

2. What fractions are labels for the point that represents the counting number *2*?

3. Halfway between the points that represent the numbers *1* and *2* there is a point that represents a measuring number. What fractions stand for this measuring number?

Multiplication of Measuring Numbers

Any two measuring numbers can be combined by multiplication to form their product. This product is another measuring number. For example, $\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$. In general, we know from arithmetic that if $\frac{a}{b}$ and $\frac{c}{d}$ are two fractions standing for measuring numbers, then $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$. That is, to find the numerator of the product of two fractions, we multiply their numerators; and to find the denominator of their product, we multiply their denominators.

Exercises:

4. Find the product of $\frac{1}{3}$ and $\frac{2}{7}$.

5. What fraction should you multiply $\frac{2}{3}$ by to obtain as the product the fraction $\frac{6}{6}$?

Ordered Pairs

To show that two numbers are to be multiplied, we write them down with a multiplication sign between them. One of the numbers is written first, to the left of the multiplication sign. The other number is written second, to the right of the multiplication sign. Because the two numbers are written in a definite order, with one of them *first* and the other one *second*, we say that they

form an *ordered pair* of numbers. In the multiplication of numbers, we always begin with an ordered pair of numbers, and then assign to this ordered pair another number called their product. If $\frac{2}{3}$ is the first member of an ordered pair of numbers, and $\frac{4}{5}$ is the second member of the ordered pair, the number that is assigned to the ordered pair by multiplication is the number $\frac{8}{15}$. To show this assignment we write $\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$. If $\frac{4}{5}$ is the first member of the ordered pair of numbers, and $\frac{2}{3}$ is the second member of the ordered pair, the number that is assigned to the ordered pair by multiplication is again the number $\frac{8}{15}$. To show this assignment we write $\frac{4}{5} \times \frac{2}{3} = \frac{8}{15}$.

To represent an ordered pair of numbers, we write them in parentheses, with the first member of the ordered pair written first, the second member written second, and a comma between them. Thus, $\left(\frac{2}{3}, \frac{4}{5}\right)$ is an ordered pair in which $\frac{2}{3}$ is the first member; $\left(\frac{4}{5}, \frac{2}{3}\right)$ is an ordered pair in which $\frac{4}{5}$ is the first member. The rule for multiplying measuring numbers assigns to the ordered pair $\left(\frac{2}{3}, \frac{4}{5}\right)$ the number $\frac{8}{15}$ as product.

A Binary Operation

A rule of combination is called a *binary operation* in a set if it assigns to *every* ordered pair of members of the set a definite third member of the same set. Since multiplication assigns to

every ordered pair of measuring numbers a definite third number, their product, that is also a measuring number, we may say that multiplication is a binary operation in the set of all measuring numbers.

If an operation is a binary operation in a set, we say that the set is *closed* with respect to the operation. Since multiplication is a binary operation in the set of all measuring numbers, the set of all measuring numbers is closed with respect to multiplication.

If a rule of combination in a set assigns a member of the set to some but not all of the ordered pairs of members of the set, then it is not entitled to be called a binary operation in the set. For example, division is a rule of combination that assigns a counting number only to some ordered pairs of counting numbers. It assigns to the ordered pair $(6, 2)$ the counting number 3, because $6 \div 2 = 3$. It does not assign a counting number to the ordered pair $(2, 6)$, because there is no counting number that is equal to $2 \div 6$. Therefore, division is not a binary operation in the set of all counting numbers, and the set of all counting numbers is not closed with respect to division. However, division does assign a quotient that is a *measuring number* to every ordered pair of *measuring numbers*. For example, since $2 \div 6 = \frac{2}{6}$, which is a measuring number, division assigns the measuring number $\frac{2}{6}$ to the ordered pair $(2, 6)$. (Remember that 2 and 6 are measuring numbers as well as counting numbers.) So, while division is not a binary operation in the set of all counting numbers, it is a binary operation in the set of all measuring numbers. The set of all measuring numbers is closed with respect to division.

Exercises:

6. Is the product of every ordered pair of counting numbers a counting number? Is multiplication a binary operation in the set

of all counting numbers? Is the set of all counting numbers closed with respect to multiplication?

7. What counting number is represented by the expression $5 - 2$? Is there any counting number that is represented by the expression $2 - 5$? Is subtraction a binary operation in the set of all counting numbers? Is the set of all counting numbers closed with respect to subtraction?

The Associative Property

The expression $\left(2 \times \frac{1}{2}\right) \times \frac{1}{3}$ gives us directions for carrying out the multiplication in two steps. It tells us to multiply first the number 2 and the number $\frac{1}{2}$, and then to multiply their product and the number $\frac{1}{3}$. When we follow these directions, we find that $\left(2 \times \frac{1}{2}\right) \times \frac{1}{3} = 1 \times \frac{1}{3} = \frac{1}{3}$.

The expression $2 \times \left(\frac{1}{2} \times \frac{1}{3}\right)$ contains the same three numbers, 2, $\frac{1}{2}$, and $\frac{1}{3}$, written in the same order. But it gives us a different set of directions. It tells us to multiply first the number $\frac{1}{2}$ and the number $\frac{1}{3}$, and then to multiply 2 and their product. When we follow these directions, we find that $2 \times \left(\frac{1}{2} \times \frac{1}{3}\right) = 2 \times \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$.

Although the two expressions give different sets of directions, both sets of directions lead to the same number, $\frac{1}{3}$. That is, both

expressions stand for the same number. Since they stand for the same number, we may write

$$2 \times \left(\frac{1}{2} \times \frac{1}{3}\right) = \left(2 \times \frac{1}{2}\right) \times \frac{1}{3}.$$

In general, if a, b, and c are any three measuring numbers, then

$$a \times (b \times c) = (a \times b) \times c. \tag{1}$$

That is, when you multiply three measuring numbers that are written in a definite order, then you get the same product whether you associate the middle number with the last number in the first step of the multiplication or you associate the middle number with the first number in the first step of the multiplication. Equation (1) is known as the *associative property* of multiplication of measuring numbers.

Not every binary operation has the associative property. For example, division is a binary operation in the set of all measuring numbers. But $(12 \div 6) \div 2$ is not equal to $12 \div (6 \div 2)$. In fact, $(12 \div 6) \div 2 = 2 \div 2 = 1$, but $12 \div (6 \div 2) = 12 \div 3 = 4$.

The parentheses in the expressions $a \times (b \times c)$ and $(a \times b) \times c$ tell us how the numbers are associated for the first step in the multiplication of the three numbers a, b, and c. But the same three numbers written in the same order have the same product no matter how they are associated. So there is no danger of confusion if we leave out the parentheses altogether. Thus we may write

$$a \times b \times c = a \times (b \times c) = (a \times b) \times c.$$

This equation tells us that when we write the product of three measuring numbers, we may write it either without parentheses, or with parentheses around any two numbers that are written next to each other in the product. It can be shown that, because of the associative property of multiplication, a similar rule also applies to the multiplication of more than three measuring numbers. We therefore have this general rule that follows from the associative property of multiplication of measuring numbers:

When we write the product of three or more measuring numbers, we may write it either without parentheses, or with parentheses around any two or more numbers that are written next to each other in the product.

According to this rule, the expressions $2 \times \frac{1}{2} \times \frac{1}{3} \times 5$, $\left(2 \times \frac{1}{2}\right) \times \left(\frac{1}{3} \times 5\right)$, $2 \times \left(\frac{1}{2} \times \frac{1}{3}\right) \times 5$, $2 \times \frac{1}{2} \times \left(\frac{1}{3} \times 5\right)$, and $\left(2 \times \frac{1}{2}\right) \times \frac{1}{3} \times 5$, all stand for the same measuring number.

Exercises:

8. What number does $8 - (5 - 2)$ stand for? What number does $(8 - 5) - 2$ stand for? Does subtraction of counting numbers have the associative property?

9. Verify by multiplication that these products are equal: $(4 \times 1) \times 2 \times 5$, $4 \times (1 \times 2) \times 5$, and $4 \times 1 \times (2 \times 5)$.

The Number One

The measuring number *1* behaves in a very special way when it is used as a multiplier, as these examples show:

$$1 \times 2 = 2. \qquad 2 \times 1 = 2.$$
$$1 \times 3 = 3. \qquad 3 \times 1 = 3.$$
$$1 \times 9 = 9. \qquad 9 \times 1 = 9.$$
$$1 \times \frac{3}{4} = \frac{3}{4}. \qquad \frac{3}{4} \times 1 = \frac{3}{4}.$$

In general, if a is any measuring number,

(2) $$1 \times a = a \times 1 = a.$$

That is, when any measuring number is multiplied by *1*, the number remains unchanged. Since multiplication of a measuring number by *1* produces a product that is identically equal to the number, we say that *1* is an *identity* element for the multiplication of measuring numbers. Since *1* is a member of the set of all measuring numbers, we may say that the set of all measuring numbers contains an identity element for the operation called multiplication. Since *1* is also a counting number, we may say that the set of all counting numbers contains an identity element for multiplication.

Exercises:

10. Consider the set of all odd counting numbers, *1*, *3*, *5*, *7*, *9*, and so on. Is the product of every ordered pair of members of this set also a member of the set? Is multiplication a binary operation in the set of all odd counting numbers? Is the set of all odd counting numbers closed with respect to multiplication? Does the set of all odd counting numbers contain an identity element for multiplication?

11. Consider the set of all even counting numbers, *2*, *4*, *6*, *8*, and so on. Is the product of every ordered pair of members of this set also a member of the set? Is multiplication a binary operation in the set of all even counting numbers? Is the set of all even counting numbers closed with respect to multiplication? Does the set of all even counting numbers contain an identity element for multiplication?

Inverses

Consider the products $\frac{2}{3} \times \frac{3}{2}$ and $\frac{3}{2} \times \frac{2}{3}$:

$$\frac{2}{3} \times \frac{3}{2} = \frac{2 \times 3}{3 \times 2} = \frac{6}{6} = 1;$$

$$\frac{3}{2} \times \frac{2}{3} = \frac{3 \times 2}{2 \times 3} = \frac{6}{6} = 1.$$

Since the product of the measuring numbers $\frac{2}{3}$ and $\frac{3}{2}$ is 1, we say that $\frac{3}{2}$ is the *multiplicative inverse* of $\frac{2}{3}$ and $\frac{2}{3}$ is the multiplicative inverse of $\frac{3}{2}$.

Similarly, $\frac{3}{4} \times \frac{4}{3} = 1$, and $\frac{4}{3} \times \frac{3}{4} = 1$, so we say that $\frac{4}{3}$ is the multiplicative inverse of $\frac{3}{4}$, and $\frac{3}{4}$ is the multiplicative inverse of $\frac{4}{3}$.

In general, if a is any measuring number, and there is another measuring number b that has the property

$$a \times b = b \times a = 1, \tag{3}$$

we say that b is the multiplicative inverse of a, and a is the multiplicative inverse of b.

If a is any measuring number, we can always find another measuring number that is its multiplicative inverse. In fact, if a is represented by the fraction $\frac{m}{n}$, then its multiplicative inverse is represented by the fraction $\frac{n}{m}$, which we get by interchanging the numerator and denominator. So the set of all measuring numbers has the property that it contains a multiplicative inverse for every one of its members.

If a is any measuring number, we use the symbol $\frac{1}{a}$ to stand for its multiplicative inverse. For example, the symbol $\frac{1}{\frac{1}{4}}$ stands for the multiplicative inverse of $\frac{1}{4}$, or the number 4. Using this notation, and keeping in mind equation (3), we may say that if a

is any measuring number, there is another measuring number $\frac{1}{a}$ that has the property that

$$a \times \frac{1}{a} = \frac{1}{a} \times a = 1. \tag{4}$$

Exercises:

12. What is the multiplicative inverse of *1*?

13. What is the multiplicative inverse of $\frac{5}{6}$?

14. What is the multiplicative inverse of $\frac{1}{2}$?

15. What is the multiplicative inverse of *5*?

16. Is the multiplicative inverse of *5* a counting number? Does the set of all counting numbers contain a multiplicative inverse for every one of its members?

17. What number is represented by the symbol $\frac{\frac{1}{3}}{5}$?

Powers of a Number

When two or more numbers are multiplied to get a product, we call each of these numbers a *factor* of the product. For example, in the product 2×3, there are two factors, *2* and *3*. In the product 2×2, there are two factors, and each of them is equal to *2*. In the product $2 \times 2 \times 2$, there are three factors, and each of them is equal to *2*. A product like 2×2, or $2 \times 2 \times 2$, in which all the factors are equal is called a *power* of that factor. Thus 2×2 and $2 \times 2 \times 2$ are both called *powers* of *2*. The

product 2×2, which has two factors, is called the *second power* of *2*. The product $2 \times 2 \times 2$, which has three factors, is called the *third power* of *2*. The product $2 \times 2 \times 2 \times 2$, which has four factors, is called the *fourth power* of *2*, and so on.

Sometimes, to write a power of a number, instead of writing all the equal factors, one by one, we merely write the factor once, and then we put next to it, but written a little higher, a number which tells us how many times the factor is being used. For example, to write $2 \times 2 \times 2$, in which the factor *2* is used three times, we write 2^3. This symbol is read as "*2* to the third power," and it is understood to mean $2 \times 2 \times 2$, or *8*. Similarly, the symbol 2^4 is read as "2 to the fourth power," and it is understood to mean $2 \times 2 \times 2 \times 2$, or *16*. The symbol 5^2 is read as "5 to the second power," and it is understood to mean 5×5, or *25*. The symbol 5^1 is read as "*5* to the first power," and is understood to mean *5*. In general, if a is any measuring number, the symbol a^n is read as "a to the nth power," and it is understood to mean the product $a \times a \times a \times \ldots \times a$, in which there are n factors that are all equal to a. In the expression a^n, a is called the base and n is called the exponent. Thus, in 5^2, the base is *5* and the exponent is *2*.

The second and third powers of a number are sometimes known by the special names "square" and "cube" of the number, respectively. Thus 5^2 may be read either as "*5* square," or as "*5* to the second power." Similarly, 5^3 may be read either as "*5* cube," or as "*5* to the third power."

The multiplicative inverse of a^n is represented by the symbol $\frac{1}{a^n}$. Thus, $\frac{1}{a^2}$ stands for the multiplicative inverse of a^2; $\frac{1}{a^3}$ stands for the multiplicative inverse of a^3; and so on.

If a is any measuring number, the set of numbers that consists of *1*, a, all the powers of a, and the multiplicative inverses of the powers of a, is called the *set of measuring numbers generated by a by multiplication*. For example, the set of measuring numbers

generated by *2* by multiplication consists of *1*, *2*, 2^2, 2^3, 2^4, and so on, and the numbers $\frac{1}{2}$, $\frac{1}{2^2}$, $\frac{1}{2^3}$, $\frac{1}{2^4}$, and so on. Since $2^2 = 2 \times 2 = 4$, $2^3 = 2 \times 2 \times 2 = 8$, $2^4 = 2 \times 2 \times 2 \times 2 = 16$, and so on, the set of measuring numbers generated by *2* by multiplication consists of *1*, *2*, *4*, *8*, *16* . . . , and $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, ... The three dots . . . used in the preceding sentence stand for the words "and so on."

Exercises:

18. What does the symbol $\left(\frac{1}{2}\right)^2$ mean? What measuring number does it stand for?

19. What is the meaning of a^4 if $a = \frac{2}{3}$?

20. What is the meaning of $\frac{1}{a^2}$ if $a = 3$?

21. List the members of the set of measuring numbers generated by *3* by multiplication. (Use three dots to stand for "and so on.") Is the product of every ordered pair of members of this set also a member of the set? Is this set closed with respect to multiplication? Is the multiplicative inverse of every member of this set also a member of this set?

Multiplying Powers of the Same Base

When we multiply two powers of the same base, the product is also a power of that base. For example, in the product $5^3 \times 5^4$, 5^3 can be replaced by $5 \times 5 \times 5$, and 5^4 can be replaced by $5 \times 5 \times 5 \times 5$. When these substitutions are made, we see that $5^3 \times 5^4 = 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 = 5^7$. Notice that the ex-

ponent *7* is the sum of the exponents *3* and *4* of the powers that were being multiplied.

In general, if a is any measuring number, and we multiply a^m and a^n, a^m contributes to the product m factors that are all equal to a, and a^n contributes to the product n factors that are all equal to a. So, altogether, the product contains $m + n$ factors that are all equal to a, and the product can be represented by the symbol a^{m+n}, in which the base is a, and the exponent is $m + n$. Thus we have this simple rule for multiplying powers of the same base: To multiply two powers of the same base, copy the base and add the exponents.

Exercises:

22. Write the product $2^6 \times 2^4$ as a power of *2*.
23. Write the product 4×4^8 as a power of *4*.

Four Properties

We have observed four properties that the set of all measuring numbers has with respect to the operation called multiplication:

I. Closure. Multiplication is a binary operation in the set of all measuring numbers. So the set of all measuring numbers is closed with respect to multiplication.

II. Associative Property. Multiplication of measuring numbers has the associative property. That is, if a, b, and c are any three measuring numbers, then

$$(1) \qquad a \times (b \times c) = (a \times b) \times c.$$

III. Identity Element. The set of all measuring numbers contains an identity element *1* for multiplication. If a is any measuring number, then

(2) $$1 \times a = a \times 1 = a.$$

IV. Inverse. The set of all measuring numbers contains a multiplicative inverse $\frac{1}{a}$ for every one of its members a, with the property

(4) $$a \times \frac{1}{a} = \frac{1}{a} \times a = 1.$$

It is interesting to see which of these properties the set of all counting numbers has with respect to multiplication:

1. Closure. We saw in exercise 6 on page 18 that multiplication is a binary operation in the set of all counting numbers. So the set of all counting numbers is closed with respect to multiplication.

2. Associative Property. Let a, b, and c be any three counting numbers. Since every counting number is also a measuring number, then a, b and c are also measuring numbers. Since multiplication of measuring numbers has the associative property, then $a \times (b \times c) = (a \times b) \times c$. That is, equation (1) is true for any three counting numbers a, b, and c. Therefore, multiplication of counting numbers has the associative property.

3. Identity Element. Since 1 is a counting number, the set of all counting numbers contains an identity element for multiplication.

4. Inverse. We saw in exercise 16 on page 24 that the set of all counting numbers does *not* contain a multiplicative inverse for every one of its members.

Consequently, with respect to the operation called multiplication, the set of all counting numbers does have properties I, II, and III, but it does not have property IV.

Exercises:

24. Consider the set of all odd counting numbers. Which of the four properties, *Closure*, *Associative Property*, *Identity Element*, and *Inverse*, does this set have with respect to multiplication?

25. Consider the set of all even counting numbers. Which of the four properties, *Closure*, *Associative Property*, *Identity Element*, and *Inverse*, does this set have with respect to multiplication?

3

Addition of Integers

In Chapter 2 we observed four properties that the set of all measuring numbers has with respect to the operation called multiplication. We called these properties *Closure*, *Associative Property*, *Identity Element*, and *Inverse*. In this chapter we turn our attention to the set of all counting numbers, and to another operation, the operation called addition. We shall examine the set of all counting numbers to see if it has, with respect to addition, properties similar to those that the set of all measuring numbers has with respect to multiplication. We shall find that the set of all counting numbers, with the operation addition, does not have all four of these properties. But if we enlarge the set by putting in more numbers, a certain enlarged set called *the set of all integers* does have these four properties with respect to addition.

Closure

In the set of all counting numbers, the operation called addition assigns to each ordered pair of counting numbers a definite third number called their sum. For example, addition assigns to the ordered pair $(2, 3)$ the sum 5. To show this assignment we write $2 + 3 = 5$.

Addition assigns a sum to *every* ordered pair of counting numbers. Moreover, the sum is itself a counting number. So the operation called addition is a binary operation in the set of all counting numbers. Using the language introduced on page 18, we may say then that the set of all counting numbers is closed with respect to

addition. Thus, the set of all counting numbers has with respect to addition the property of *Closure*.

The Associative Property

The expression $(2 + 3) + 7$ gives us directions for carrying out the addition in two steps. It tells us to add first the numbers *2* and *3*, and then to add their sum and the number *7*. When we follow these directions, we find that $(2 + 3) + 7 = 5 + 7 = 12$.

The expression $2 + (3 + 7)$ contains the same three numbers, *2*, *3* and *7*, written in the same order. But it gives us a different set of directions. It tells us to add first the numbers *3* and *7*, and then to add *2* and their sum. When we follow these directions, we find that $2 + (3 + 7) = 2 + 10 = 12$.

Although the two expressions give different sets of directions, both sets of directions lead to the same number, *12*. That is, both expressions stand for the same number. Since they stand for the same number, we may write

$$2 + (3 + 7) = (2 + 3) + 7.$$

In general, if a, b, and c are any three counting numbers, then

$$a + (b + c) = (a + b) + c. \tag{5}$$

Equation (5) is like equation (1) in Chapter 2, with plus signs written in equation (5) where multiplication signs are written in equation (1). It expresses the fact that addition of counting numbers has the *Associative Property*.

The parentheses in the expressions $a + (b + c)$ and $(a + b) + c$ tell us how the numbers are associated for the first step in the addition of the numbers a, b and c. But the same three numbers written in the same order have the same sum no matter how they are associated. So there is no danger of confusion if we leave out the parentheses altogether. Then we may write

$$a + b + c = a + (b + c) = (a + b) + c.$$

Exercises:

1. Carry out the addition $(9 + 16) + 43$ in two steps.

2. Carry out the addition $9 + (16 + 43)$ in two steps. Verify that exercises 1 and 2 both have the same answer.

3. On page 21 there is a rule about the use of parentheses in writing the product of three or more measuring numbers. This rule is based on the fact that multiplication of measuring numbers has the associative property. Since addition of counting numbers also has the associative property, a similar rule exists for writing the sum of three or more counting numbers. Write this rule.

Identity Element

The number *1* is an identity element for multiplication in the set of all measuring numbers. It has the property that when any number in the set is multiplied by *1*, the number remains unchanged. A member of a set of numbers is called an identity element for addition if it behaves with respect to addition the way the number *1* behaves with respect to multiplication. That is, a member of a set of numbers is an identity element for addition if, when it is added to any number in the set, the number remains unchanged. In the set of all counting numbers there is no number that behaves this way with respect to addition. If you add *1* to any counting number, you get the next higher number, not the same number. If you add *2* to any counting number, you get a higher number. If you add *3* to any counting number, you get a higher number; and so on. So the set of all counting numbers does not have an identity element for addition.

Suppose we enlarge the set of numbers we use by putting into the set the number *0* as well as all the counting numbers. This enlarged set, which consists of the numbers *0*, *1*, *2*, *3*, *4*, and so on, is called the *set of all whole numbers*. The number *0* has the prop-

erty that when it is added to any whole number, that number remains unchanged. For example, $0 + 5 = 5 + 0 = 5$. In general, if a is any whole number,

$$0 + a = a + 0 = a. \tag{6}$$

Equation (6) is like equation (2) in Chapter 2, with *0* written in equation (6) where *1* is written in equation (2), and with a plus sign written in equation (6) where there is a multiplication sign in equation (2). Since *0* has this property, it is an identity element for addition in the set of all whole numbers. Thus, while the set of all counting numbers does not have an identity element for addition, the set of all whole numbers does.

We may picture the set of all whole numbers as a set of points on a number line by extending the picture given on page 13 for the set of all counting numbers. We let the starting point stand for the number *0*, and we let the equally spaced points to the right of the starting point stand for the counting numbers, as they do in the picture on page 13. Thus we get this picture of the set of all whole numbers:



The Whole Numbers as Points on the Number Line

To add two whole numbers we follow these rules: If neither of them is *0*, then they are both counting numbers, and we add them as we add counting numbers. If one of them is *0* or both of them are *0*, then we add them by the rule given in equation (6).

Exercises:

4. If a and b are whole numbers, and neither of them is *0*, is $a + b$ a whole number? Why?

5. If a and b are whole numbers, and $a = 0$ but b is not 0, is $a + b$ a whole number? Why?

6. If a and b are whole numbers, and a is not 0, but $b = 0$, is $a + b$ a whole number? Why?

7. If a and b are whole numbers, and $a = 0$ and $b = 0$, is $a + b$ a whole number? Why?

8. What conclusion can you draw from the results of exercises 4, 5, 6, and 7?

9. If a, b, and c are whole numbers, and neither of them is 0, use the rules for adding whole numbers to prove that $a + (b + c) = (a + b) + c$.

10. If a, b, and c are whole numbers, and $a = 0$, prove that $a + (b + c) = (a + b) + c$.

11. If a, b, and c are whole numbers, and $b = 0$, prove that $a + (b + c) = (a + b) + c$.

12. If a, b, and c are whole numbers, and $c = 0$, prove that $a + (b + c) = (a + b) + c$.

13. What conclusion can you draw from the results of exercises 9, 10, 11 and 12?

Inverse

In exercise 8 we found out that the set of all whole numbers is closed with respect to addition. In exercise 13 we found out that addition of whole numbers has the associative property. Equation (6) tells us that the set of all whole numbers contains an identity element for addition. So the set of all whole numbers has with respect to addition the properties *Closure*, *Associative Property*, and *Identity Element*. Let us see now whether it also has the *Inverse* property.

The first thing we have to do is state clearly what we mean by the *Inverse* property with respect to the operation addition. On

page 23, where we were talking about the multiplication of two numbers a and b, we said that b is the multiplicative inverse of a, and a is the multiplicative inverse of b if a and b have the property

$$a \times b = b \times a = 1 \tag{3}$$

where 1 is the identity element for multiplication. We can change this definition into one that applies to the addition of two numbers a and b by replacing the word *multiplicative* by the word *additive*, replacing the sign $\times$ by the sign $+$, and replacing 1, which is the identity element for multiplication, by 0, which is the identity element for addition. We obtain in this way the following definition of additive inverse: *The number b is the additive inverse of a, and a is the additive inverse of b if a and b have the property*

$$a + b = b + a = 0. \tag{7}$$

That is, if the sum of two numbers is 0, then each is the additive inverse of the other.

We said on page 23 that the set of all measuring numbers has the *Inverse* property with respect to *multiplication* because it contains a *multiplicative* inverse for every one of its members. Similarly, we shall say that a set of numbers has the *Inverse* property with respect to *addition* if it contains an *additive* inverse for every one of its members.

Let us see then if the set of all whole numbers contains an additive inverse for every one of its members. To do so, we examine each of the numbers 0, 1, 2, 3, 4, and so on, in turn, to see if there is a whole number that is its additive inverse.

We ask first if there is any whole number that we can add to 0 to obtain 0 as their sum. We know there is, because $0 + 0 = 0$. So 0 is the additive inverse of 0.

We ask if there is a whole number that we can add to 1 to obtain 0 as their sum. To answer this question, let us add each whole number in turn to 1:

$$1 + 0 = 1;$$
$$1 + 1 = 2;$$
$$1 + 2 = 3;$$

and so on.

This shows that if we add a whole number to *1*, their sum is not less than *1*. There is no whole number we can add to *1* to obtain *0* as their sum. So there is no whole number that is the additive inverse of *1*.

Similarly, if we add a whole number to *2*, their sum is not less than *2*. There is no whole number we can add to *2* to obtain *0* as their sum. So there is no whole number that is the additive inverse of *2*. In the same way, we find that there is no whole number that is the additive inverse of *3*, or *4*, or *5*, and so on.

Thus we see that the set of all whole numbers contains an additive inverse for *0*, but it does not contain an additive inverse for any other of its members. Since it does not contain an additive inverse for every one of its members, it does not have the *Inverse* property with respect to addition.

To obtain a set of numbers that includes all the whole numbers and also contains an additive inverse for every one of its members, we have to enlarge the set of numbers again.

The Other Side of the Number Line

The way in which we should enlarge the set of numbers we are using is suggested by the way in which we have pictured the whole numbers as points on the number line. On the number line, the starting point stands for *0*, and equally spaced points to the right of the starting point stand for the other whole numbers. We can get some more equally spaced points on the number line by putting them in to the left of the starting point, or zero point. This suggests that we let these points to the left of the zero point stand

for some new numbers. The point that is one unit to the right of the zero point stands for the number *1*. We let the point that is one unit to the left of the zero point stand for a number we call *negative one* and write as -1. The point that is two units to the right of the zero point stands for the number *2*. We let the point that is two units to the left of the zero point stand for a number we call *negative two* and write as -2. In general, if n is any whole number that is not *0*, we let the point that is n units to the left of the zero point stand for a number we call *negative n* and write as $-n$. The enlarged set of numbers that we obtain in this way



The Integers as Points on the Number Line

is called the *set of all integers*. The numbers *1*, *2*, *3*, . . . that are represented by points to the right of the zero point are called *positive integers*. The numbers -1, -2, -3, . . . that are represented by points to the left of the zero point are called *negative integers*. The positive integers are the same as the counting numbers.

Numbers as Motions

We have extended the system of whole numbers in the hope that the extended system, the system of integers, will contain an additive inverse for every one of its members. Before there can be any such thing as an *additive inverse* for an integer, there must be an operation of *addition* of integers. To see how we should define addition of integers, we first examine some new ways in which we may picture the integers on the number line.

We have already pictured each integer as a point on the number line. If we draw an arrow to that point from the zero point, we

get a second picture that can stand for the integer. Thus, an arrow drawn with its tail at the zero point and its head at the *1* point stands for *1*. An arrow drawn with its tail at the zero point and its head at the -1 point stands for -1. An arrow drawn with its tail at the zero point and its head at the *2* point stands for *2*. An arrow drawn with its tail at the zero point and its head at the -2 point stands for -2, and so on. In this way of picturing integers as arrows whose tails are at the zero point, the arrow that stands for a positive integer n has a length of n units and points to the right. The arrow that stands for a negative integer $-n$ has

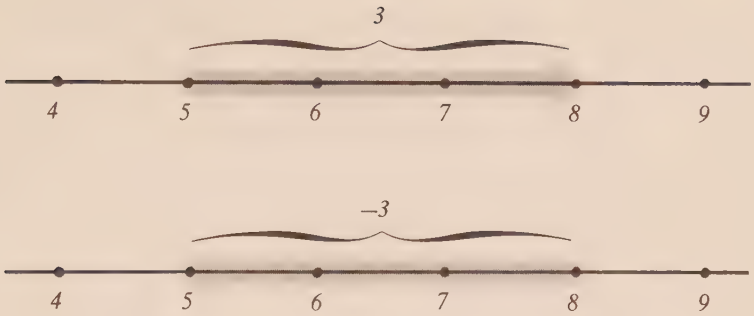


Integers Pictured as Arrows Attached to the Zero Point

a length of n units and points to the left. The arrow that stands for *0* has a length of *0* units and may be thought of as pointing either to the right or to the left.

In this second way of picturing an integer, it is represented by an arrow whose tail is attached to the zero point. We get a third way of picturing the integer by detaching the arrow from the zero point and putting it anywhere on the number line, as long as we do not change its length and direction. Then, any arrow whose length is *1* unit and that points to the right stands for the integer *1*. Any arrow whose length is *1* unit and points to the left stands for the integer -1. Any arrow whose length is *2* units and points to the right stands for the integer *2*. Any arrow whose length is *2* units and points to the left stands for the integer -2, and so on. For example, an arrow whose tail is at the *5* point and whose head is at the *8* point is *3* units long and points to the right. So it stands for the positive integer *3*. An arrow whose tail is at the *8* point and whose head is at the *5* point is *3* units long and points to the left. So it stands for the negative integer -3.

The arrow picture of an integer suggests a fourth way of pic-



Integers Pictured as Detachable Arrows

turing the integer. We may think of the integer as a *motion* along the number line from the tail of the arrow to the head of the arrow. Such a motion from one place to another place is called a *displacement*. Since the tail of an arrow may be put at any point on the number line, the motion or displacement may start from any point on the number line. In this system for picturing integers, a positive integer n is pictured as a motion to the right through a distance of n units. A negative integer $-n$ is pictured as a motion to the left through a distance of n units. For example, the integer *2* may be thought of as a motion from the *0* point to the *2* point, or from the *1* point to the *3* point, or from the *2* point to the *4* point, or from the -1 point to the *1* point, or from the -2 point to the *0* point, and so on.

Exercises:

14. What integer is represented by an arrow whose tail is at the *6* point and whose head is at the *10* point?

15. What integer is represented by an arrow whose tail is at the -1 point and whose head is at the *2* point?

16. What integer is represented by an arrow whose tail is at the -4 point and whose head is at the -3 point?

17. What integer is represented by an arrow whose tail is at the *5* point and whose head is at the *3* point?

18. What integer is represented by an arrow whose tail is at the *1* point and whose head is at the -4 point?

19. What integer is represented by an arrow whose tail is at the -2 point and whose head is at the -3 point?

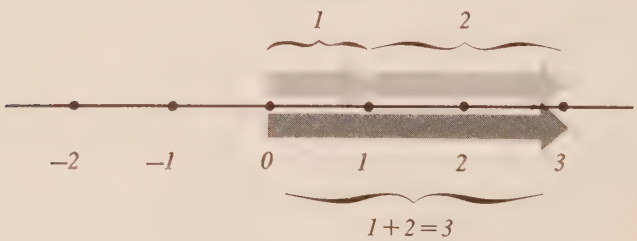
20. If the integer *6* is represented by an arrow with its tail at the -2 point, where is its head?

21. If the integer -3 is represented by an arrow with its tail at the -2 point, where is its head?

22. What integer is represented by a motion from the *8* point to the *10* point? From the *10* point to the *8* point?

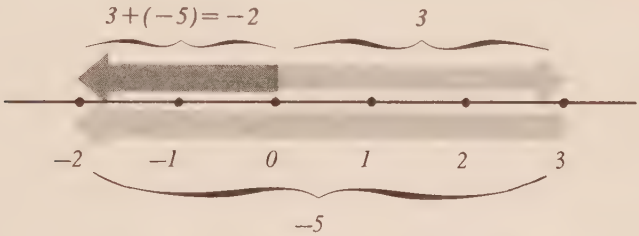
Combining Motions

We are now ready to define addition of integers. We add integers by combining the motions or displacements that represent them. The motions are combined by performing one motion after the other. The sum of the two motions is the single motion that has the same effect as the combined motions. For example, to find the sum $1 + 2$, we picture *1* as a motion *1* unit to the right, and we picture *2* as a motion *2* units to the right. To combine the motions, move a pointer on the number line, first *1* unit to the right and then, after that, *2* units to the right. The combined motions have the same effect as a single motion *3* units to the right. So $1 + 2 = 3$. To find the sum $3 + (-5)$, we picture *3* as



a motion *3* units to the right, and we picture -5 as a motion *5* units to the left. To combine the motions, move a pointer on the number line, first *3* units to the right and then, after that, *5* units to the left. The combined motions have the same effect as a single motion *2* units to the left. So, $3 + (-5) = -2$.

When we add two integers by combining the motions that represent them, we can get the answer automatically *by starting the first motion at the zero point.* Then the point where the second motion ends represents the sum. For example, to find the sum $3 + (-5)$, start at the zero point and move three units to the right, and then move *5* units to the left. The first motion is represented in the diagram below by an arrow whose tail is at the *0* point and whose head is at the *3* point. The second motion is represented by an arrow whose tail is at the *3* point and whose



head is at the -2 point. The single motion that has the same effect as these two motions combined is a motion from the *0* point to the -2 point, and it is represented by the arrow whose tail is at the *0* point and whose head is at the -2 point. But in our second system for picturing integers, this arrow represents the integer -2. So the point where the second motion ends actually represents the result of combining the two motions, or the sum of the two integers.

Exercises:

23. By starting at the zero point and combining two motions, find the sum $2 + (-4)$.

24. By starting at the zero point and combining two motions, find the sum $6 + (-2)$.

25. By starting at the zero point and combining two motions, find the sum $5 + (-5)$.

The Four Properties

Now that we have defined addition of integers, we can examine the set of all integers to see if it has with respect to addition the four properties, *Closure*, *Associative Property*, *Identity Element* and *Inverse*.

I. Closure. To add any two integers, first we picture each integer as a motion through a whole number of units, to the right or to the left, on the number line. Then we combine the two motions to see what single motion has the same effect. The single motion that has the same effect as the combined motions is itself a motion through a whole number of units on the number line. But every such motion stands for an integer. So the sum of any two integers is an integer. Therefore, the set of all integers is closed with respect to addition.

II. Associative Property. Let a, b, and c be any three integers. The expression $a + (b + c)$ stands for the single motion that has the same effect as the motion a followed by the motion $b + c$. The motion $b + c$ is the single motion that has the same effect as the motion b followed by the motion c. So $a + (b + c)$ has the same effect as the motion a followed by the motion b followed by the motion c. The expression $(a + b) + c$ stands for the single motion that has the same effect as the motion $a + b$ followed by the motion c. The motion $a + b$ is the single motion that has the same effect as the motion a followed by the motion b. So $(a + b) + c$ has the same effect as the motion a followed by

the motion b followed by the motion c. Since $a + (b + c)$ and $(a + b) + c$ have the same effect, they are equal. That is,

$$a + (b + c) = (a + b) + c. \tag{5}$$

So addition of integers has the associative property.

III. Identity Element. Since the integer 0 is pictured as a motion of 0 units to the right or left, it is obvious that $0 + a$ and $a + 0$ stand for the same motion as a itself. So, for every integer a,

$$0 + a = a + 0 = a. \tag{6}$$

That is, the set of all integers contains an identity element 0 for addition.

IV. Inverse. If a is any integer, it stands for a motion of some whole number of units to the right or to the left. Then there is another integer b that stands for a motion of the same number of units in the opposite direction. If we combine two motions of the same number of units in opposite directions they cancel each other. That is,

$$a + b = b + a = 0. \tag{7}$$

This means that the set of all integers contains an additive inverse for every one of its members.

If a is any integer, we use the symbol $-a$ to stand for its additive inverse. For example, if a is the positive integer 2, its additive inverse, $-a$, is the negative integer -2. If a is the negative integer -3, its additive inverse, $-a$, is the positive integer 3. That is, $-(-3) = 3$. Using this notation, and keeping in mind equation (7), we can say that if a is any integer, there is another integer $-a$ that has the property that

$$a + (-a) = (-a) + a = 0. \tag{8}$$

So, the set of all integers has with respect to addition the same four properties, *Closure*, *Associative Property*, *Identity Element*, and *Inverse*, that the set of all measuring numbers has with respect to multiplication.

Exercises:

26. What is the additive inverse of 6? of -8?
27. Find $3 + ((-2) + 6)$ in two steps by first finding $(-2) + 6$.
28. Find $(3 + (-2)) + 6$ in two steps by first finding $3 + (-2)$.
29. Use the results of exercises 27 and 28 to verify that $3 + ((-2) + 6) = (3 + (-2)) + 6$.
30. What integer does $-(-5)$ stand for?

Zero as an Exponent

On page 25 we said that, if a is any measuring number, the symbol a^n is understood to mean the product $a \times a \times \ldots \times a$ in which there are n factors that are all equal to a. This definition assigns a meaning to a^n only when the exponent n is a counting number or a positive integer. We shall now extend the definition by assigning a meaning to a^n when n is 0. That is, we shall now assign a meaning to a^0.

On page 27 we obtained this rule for multiplying powers of the same base: To multiply two powers of the same base, copy the base and add the exponents. This is a convenient rule to have. We shall therefore let a^0 stand for a measuring number chosen in such a way that this rule is still obeyed.

Suppose we multiply a^0 by a^n. If the rule for multiplying powers of the same base is obeyed, we have

$$a^n \times a^0 = a^{n+0}.$$

Since *0* is the identity element for the addition of integers, $n + 0 = n$. Therefore,

$$a^n \times a^0 = a^n.$$

If we multiply both sides of this equation by $\frac{1}{a^n}$, the multiplicative inverse of a^n, we get

$$\frac{1}{a^n} \times a^n \times a^0 = \frac{1}{a^n} \times a^n.$$

But $\frac{1}{a^n} \times a^n = 1$. So we have

$$1 \times a^0 = 1.$$

Since *1* is the identity element for the multiplication of measuring numbers, $1 \times a^0 = a^0$. So we have $a^0 = 1$. Then this is the way we must define a^0 if we want the rule for multiplying powers of the same base to be obeyed: *The zero power of any measuring number is defined to be equal to 1.*

Negative Integers as Exponents

We now extend further the definition of a^n by assigning a meaning to a^n when n is a negative integer. Again we do it in such a way that the rule for multiplying powers of the same base is still obeyed.

Suppose n is the negative integer $-m$, assume that a^{-m} stands for a measuring number, and multiply a^{-m} by a^m. If the rule for multiplying powers of the same base is obeyed, we have

$$a^m \times a^{-m} = a^{m+(-m)}.$$

Since $-m$ is the additive inverse of m, $m + (-m) = 0$. So we have

$$a^m \times a^{-m} = a^0 = 1.$$

Similarly,

$$a^{-m} \times a^m = a^{(-m)+m} = a^0 = 1.$$

For these equations to be true, a^{-m} must be the multiplicative inverse of a^m. We have already agreed to represent the multiplicative inverse of a^m by the symbol $\frac{1}{a^m}$. So we define a^{-m} to mean $\frac{1}{a^m}$. Thus a^{-1} means $\frac{1}{a^1}$, or $\frac{1}{a}$, a^{-2} means $\frac{1}{a^2}$, a^{-3} means $\frac{1}{a^3}$, etc.

Exercises:

31. What number does 5^0 stand for?
32. What number does $\left(\frac{1}{2}\right)^0$ stand for?
33. What number does 3^{-2} stand for?
34. What number does 2^{-5} stand for?
35. If a is a measuring number, write as powers of a the members of the set of numbers generated by a.
36. Multiply: $a^{-2} \times a^2$; $a^{-3} \times a^3$.
37. Multiply: $a^{-2} \times a^{-5}$; $a^{-2} \times a^5$; $a^{-2} \times a$.
38. Multiply: $a \times a^2$; $a \times a^3$; $a \times a^4$; $a \times a^{-4}$.
39. Multiply: $a \times a^n$.

4

The Idea of a Group

In Chapter 2 we found that the set of all measuring numbers has with respect to the operation called multiplication the four properties *Closure*, *Associative Property*, *Identity Element*, and *Inverse*. In Chapter 3 we found that the set of all integers has the same four properties with respect to the operation called addition. In order to emphasize this resemblance even more, we introduce a slight change in notation.

We have been using the symbol $\times$ to stand for the operation multiplication in the set of all measuring numbers. Let us temporarily replace this symbol by the symbol $\circ$, the initial letter of the word "operation." Read this symbol as "oh." Then the symbol $\circ$ stands for a binary operation in the set of all measuring numbers, and if (a, b) is an ordered pair of measuring numbers, $a \circ b$ stands for the measuring number that the operation $\circ$ assigns to this ordered pair. We have been using the symbol I to stand for the measuring number that is the identity element for multiplication. Let us temporarily replace this symbol by the symbol i, the initial letter of the word "identity." Then we can restate in terms of these new symbols the four properties that the set of all measuring numbers has with respect to multiplication:

I. Closure. The operation $\circ$ is a binary operation in the set. The set is closed with respect to $\circ$.

II. Associative Property. If a, b, and c are any three members of the set, they satisfy equation (1) on page 20. When we replace $\times$ by $\circ$, equation (1) takes the form of equation (9):

(9) $$a \circ (b \circ c) = (a \circ b) \circ c.$$

III. Identity Element. If a is any member of the set, it satisfies equation (2) on page 21. When we replace $\times$ by $\circ$, and replace *1* by i, equation (2) takes this form:

(10) $$i \circ a = a \circ i = a.$$

IV. Inverse. If a is any member of the set, there is another member b in the set that satisfies equation (3) on page 23. When we replace $\times$ by $\circ$, and replace *1* by i, equation (3) takes this form:

(11) $$a \circ b = b \circ a = i.$$

Now let us introduce the same change in notation for the operation addition in the set of all integers. Let us temporarily replace $+$ by $\circ$, and temporarily replace *0* by i. Then we can restate in terms of these new symbols the four properties that the set of all integers has with respect to addition:

I. Closure. The operation $\circ$ is a binary operation in the set. The set is closed with respect to $\circ$.

II. Associative Property. If a, b, and c are any three members of the set, they satisfy equation (5) on page 31. When we replace $+$ by $\circ$, equation (5) takes this form:

(9) $$a \circ (b \circ c) = (a \circ b) \circ c.$$

III. Identity Element. If a is any member of the set, it satisfies equation (6) on page 33. When we replace $+$ by $\circ$, and replace *0* by i, equation (6) takes this form:

(10) $$i \circ a = a \circ i = a.$$

IV. Inverse. If a is any member of the set, there is another member b in the set that satisfies equation (7) on page 35. When we replace $+$ by $\circ$, and replace 0 by i, equation (7) takes this form:

$$a \circ b = b \circ a = i. \tag{11}$$

Thus we see that when we use this new notation, the four properties that the set of all measuring numbers has with respect to multiplication, and the four properties that the set of all integers has with respect to addition can be expressed in exactly the same language. In both cases, we have a set of elements, a binary operation $\circ$ in the set and an element i in the set, with these properties:

I. Closure. The set is closed with respect to $\circ$.

II. Associative Property. If a, b, and c are any three members of the set, they satisfy equation (9).

III. Identity Element. If a is any member of the set, it satisfies equation (10).

IV. Inverse. If a is any member of the set, there is another member b in the set that satisfies equation (11).

Definition of a Group

Whenever a set, taken with a particular binary operation $\circ$ in the set, has these four properties, we call the set a *group* with respect to this operation. Thus we see that the set of all measuring numbers is a group with respect to multiplication, and the set of all integers is a group with respect to addition. We shall get acquainted with many other groups in the chapters that follow.

In a group, the element i that satisfies equation (9) is called the *identity element* of the group. It can be shown that a group has only one identity element. (See page 96.) If a is an element of a group, the element b that satisfies equation (11) is called the *inverse of* a. It can be shown that an element of a group has only one inverse. (See page 97.)

In the group of measuring numbers, with the operation multiplication represented by the symbol $\times$, we represented the multiplicative inverse of a by the symbol $\frac{1}{a}$. When this symbol is used, equation (3) takes the form of equation (4) on page 28. In the group of integers, with the operation addition represented by the symbol $+$, we represented the additive inverse of a by the symbol $-a$. When this symbol is used, equation (7) takes the form of equation (8) on page 43. When we use the symbol $\circ$ to stand for the group operation, we write the inverse of an element of the group in still another way. We take our cue from page 46, where we found that the symbol a^{-1} is another way of writing $\frac{1}{a}$. When the operation in a group is represented by the symbol $\circ$, we represent the inverse of a by a^{-1}. Then equations (3), (4), (7), (8), and (11) take this form:

$$a \circ a^{-1} = a^{-1} \circ a = i. \tag{12}$$

Simplified Notation

In any group, the operation $\circ$ is like the operation multiplication in the set of all measuring numbers. For this reason it is often called the *multiplication* operation in the group, $a \circ b$ is called the product of a and b, and a and b are called the factors of $a \circ b$. In order to simplify the writing of products, it is customary to leave out the symbol $\circ$ and simply write the factors right next to each other. That is, we write ab for the product of a and b, and

understand it to mean $a \circ b$. Using this simplified notation, we can restate the definition of a group as follows:

A group is a set of elements with a binary operation in the set with these four properties:

I. Closure. If (a, b) is an ordered pair of members of the set, the operation assigns to this ordered pair a member of the set denoted by ab. The set is closed with respect to the operation.

II. Associative Property. If a, b and c are any three members of the set, they satisfy

$$a(bc) = (ab)c. \tag{13}$$

III. Identity Element. The set contains an element i, called the identity element, with the property that, if a is any member of the set,

$$ia = ai = a. \tag{14}$$

IV. Inverse. If a is any member of the set, there is in the set a member a^{-1}, called the inverse of a, with the property that

$$aa^{-1} = a^{-1}a = i. \tag{15}$$

Not every set with a binary operation is a group. For example, consider the set of all counting numbers with the operation $\times$. The set is closed with respect to the operation, the operation has the associative property, and the set contains an identity element for the operation, namely the number *1*. So the set of all counting numbers with the operation $\times$ has properties I, II and III of a group. However, the set does not contain a multiplicative inverse for every one of its members. For example, there is no counting number that you can multiply by *3* to get *1* as a product. So the set of all counting numbers with the operation $\times$ does not have property IV of a group. Therefore it is not a group.

Exercises:

1. Which of the four properties of a group are possessed by the set of all whole numbers with the operation $+$? Is the set of all whole numbers a group with respect to addition?

2. Consider the set of all measuring numbers with the operation $\div$. Is the set closed with respect to the operation? Does the operation have the associative property? Is the number *1* an identity element with respect to the operation? Is the set of all measuring numbers with the operation $\div$ a group?

3. Consider the set of all even integers whose members are *0, 2, 4, 6,* . . . and $-2, -4, -6, \ldots$, and use the operation $+$. Which of the four properties of a group are possessed by this set taken with this operation? Is the set of all even integers a group with respect to addition?

5

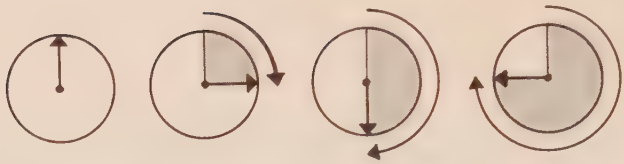
The Turns of a Wheel

In the preceding chapters we were introduced to two groups, the set of all measuring numbers taken with the operation multiplication, and the set of all integers taken with the operation addition. In each of these groups the members of the group are numbers. There are also groups whose members are not numbers. As examples of these we shall examine some groups whose members are the turns of a wheel.

The Measure of a Turn

Consider a wheel that is free to turn on its axle. To help us see what happens as the wheel turns, let us imagine that there is an arrow drawn on the wheel from the axle to the rim of the wheel. The wheel is free to turn in either of two directions. If it turns in the clockwise direction, the arrow moves like a hand of a clock. If it turns in the counterclockwise direction, the arrow moves the other way.

Suppose we start with the arrow in any position, and then start turning the wheel clockwise, as shown in the drawings on page 54. As the wheel turns, the arrow takes on different positions, until finally it arrives at its original position again. If the wheel is allowed to turn for a long time, the arrow goes through the same series of positions over and over again, and returns to its starting position many times. The amount of turning that it takes to bring the arrow back for the first time to its starting position



is called *one complete turn*. Other amounts of turning may be measured by using one complete turn as a unit of measure. The turn shown in the second diagram above is one-fourth of a complete turn. The turn shown in the third diagram is one-half of a complete turn. The turn shown in the fourth diagram is three-fourths of a complete turn. A turn that brings the arrow back for the second time to its starting position is two complete turns.

For measuring turns that are less than a complete turn, it is convenient to use a smaller unit of measure. The unit that is most commonly used is called a *degree*, and it is represented by the symbol °. It is the amount of turning which, when repeated *360* times, produces exactly one complete turn. Expressed in terms of this unit of measure, one complete turn is *360°* (read as "*360* degrees"). One-fourth of a complete turn is *90°*. Two complete turns is *720°*.

Exercises:

1. How many degrees are there in one-half of a complete turn? in three-fourths of a complete turn? in one-third of a complete turn? in three complete turns?
2. What fraction of a complete turn is *60°*?
3. How many complete turns are there in *1440°*?

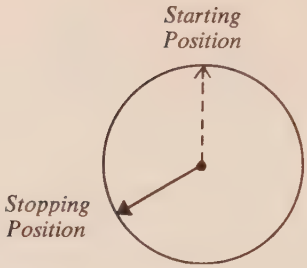
Turns That Have the Same Effect

Every turn of the wheel moves the arrow on the wheel from one position to another. There are many different amounts of

turning that can bring the arrow from a fixed starting position to any given stopping position. For example, suppose we start with the arrow pointing up, and stop with the arrow pointing to the right. We can reach this stopping position by turning the wheel $90°$ clockwise. However, we can also reach it by turning the wheel $450°$ clockwise, because $450° = 90° + 360°$, and the first $90°$ of turning bring the arrow to the position where it points to the right, and the next $360°$ of turning bring the arrow back to the same position. For the same reason, a clockwise turn of $90° + 360° + 360°$, or $810°$, will also bring the arrow to the same stopping position.

We shall consider two turns to be equal if they have the same effect in moving an arrow drawn from the center of the wheel. That is, we shall say two turns are equal if they carry the arrow from a fixed starting position to the same stopping position. Then a clockwise turn of $810°$ is equal to a clockwise turn of $90°$. A clockwise turn of $360°$ is equal to a turn of $0°$. If the number of degrees in a clockwise turn is 360 or more, we can find an equal clockwise turn with less than $360°$ by simply subtracting 360 over and over again from this number until the remainder is less than 360. For example, if we subtract 360 twice from 730, the remainder is 10. So a clockwise turn of $730°$ is equal to a clockwise turn of $10°$. Thus every clockwise turn can be expressed as a clockwise turn of less than $360°$. Similarly, every counterclockwise turn can be expressed as a counterclockwise turn of less than $360°$.

Suppose we want to move the arrow on the wheel from the starting position to the stopping position shown in the diagram on p. 56. We can do so by turning the wheel either clockwise or counterclockwise. If the clockwise and counterclockwise turns are both less than $360°$, then the diagram shows that the sum of their measures is $360°$. Because of this fact, every counterclockwise turn can be expressed as a clockwise turn of less than $360°$. For example, a counterclockwise turn of $410°$ is equal to a counterclockwise turn of $50°$. But a counterclockwise turn of $50°$ is equal



to a clockwise turn of *310°*. So a counterclockwise turn of *410°* is equal to a clockwise turn of *310°*.

Because of the facts observed in the last two paragraphs, every turn can be expressed as a clockwise turn of less than *360°*.

Exercises:

4. Express as a clockwise turn of less than *360°*: a) a clockwise turn of *380°*; b) a clockwise turn of *1500°*; c) a counterclockwise turn of *80°*; d) a counterclockwise turn of *400°*; e) a counterclockwise turn of *0°*.

Since every counterclockwise turn can be expressed as a clockwise turn, we shall lose nothing by considering only clockwise turns. So, from now on, we shall speak only of clockwise turns. Since all the turns we speak of shall be clockwise, it will not be necessary for us to keep mentioning this fact. So, in the remainder of this chapter, wherever a turn is mentioned, it will be understood to be a clockwise turn.

Combining Turns

We can combine two turns to form a single turn by performing the turns one right after the other. For example, to combine a *10°* turn and a *20°* turn, first turn the wheel *10°* and then turn it another *20°*. The result is a turn of *30°*. To combine a turn of *100°*

and a turn of $300°$, first turn the wheel $100°$ and then turn it another $300°$. The result is a turn of $400°$, which is equal to a turn of $40°$.

Now consider the set of all turns that are less than $360°$. Let a and b be any two turns in the set. Denote by ab the turn that results from combining them by the rule given above. We shall call ab the product of a and b. To identify the turn ab, we find the sum of the number of degrees in a and the number of degrees in b, and then find an equal turn that is less than $360°$. Then the product ab is also a member of the set of all turns that are less than $360°$. Thus the rule given above for multiplying turns defines a binary operation in this set.

Exercises:

5. What is the product of a $40°$ turn and a $50°$ turn?

6. What is the product of a $200°$ turn and a $300°$ turn?

7. What is the product of a $0°$ turn and a $40°$ turn? of a $0°$ turn and a $50°$ turn? of a $60°$ turn and a $0°$ turn?

8. What is the product of a $20°$ turn and a $340°$ turn? of a $12°$ turn and a $348°$ turn? of a $5\frac{1}{2}°$ turn and a $354\frac{1}{2}°$ turn?

9. What turn multiplied by a $60°$ turn will produce a product that is equal to a $0°$ turn?

10. If x is not 0, what turn multiplied by an $x°$ turn will produce a product that is equal to a $0°$ turn?

The Group of All Turns

We have introduced a multiplication operation in the set of all turns that are less than $360°$. Let us examine the set, taken

with this operation, to see if it has the four properties that define a group.

I. Closure. We have already seen that the product of any two turns that are less than $360°$ is equal to a turn that is less than $360°$. So the set is closed with respect to multiplication of turns.

II. Associative Property. Let a, b, and c be any three turns. The expression $a(bc)$ stands for the single turn that has the same effect as the turn a followed by the turn bc. The turn bc is the single turn that has the same effect as the turn b followed by the turn c. So $a(bc)$ has the same effect as the turn a followed by the turn b followed by the turn c. The expression $(ab)c$ stands for the single turn that has the same effect as the turn ab followed by the turn c. The turn ab is the single turn that has the same effect as the turn a followed by the turn b. So $(ab)c$ has the same effect as the turn a followed by the turn b followed by the turn c. Since $a(bc)$ and $(ab)c$ have the same effect, they are equal. That is, $a(bc) = (ab)c$, or the multiplication of turns has the associative property. Notice that the proof given here is like the proof given on page 42, where we showed that addition of integers has the associative property.

III. Identity Element. Let i stand for a turn of $0°$, and let a stand for any turn. Let x be the number of degrees in the turn a. Then the number of degrees in the turn ia is $0° + x°$, which is equal to $x°$. Then the turn ia is equal to the turn a, since they both have the same number of degrees. Similarly, the turn ai has $x° + 0°$, which is equal to $x°$. So the turn ai is also equal to the turn a. Since $ia = ai = a$ for every turn a, i is the identity element for the set of all turns.

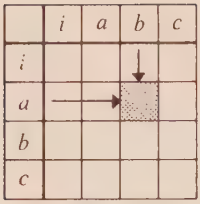
IV. Inverse. Let a be any turn, and let x be the number of degrees in it. If $a = i$, then $x = 0$. The turn i is its own inverse, because $ii = i$, since $0° + 0° = 0°$. If a is not equal to i, then x

is not *0*, and *360 − x* is less than *360*. Then there is a turn *b* for which the number of degrees is *360 − x*. The sum of *x* and *360 − x* is *360*. So *ab* = a turn of *360°* = a turn of *0°* = *i*. Since the sum of *360 − x* and *x* is also *360*, *ba* = a turn of *360°* = a turn of *0°* = *i*. Since *ab* = *ba* = *i*, *b* is the inverse of *a*.

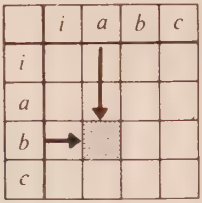
The set of all turns that are less than *360°*, taken with the operation multiplication, has all the four properties that define a group. Therefore, it is a group with respect to the multiplication of turns.

A Group with Four Members

The group of all turns that are less than *360°* contains within itself many other groups. As an example, let us consider the set of turns each of which is a whole number multiple of *90°*. This set has only four members in it, the turn of *0°*, the turn of *90°*, the turn of *180°*, and the turn of *270°*. Let us use as names for these turns the symbols *i*, *a*, *b*, and *c* respectively. We shall show that the set whose members are *i*, *a*, *b*, and *c* is a group with respect to the multiplication of turns. To prove this, we first have to find the product of every pair of members of the set. We shall record these products in a table arranged in the following way. We make a table with four rows and four columns. We label the four rows, *i*, *a*, *b*, and *c* respectively, by writing one of these symbols to the left of each row. We also label the four columns *i*, *a*, *b*, and *c* respectively, by writing one of these symbols at the top of each column. The rows and columns divide the table into *16* boxes. Each box is in a definite row and a definite column. We shall write in each box the product we get when we multiply the turn that labels its row and the turn that labels its column. For example, the product *ab* will be written in the box that is in the *a* row and the *b* column. The product *ba*, on the other hand, will be written in the box that is in the *b* row and the *a* column. These two boxes are pointed out by the arrows on page 60.



The Box in the a Row and b Column



The Box in the b Row and a Column

To multiply any two turns in the set, we use the rule for multiplication of turns given on page 56. For example, since a is a $90°$ turn and b is a $180°$ turn, ab is equal to a turn of $90° + 180°$, or $270°$. But c is a turn of $270°$. So $ab = c$, and we write c in the box that is in the a row and the b column. cc is equal to a turn of $270° + 270°$, or $540°$. But a turn of $540°$ is equal to a turn of $180°$. For this reason, $cc = b$, and we write b in the box that is in the c row and the c column. The completed table is shown below.

	i	a	b	c
i	i	a	b	c
a	a	b	c	i
b	b	c	i	a
c	c	i	a	b

We now examine the products to see if the set whose members are i, a, b and c is a group with respect to multiplication of turns:

I. Closure. A glance at the table shows that the product of two members of the set is either i or a or b or c. That is, every product of members of the set is also a member of the set. So the set is closed with respect to multiplication.

II. Associative Property. i, a, b, and c are members of the set of all turns that are less than $360°$. In this set of all turns,

multiplication has the associative property no matter which turns we multiply. So it certainly has this property when we multiply turns selected from among *i*, *a*, *b* and *c*.

III. Identity Element. Since *i* is the identity element for the set of all turns, multiplication of any turn by *i* leaves it unchanged. So, multiplication of either *i* or *a* or *b* or *c* by *i* leaves it unchanged. Then *i* is an identity element for multiplication in the set whose only members are *i*, *a*, *b* and *c*.

IV. Inverse. The table shows that the set whose members are *i*, *a*, *b* and *c* contains an inverse for every one of its members. The inverse of *i* is *i*, since $ii = i$. The inverse of *a* is *c*, and the inverse of *c* is *a*, since $ac = ca = i$. The inverse of *b* is *b*, since $bb = i$.

Thus we see that the set whose members are *i*, *a*, *b* and *c*, taken with the operation multiplication, has all four of the properties that define a group. Therefore it is a group with respect to the multiplication of turns.

Exercises:

11. Let *i* stand for a turn of *0°*, and let *b* stand for a turn of *180°*. Consider the set whose only members are *i* and *b*. a) Construct a multiplication table for this set, using two rows labeled *i* and *b* respectively, and two columns labeled *i* and *b* respectively. b) Is the set whose members are *i* and *b* closed with respect to multiplication of turns? c) Does multiplication in this set have the associative property? Why? d) Is *i* an identity element for multiplication in this set? Why? e) Does the set contain an inverse for each of its members? f) Is the set whose members are *i* and *b* a group with respect to multiplication of turns?

12. Let *i* stand for a turn of *0°*, let *p* stand for a turn of *120°*, and let *q* stand for a turn of *240°*. Consider the set whose only

members are i, p, and q. a) Construct a multiplication table for this set, using three rows labeled i, p, and q respectively, and three columns labeled i, p, and q respectively. b) Is the set whose members are i, p, and q closed with respect to multiplication of turns? c) Does multiplication in this set have the associative property? Why? d) Is i an identity element for multiplication in this set? Why? e) What are the inverses of i, p, and q respectively? Does the set contain an inverse for each of its members? f) Is the set whose members are i, p, and q a group with respect to multiplication of turns?

13. Let i stand for a turn of $0°$, and consider the set whose only member is i. a) Construct a multiplication table for this set, using one row labeled i, and one column labeled i. b) Is the set whose only member is i closed with respect to multiplication of turns? c) Does multiplication in this set have the associative property? That is, is $i(ii)$ equal to $(ii)i$? d) Is i an identity element for multiplication in this set? e) Does the set contain an inverse for each of its members? f) Is the set whose only member is i a group with respect to multiplication of turns?

Finite and Infinite Groups

The group whose only members are i, a, b and c, described above, has exactly four members. The number *four* is a counting number. If the number of members in a set is a counting number, we say that the set has *finitely many* members and is a *finite* set. A group whose membership is a finite set is called a *finite group.* The number of members in a finite group is called the *order* of the group.

The set of all integers is not a finite set. No matter how many of its members we might count, there would always be some members left uncounted. A set like this, which has more members than can be counted, is said to have *infinitely many* members, and is called an *infinite* set. A group whose membership is an

infinite set is called an *infinite group*. The group of all integers, used with the operation addition, is an infinite group.

A multiplication table can always be constructed for a finite group. Then all the properties of the group can be discovered by examining the products recorded in the table. Nearly all of the groups we shall talk about in the rest of this book are finite groups. We shall find it helpful to construct a multiplication table for each of them.

Exercises:

14. The set of all measuring numbers, used with the operation multiplication, is a group. Is this group finite or infinite?

15. Is the group described in exercise 11 on page 61 finite or infinite?

16. The group of all turns that are less than $360°$ includes among its members a turn of 1 degree, a turn of $\frac{1}{2}$ degree, a turn of $\frac{1}{3}$ degree, a turn of $\frac{1}{4}$ degree, a turn of $\frac{1}{5}$ degree, and so on. In general, if n is any counting number, the group includes among its members a turn of $\frac{1}{n}$ degree. Is the group of all turns that are less than $360°$ finite or infinite?

17. What is the least number of members that a group may have?

18. What is the order of the group described in exercise 12 on page 61?

6

Clock-Face Numbers

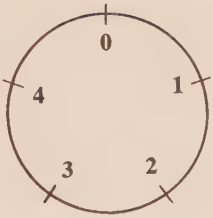
Numbers on a Circle

In Chapter 2 we saw that the set of all integers, which is an infinite set, can be pictured as a set of equally spaced points on a straight line. In this chapter we shall get acquainted with some finite sets that may be pictured as sets of equally spaced points on a circle.

We shall introduce in each of these sets an operation called *plus*, represented by the sign $+$, and an operation called *times*, represented by the sign $\times$. With these operations, each of these sets behaves in some ways like the system of integers. Because of this resemblance, we call the members of these sets numbers. Since they are arranged on a circle, like the numbers on the face of a clock, they are often called *clock-face numbers*. We shall identify some sets of clock-face numbers that are groups with respect to the operation *plus*, and some that are groups with respect to the operation *times*.

A Five-Point Number System

As a first example of a clock-face number system, we construct one that has five members. Draw a circle, and place on it five points that divide the circle into five equal parts. Using one of these points as starting point, and going around the circle in the clockwise direction, we assign to these points the names *zero*, *one*, *two*, *three* and *four*, and represent them by the symbols *0*, *1*, *2*, *3*



and *4* respectively. Although the names of these numbers are the same as the names of the first five whole numbers, we should not confuse them with the whole numbers. These numbers differ from the whole numbers in many ways, as we shall see.

Addition of Clock-Face Numbers

To define addition for these clock-face numbers, we take our cue from the addition of integers. In Chapter 3 we saw that we can interpret each integer as a motion on the number line, and we can add two integers by combining the motions that they represent. We saw, too, on page 41, that if the first motion starts at the zero point, the second motion ends at the point which represents their sum. This suggests that we define addition of clock-face numbers in the following way: The numbers divide the circle into equal parts. Call each of these parts a unit. Interpret each number to stand for a clockwise motion of a whole number of units around the circle. The number *0* stands for a motion of zero units, the number *1* stands for a motion of *1* unit, the number *2* stands for a motion of *2* units, and so on. To add two numbers, start at zero, and carry out, one after the other, the two motions that the numbers represent. Then the point where the second motion ends will be the sum of the two numbers. For example, to find the sum $1 + 2$, start at zero, and move *1* unit clockwise to the point *1*. Then, from there, move *2* units clockwise. This second motion ends at the point *3*. So $1 + 2 = 3$. To find the sum $2 + 3$, start at zero, and move *2* units clockwise to the point *2*. Then, from

there, move *3* units clockwise. This second motion ends at the point *0*. So $2 + 3 = 0$. Similarly, $2 + 4 = 1$.

To record the sums of these clock-face numbers, we prepare an addition table similar to the multiplication tables used in Chapter 5. We make a table with five rows and five columns. The rows are labeled *0*, *1*, *2*, *3* and *4* respectively at the left, and the columns are labeled *0*, *1*, *2*, *3* and *4* respectively at the top. If a and b are any two of the clock-face numbers, the sum $a + b$ is written in the box that is in the a row and the b column. The sums $1 + 2$ and $2 + 3$ are shown in the proper boxes in the table below.

+	0	1	2	3	4
0					
1			3		
2				0	
3					
4					

Exercises:

1. Copy the addition table for the five-point clock-face number system. Find the sums of all ordered pairs of numbers in the system, and enter them in the appropriate boxes.

2. Draw a circle and put four points on it that divide the circle into four equal parts. Label these points *0*, *1*, *2* and *3*, arranged in the clockwise direction. They are the members of a four-point clock-face number system in which addition is done, as in the five-point system, by combining motions around the circle. Prepare and complete an addition table for the four-point clock-face number system.

3. Draw a circle and put three points on it that divide the circle into three equal parts. Label these points *0*, *1* and *2*, arranged in the clockwise direction. Prepare and complete an addition table for the three-point clock-face number system.

4. Draw a circle and put two points on it that divide the circle into two equal parts. Label these points *0* and *1*. Prepare and complete an addition table for the two-point clock-face number system.

Multiplication of Clock-Face Numbers

To define multiplication for clock-face numbers, we take our cue from the multiplication of counting numbers. In the system of counting numbers, 2×3 means $3 + 3$, 3×3 means $3 + 3 + 3$, 4×3 means $3 + 3 + 3 + 3$, and so on. In general, if a and b are counting numbers, $a \times b$ means the sum of a numbers each of which is equal to b. If a and b are members of a clock-face number system, we define $a \times b$ in the same way if a is not 0. For example, to find 4×3, we must add $3 + 3 + 3 + 3$. So we start at zero, and carry out in succession 4 motions of *3* units each. Then the point where the last motion ends is the value of the product. In the five-point clock-face number system, $4 \times 3 = 2$.

If $a = 0$, we define $0 \times b = 0$ for all values of b, as we do in the system of whole numbers. Thus, in the five-point clock-face number system, $0 \times 0 = 0$, $0 \times 1 = 0$, $0 \times 2 = 0$, $0 \times 3 = 0$, and $0 \times 4 = 0$. These products are all entered in the appropriate boxes in the multiplication table below.

×	0	1	2	3	4
0	0	0	0	0	0
1					
2					
3					
4				2	

Exercises:

5. Copy and complete the multiplication table for the five-point clock-face number system.

6. Prepare and complete a multiplication table for the four-point clock-face number system.

7. Prepare and complete a multiplication table for the three-point clock-face number system.

8. Prepare and complete a multiplication table for the two-point clock-face number system.

A Group with Respect to +

We now examine the set of all members of the five-point clock-face number system, taken with the operation +. Let us see whether or not it has the four properties of a group. The complete addition table for this set is shown below.

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Addition Table for the Five-point Clock-face Number System

I. Closure. The sum of any two of the numbers *0*, *1*, *2*, *3* and *4* is one of these numbers. So the set is closed with respect to the operation +.

II. Associative Property. An argument like that given for addition of integers shows that addition of clock-face numbers has the associative property. (See page 42.)

III. Identity Element. The first row and the first column of the table show that if a is any one of the numbers 0, 1, 2, 3 and 4, $0 + a = a + 0 = a$. So 0 is an identity element for addition.

IV. Inverse. The table shows that there is an additive inverse for every one of the numbers 0, 1, 2, 3 and 4. The inverse of 0 is 0, since $0 + 0 = 0$. The inverse of 1 is 4, and the inverse of 4 is 1, since $1 + 4 = 4 + 1 = 0$. The inverse of 2 is 3, and the inverse of 3 is 2, since $2 + 3 = 3 + 2 = 0$.

Thus we see that the set of all members of the five-point clock-face number system is a group with respect to the operation $+$. In this regard, the five-point clock-face number system resembles the system of integers.

If n is any whole number greater than or equal to 2, there is a clock-face number system with n points. It is easy to see that the set of all members in any one of these clock-face number systems is a group with respect to the operation $+$. We shall call this group the *additive group* of the clock-face number system.

Exercises:

9. In the additive group of the four-point clock-face number system, what are the inverses of 0, 1, 2 and 3 respectively?

10. In the additive group of the three-point clock-face number system, what are the inverses of 0, 1 and 2 respectively?

11. In the additive group of the two-point clock-face number system, what are the inverses of 0 and 1 respectively?

A Group with Respect to $\times$

We now examine the set of all members of the five-point clock-face number system, taken with the operation $\times$. We shall try to

find within it a group with respect to the operation $\times$. The complete multiplication table for this set is shown below.

$\times$	*0*	*1*	*2*	*3*	*4*
0	*0*	*0*	*0*	*0*	*0*
1	*0*	*1*	*2*	*3*	*4*
2	*0*	*2*	*4*	*1*	*3*
3	*0*	*3*	*1*	*4*	*2*
4	*0*	*4*	*3*	*2*	*1*

Multiplication Table for the Five-point Clock-face Number System

The second row and the second column of the table show that the number *1* is an identity element for the operation $\times$. So, if a number has a multiplicative inverse, the product of the number and its multiplicative inverse must be *1*. From this fact we can see immediately that the number *0* cannot have a multiplicative inverse, because the product of *0* and any number is *0*, as shown in the first row and the first column of the table. Since *0* cannot have a multiplicative inverse, it cannot belong to any group with respect to the operation $\times$. We therefore begin by excluding *0*, and we consider only the set of all non-zero members of the five-point clock-face number system. To get the multiplication table for this set, we merely leave out the first row and the first column of the multiplication table for the whole system. The result is shown on page 71.

Now let us see whether this smaller set is a group with respect to the operation $\times$.

I. Closure. The product of any two of the numbers *1*, *2*, *3* and *4* is one of these numbers. So the set is closed with respect to the operation $\times$.

×	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

Multiplication Table
for the Set of all Non-zero Numbers
of the Five-point Clock-face Number System

II. Associative Property. Let us compare the two products $2 \times (4 \times 3)$ and $(2 \times 4) \times 3$. We see in the table that $4 \times 3 = 2$. So $2 \times (4 \times 3) = 2 \times 2$. We see in the table that $2 \times 2 = 4$. Therefore $2 \times (4 \times 3) = 4$. We see in the table that $2 \times 4 = 3$. So $(2 \times 4) \times 3 = 3 \times 3$. We see in the table that $3 \times 3 = 4$. Therefore $(2 \times 4) \times 3 = 4$. So $2 \times (4 \times 3) = (2 \times 4) \times 3$, since both products are equal to 4. If a, b and c are any three of the numbers 1, 2, 3 and 4, we can show in the same way, by using the table, that $a \times (b \times c) = (a \times b) \times c$. So multiplication in this set has the associative property.

III. Identity Element. The first row and the first column of the table show that if a is any one of the numbers 1, 2, 3 and 4, $1 \times a = a \times 1 = a$. So 1 is an identity element for multiplication.

IV. Inverse. The table shows that there is a multiplicative inverse for every one of the numbers 1, 2, 3 and 4. The inverse of 1 is 1, since $1 \times 1 = 1$. The inverse of 2 is 3, and the inverse of 3 is 2, since $2 \times 3 = 3 \times 2 = 1$. The inverse of 4 is 4, since $4 \times 4 = 1$.

Thus we see that the set of all non-zero members of the five-point clock-face number system is a group with respect to the operation $\times$. In this regard it resembles the system of measuring numbers.

Exercises:

12. Use the multiplication table for the set of all non-zero members of the five-point clock-face number system to find and compare the following pairs of products:

a) $1 \times (2 \times 4)$ and $(1 \times 2) \times 4$;
b) $2 \times (2 \times 4)$ and $(2 \times 2) \times 4$;
c) $4 \times (3 \times 2)$ and $(4 \times 3) \times 2$;
d) $3 \times (3 \times 2)$ and $(3 \times 3) \times 2$.

Let us examine next the set of non-zero members of the four-point clock-face number system. The multiplication table for this set is shown below.

$\times$	1	2	3
1	1	2	3
2	2	0	2
3	3	2	1

The table shows that *1* is an identity element in this set for the operation $\times$. So if a number in the set has a multiplicative inverse the product of the number and its multiplicative inverse must be *1*. From this fact we see immediately that the number *2* in the four-point clock-face number system does not have a multiplicative inverse, because the product of *2* and any number is either *2* or *0*, as shown in the second row and the second column of the table above. Therefore the set of non-zero members of the four-point clock-face number system is not a group with respect to the operation $\times$.

Thus we see that the set of non-zero members of a clock-face number system is a group with respect to $\times$ in some clock-face number systems, and is not a group with respect to $\times$ in some other clock-face number systems. In those systems in which it is a group, it is called the *multiplicative group* of the system.

Exercises:

13. a) Construct the multiplication table for the non-zero members of the three-point clock-face number system. b) What are the multiplicative inverses of *1* and *2* respectively? c) Is the set of all non-zero members of the three-point clock-face number system a group with respect to $\times$?

14. a) Construct the multiplication table for the non-zero members of the six-point clock-face number system. b) Which numbers in the table do not have multiplicative inverses? c) Is the set of all non-zero members of the six-point clock-face number system a group with respect to $\times$?

15. a) Construct the multiplication table for the non-zero members of the seven-point clock-face number system. b) What are the multiplicative inverses of *1*, *2*, *3*, *4*, *5* and *6* respectively? c) Is the set of all non-zero members of the seven-point clock-face number system a group with respect to $\times$?

Zero Divisors

In the multiplication table for the non-zero members of the six-point clock-face number system we see that $2 \times 3 = 0$, and $3 \times 2 = 0$. Both of the numbers *2* and *3* are non-zero numbers, yet their product is zero. If the product of two non-zero numbers is zero in some number system, the two numbers are called *zero divisors* in that system. So the numbers *2* and *3* are zero divisors in the six-point clock-face number system.

We shall show that because *2* is a zero divisor in the six-point clock-face number system, it cannot have a multiplicative inverse in that system. If it did have a multiplicative inverse b, then we would have $2 \times b = 1$. If we multiply both sides of this equation

by *3*, we get $3 \times (2 \times b) = 3 \times 1$. But this is impossible, because $3 \times (2 \times b) = (3 \times 2) \times b = 0 \times b = 0$, while $3 \times 1 = 3$, which is not zero. Since assuming that *2* has a multiplicative inverse leads to an impossible result, we must conclude that *2* does not have a multiplicative inverse.

Exercises:

16. Use a proof like that given above to show that because *3* is a zero divisor in the six-point clock-face number system, it cannot have a multiplicative inverse in that system.

17. In the four-point clock-face number system, *2* is a zero divisor, because *2* is not zero and $2 \times 2 = 0$. Use a proof like that given above to show that *2* cannot have a multiplicative inverse in the four-point clock-face number system.

18. a) Construct the multiplication table for the non-zero members of the eight-point clock-face number system. b) Find a number in this system that is a zero divisor. c) Use a proof like that given above to show that this number cannot have a multiplicative inverse in the eight-point clock-face number system.

19. A counting number n is called a *composite* number if there are two counting numbers a and b, each smaller than n, such that $a \times b = n$. If n is composite, then, in the n-point clock-face number system these numbers a and b are zero divisors, because each is different from zero, yet $a \times b = b \times a = 0$. Use a proof like that given above to show that b cannot have a multiplicative inverse c in the n-point clock-face number system.

20. Which of the counting numbers *9*, *10*, *11*, *12* and *13* is composite?

21. If n is a composite counting number, is the set of non-zero members of the n-point clock-face number system a group with respect to the operation $\times$? Explain.

7

Symmetries of Plane Figures

Moved but Not Changed

A plane figure is a figure that is drawn on a flat surface. For example, the triangle, rectangle, and square drawn below on the surface of this page are all plane figures. A *symmetry* of a plane figure is a motion that puts the figure into a new position where it fits exactly on itself. For example, a 90° clockwise turn of a square around its center moves each side to a place where a side was



Triangle *Rectangle* *Square*

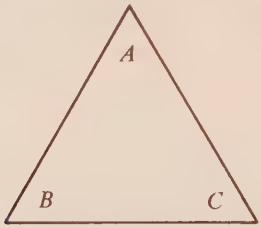
before. As a result, the square in its new position fits exactly over the square in its old position. Then, although the square has been moved, its appearance to the eye has not been changed. So a 90° clockwise turn of a square around its center is a symmetry of the square.

In this chapter we shall examine some groups whose members are the symmetries of a plane figure.

The Symmetries of an Equilateral Triangle

An equilateral triangle has equal sides and equal angles. To find the symmetries of an equilateral triangle, first cut one out of paper,

and label its vertices A, B, and C, as shown in the diagram below. Put the labels on the reverse side of the paper, too, so that you can read the labels even when the triangle is turned over.

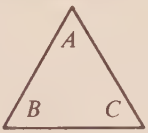
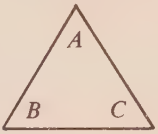
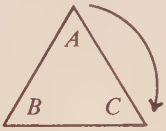
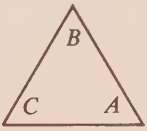
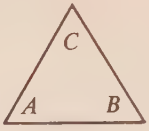
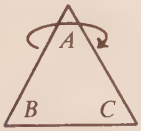
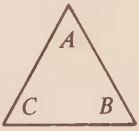
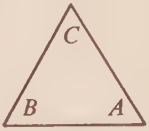
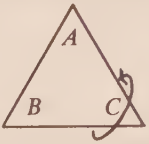
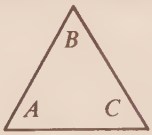


In order to have a definite starting position for the triangle, put it on a flat surface so that the side BC extends from left to right, and the opposite vertex A lies above BC, as shown in the diagram above. Now let us look for motions that are symmetries of this figure. As we find them, we shall assign a name to each one.

Our experience with turning a wheel suggests that we first try turning the triangle around its center. If we turn the triangle one third of a complete turn, or $120°$, clockwise, A moves to where C is, C moves to where B is, and B moves to where A is. The triangle in its new position fits exactly over the triangle in its old position. So a clockwise turn of $120°$ is a symmetry of the triangle. Let us call this symmetry p. If we turn the triangle $240°$ clockwise, A moves to where B is, C moves to where A is, and B moves to where C is. So a clockwise turn of $240°$ is also a symmetry of the triangle. Let us call it q. A clockwise turn of $360°$ is also a symmetry of the triangle, because it brings each of the points A, B and C back to its original position. Let us call this symmetry i. There are other motions that have the same effect as p, q or i. For example, a clockwise turn of $480°$ has the same effect as a clockwise turn of $120°$. A counterclockwise turn of $120°$ has the same effect as a clockwise turn of $240°$. A turn of $0°$, which does not move the triangle at all, has the same effect as a clockwise turn of $360°$. As in Chapter 5, we shall consider two motions to be equal if they have the same effect. So a clockwise turn of $480°$ is equal to p, a counterclockwise turn of $120°$ is equal to q, and a turn

of $0°$ is equal to i. It is easy to see that any turn around the center of the triangle that is a symmetry of the triangle must be equal to either p, q or i.

However, there are three more symmetries of the triangle that

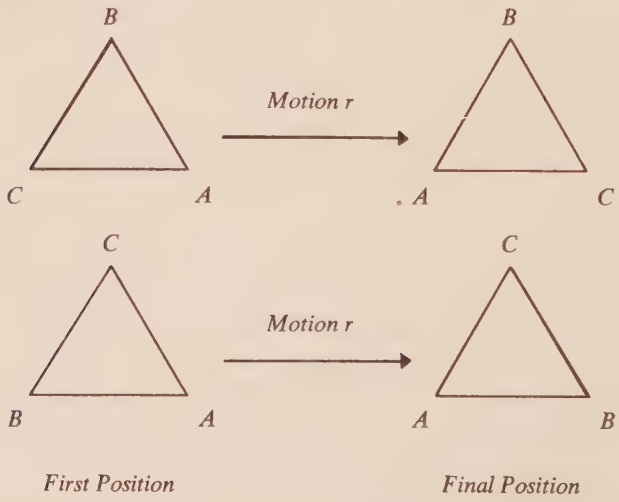
Motion	*Symbol*	*First Position*	*Final Position*
No Motion	i		
Rotate 120° Clockwise	p		
Rotate 240° Clockwise	q		
Flip Over, Keeping Top Vertex Fixed	r		
Flip Over, Keeping Left Vertex Fixed	s		
Flip Over, Keeping Right Vertex Fixed	t		

Symmetries of an Equilateral Triangle

are not turns around its center. Each of them is a motion in which we flip the triangle over to bring its bottom face up, while leaving one of the vertices where it is. If we flip the triangle over with the top vertex fixed, then the left and right vertices change places. Let us call this symmetry *r*. If we flip the triangle over with the left vertex fixed, then the top and right vertices change places. Let us call this symmetry *s*. If we flip the triangle over with the right vertex fixed, then the top and left vertices change places. Let us call this symmetry *t*. The complete list of symmetries of the equilateral triangle is shown in the chart on page 77. The chart also shows the new position to which each symmetry moves the vertices of the triangle if we start with *B* at the left, *C* at the right, and *A* at the top.

Other First Positions

The first positions shown in the chart are not the only first positions the triangle may have. Any position of the triangle may be used as its first position. Of course, the final position the tri-



angle has after it is moved depends on what the first position was. The diagrams on page 78 show two different first positions, and the effect of using motion r to move the triangle from these first positions.

Exercises:

1. If the triangle in its first position has B at the top, C at the left, and A at the right, find the final position that results from using each of the symmetries i, p, q, r, s and t.

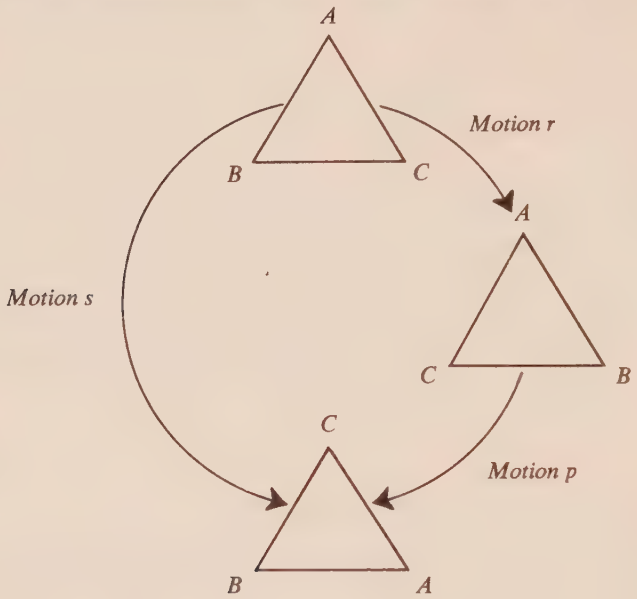
2. If the triangle in its first position has C at the top, B at the left, and A at the right, find the final position that results from using each of the symmetries i, p, q, r, s and t.

The Product of Two Symmetries

Since a symmetry of a plane figure is a motion, we can combine two symmetries by performing one motion right after the other. The single motion that has the same effect is called the product of the two symmetries. Since each of the symmetries leaves the appearance of the figure unchanged, so does their product. So the product of two symmetries is also a symmetry.

To find the product of two symmetries of an equilateral triangle, use as the first position of the triangle the one shown in the chart on page 77. Apply the first of the two symmetries to this first position to produce a second position. Then apply the second of the two symmetries to this second position to produce a final position. Then see in the chart which single symmetry produces the same final position as do the two symmetries combined. This single symmetry is the product of the two symmetries. For example, when we start with A at the top, B at the left, and C at the right, motion r leaves A fixed and interchanges B and C. Motion p then turns the triangle $120°$, to produce the final position in which

C is at the top, B is at the left, and A is at the right. The single motion that produces the same final position is s. So $rp = s$.



$rp=s$

Exercises:

3. Find the products st and ts.

4. Find the products r^2, s^2 and t^2. (r^2 means rr, as in the multiplication of numbers. See page 24.)

5. Prepare a multiplication table with six rows labeled i, p, q, r, s and t respectively, and with six columns labeled i, p, q, r, s and t respectively. Find the product of each possible ordered pair of symmetries and write it in the appropriate box.

6. Use the table to help you find $p(st)$ and $(ps)t$.

Group Properties

The multiplication table for the set of all symmetries of an equilateral triangle is shown below.

	i	p	q	r	s	t
i	i	p	q	r	s	t
p	p	q	i	t	r	s
q	q	i	p	s	t	r
r	r	s	t	i	p	q
s	s	t	r	q	i	p
t	t	r	s	p	q	i

Multiplication Table
for the Symmetries of an Equilateral Triangle

An examination of the products in this table shows that the set of all symmetries of an equilateral triangle, with the operation multiplication, has all four properties of a group:

I. Closure. Every product of two symmetries is also a symmetry. So the set of all symmetries of an equilateral triangle is closed with respect to multiplication.

II. Associative Property. Multiplication of motions has the associative property. (See page 42.)

III. Identity Element. The entries in the first row and in the first column of the table show that i is an identity element for multiplication of symmetries.

IV. Inverse. The table shows that every symmetry of an equilateral triangle has an inverse which is also a symmetry of the triangle. For example, the inverse of p is q, since $pq = qp = i$.

Therefore, the set of all symmetries of an equilateral triangle is a group with respect to multiplication of symmetries.

Exercises:

7. What are the inverses of i, p, q, r, s and t respectively?

8. In the multiplication table for the symmetries of an equilateral triangle, leave out the r, s and t rows and the r, s and t columns. What is left is a multiplication table for the set whose only members are i, p, and q. a) Is this set closed with respect to multiplication? b) Does it contain the inverse of each of its members? c) Is this set a group with respect to multiplication?

9. In the multiplication table for the symmetries of an equilateral triangle, leave out the i, p and q rows and the i, p and q columns. What is left is a multiplication table for the set whose only members are r, s and t. a) Is this set closed with respect to multiplication? b) Does it contain an identity element for multiplication? c) Does it contain the inverse of each of its members? d) Is this set a group with respect to multiplication?

10. Construct a multiplication table for the set whose only members are i and r. Is this set a group with respect to multiplication?

The Order of Factors Counts

We see in the multiplication table for the symmetries of an equilateral triangle that $st = p$, while $ts = q$. So the products st and ts are not equal to each other. Thus the order in which the factors are written in a product may make a difference in the value of the product. In this respect, multiplication of symmetries is different from multiplication of counting numbers. In multiplication of counting numbers, the product 2×3 and the product 3×2 are both equal to 6, so $2 \times 3 = 3 \times 2$. In general, if a and b are any two counting numbers, $a \times b = b \times a$. This is known

as the *commutative property* of multiplication of counting numbers, because the counting numbers in any product may *commute* or change places without changing the value of the product. Addition of integers also has the commutative property, because if a and b are any two integers, $a + b = b + a$. However, as we see, multiplication of the symmetries of an equilateral triangle does not have the commutative property, because there are some products of symmetries in which interchanging the factors changes the value of the product. Thus there are some groups in which the group operation has the commutative property, and there are some groups in which it does not. A group in which the operation has the commutative property is called a *commutative group*. A commutative group is also called an *abelian group*, after Niels Henrik Abel, one of the pioneers in the study of groups.

Exercises:

11. In which of the products *pq*, *ir*, and *rs* may the factors change places without changing the value of the product?

12. Does multiplication of measuring numbers have the commutative property?

13. Is the additive group in the five-point clock-face number system a commutative group?

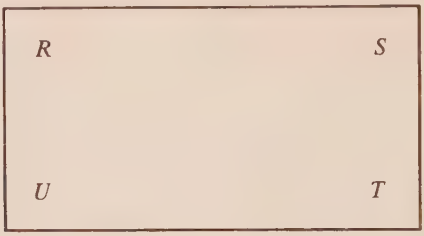
14. In the five-point clock-face number system, is the multiplicative group whose members are *1*, *2*, *3* and *4* a commutative group?

15. We saw in exercise 8 on page 82 that the group of symmetries of an equilateral triangle contains a smaller group whose only members are *i*, *p* and *q*. Is this smaller group a commutative group?

The Symmetries of a Rectangle

The four-sided figure on page 84 is a rectangle in which the length *RS* and the width *RU* are not equal. To find the symmetries

of such a rectangle, first cut one out of paper and label the vertices as shown in the diagram. Put the labels on the reverse side of the



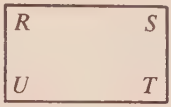
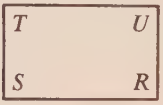
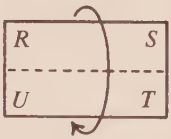
paper, too, as we did with the equilateral triangle. Then put the rectangle on a flat surface, with *RS* at the top, *UT* at the bottom, *RU* at the left, and *ST* at the right.

A rotation of $0°$ around the center of the rectangle does not move it at all, so it is a symmetry of the rectangle. Let us call it *i*. A clockwise rotation of $180°$ moves *R* to where *T* is, moves *S* to where *U* is, moves *T* to where *R* is, and moves *U* to where *S* is. So it, too, is a symmetry of the rectangle. Let us call it *a*.

A line drawn halfway between the horizontal sides of the rectangle is called a *horizontal median* of the rectangle. A line drawn halfway between the vertical sides of the rectangle is called a *vertical median*. A motion which flips the rectangle over while keeping a median fixed is obviously a symmetry of the rectangle. Let us denote by *b* the motion that flips the rectangle over while keeping the horizontal median fixed. Let us denote by *c* the motion that flips the rectangle over while keeping the vertical median fixed. The motions *i*, *a*, *b* and *c* are the only symmetries of the rectangle. The chart on page 85 shows the new position to which each symmetry moves the vertices of the rectangle if we begin with the first position shown in the chart.

Exercises:

16. If the rectangle in its first position has *SR* at the top, *TU* at the bottom, *ST* at the left, and *RU* at the right, find the final

Motion	*Symbol*	*First Position*	*Final Position*
No Motion	*i*		
Rotate 180° Clockwise	*a*		
Flip Over, Keeping Horizontal Median Fixed	*b*		
Flip Over, Keeping Vertical Median Fixed	*c*		

The Symmetries of a Rectangle

position that results from using each of the symmetries *i*, *a*, *b* and *c*.

17. If the rectangle in its first position has *TU* at the top, *SR* at the bottom, *TS* at the left, and *UR* at the right, find the final position that results from using each of the symmetries *i*, *a*, *b* and *c*.

18. Prepare a multiplication table for the symmetries of a rectangle, with four rows labeled *i*, *a*, *b* and *c* respectively, and with four columns labeled *i*, *a*, *b* and *c* respectively. Find the product of each possible ordered pair of symmetries and write it in the appropriate box.

19. Use the table to help you find $a(bc)$ and $(ab)c$.

The Group of Symmetries of a Rectangle

The multiplication table for the set of all symmetries of a rectangle whose length and width are unequal is shown below. An

	i	a	b	c
i	i	a	b	c
a	a	i	c	b
b	b	c	i	a
c	c	b	a	i

Multiplication Table
for the Symmetries of a Rectangle

examination of the products in this table shows that the set of all symmetries of the rectangle has all four properties of a group with respect to multiplication of symmetries.

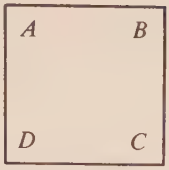
Exercises:

20. In the group of symmetries of the rectangle, a) what motion is the identity element? b) what are the inverses of i, a, b and c respectively?

21. a) Compare the products ab and ba. b) Compare the products ac and ca. c) Compare the products bc and cb. d) Is the group of symmetries of the rectangle a commutative group?

Symmetries of a Square

A square is a rectangle in which the length and width are equal. To find the symmetries of a square, first cut one out of paper and



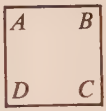
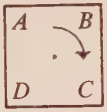
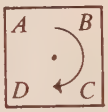
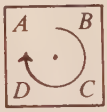
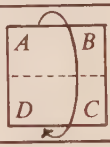
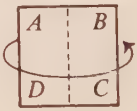
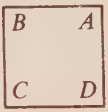
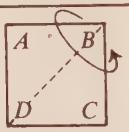
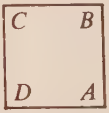
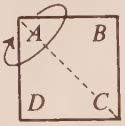
label the vertices as shown in the diagram above. Put the labels on the reverse side, too, so that each vertex may be identified when the paper is turned over. Then put the square on a flat surface, with AB at the top, DC at the bottom, AD at the left, and BC at the right.

We can make the square fit on itself if we turn it $0°$, $90°$, $180°$, or $270°$ clockwise around its center. Let us call these four rotations i, p, q and r respectively. We can also make the square fit on itself if we flip it over while keeping the horizontal median or the vertical median fixed. Let us call these two flips s and t respectively. We can make the square fit on itself, too, if we flip it over while keeping fixed a diagonal line which joins opposite vertices. Let us call u the motion that flips the square over while keeping fixed the diagonal from the lower left vertex to the upper right vertex. Let us call v the motion that flips the square over while keeping fixed the diagonal from the upper left vertex to the lower right vertex. The motions i, p, q, r, s, t, u and v are the only symmetries of the square. The chart on page 88 shows how they change the positions of the vertices of the square.

Exercises:

22. Prepare a multiplication table for the symmetries of the square, with eight rows labeled i, p, q, r, s, t, u and v respectively, and with eight columns labeled i, p, q, r, s, t, u and v respectively. Find the product of each possible ordered pair of symmetries and write it in the appropriate box.

23. Use the table to help you find $s(tu)$ and $(st)u$.

Motion	*Symbol*	*First Position*	*Final Position*
No Motion	*i*		
Rotate 90° Clockwise	*p*		
Rotate 180° Clockwise	*q*		
Rotate 270° Clockwise	*r*		
Flip Over, Keeping Horizontal Median Fixed	*s*		
Flip Over, Keeping Vertical Median Fixed	*t*		
Flip Over, Keeping Fixed the Diagonal from Lower Left Vertex to Upper Right Vertex	*u*		
Flip Over, Keeping Fixed the Diagonal from Upper Left Vertex to Lower Right Vertex	*v*		

The Symmetries of a Square

The Group of Symmetries of a Square

The multiplication table for the set of all symmetries of a square is shown below. An examination of the products in this table shows that the set of all symmetries of a square has all four properties of a group with respect to multiplication of symmetries.

	i	p	q	r	s	t	u	v
i	i	p	q	r	s	t	u	v
p	p	q	r	i	u	v	t	s
q	q	r	i	p	t	s	v	u
r	r	i	p	q	v	u	s	t
s	s	v	t	u	i	q	r	p
t	t	u	s	v	q	i	p	r
u	u	s	v	t	p	r	i	q
v	v	t	u	s	r	p	q	i

Multiplication Table
for the Symmetries of a Square

Exercises:

24. In the group of symmetries of a square, a) what is the identity element? b) what are the inverses of i, p, q, r, s, t, u and v respectively?

25. Compare the products pu and up. Is the group of symmetries of a square a commutative group?

26. In the multiplication table for the symmetries of a square, leave out the s, t, u and v rows and columns. What is left is a multiplication table for the set whose only members are i, p, q and r. a) Is this set closed with respect to multiplication? b) Does it contain the inverse of each of its members? c) Is this set a group with respect to multiplication?

27. In the multiplication table for the symmetries of a square, leave out the p, r, u and v rows and columns. What is left is a multiplication table for the set whose only members are i, q, s and t. a) Is this set closed with respect to multiplication? b) Does it contain the inverse of each of its members? c) Is this set a group with respect to multiplication?

28. Construct a multiplication table for the set whose only members are s, t, u and v. Is this set closed with respect to multiplication?

Letters of the Alphabet

Some of the letters of the alphabet are made with straight lines only. Each of them can be drawn inside a square, as shown in the drawing below.



Consider the letter A drawn in a square. Any motion that carries the letter into a new position where it fits exactly on itself also makes the square fit exactly on itself. So any symmetry of the letter A must also be a symmetry of the square. So, to find the symmetries of the letter A, we need examine only the motions i, p, q, r, s, t, u and v, shown on page 88, to see which ones make the letter A fit on itself. It is not hard to see that the only symmetries of the letter A are the motions i and t. The multiplication

table for the set whose only members are i and t shows that this set is a group with respect to multiplication.

Similarly, the symmetries of any letter drawn in a square are among the symmetries of the square. For each of these letters, the set of all symmetries of that letter is a group with respect to multiplication.

Exercises:

29. a) Which other straight-line letters drawn in a square have the same set of symmetries as the letter A? b) Construct the multiplication table for these symmetries.

30. a) Find the symmetries of the letter H drawn in a square. b) Construct the multiplication table for these symmetries. c) Which other straight-line letters drawn in a square have the same set of symmetries as the letter H?

31. a) Find the symmetries of the letter Z drawn in a square. b) Construct the multiplication table for these symmetries. c) Which other straight-line letters drawn in a square have the same set of symmetries as the letter Z?

32. a) Find the symmetries of the letter L drawn in a square. b) Construct the multiplication table for these symmetries. c) Which other straight-line letters drawn in a square have the same set of symmetries as the letter L?

33. a) Find the symmetries of the letter X drawn in a square. b) Construct the multiplication table for these symmetries. c) Which other straight-line letters drawn in a square have the same set of symmetries as the letter X?

34. a) Find the symmetries of the letter F drawn in a square. b) Construct the multiplication table for these symmetries. c) Which other straight-line letters drawn in a square have the same set of symmetries as the letter F?

35. In the game called "Simon Says," a leader calls out any one of the following commands: right face; about face; left face; Simon says "right face"; Simon says "about face"; Simon says "left face." The players obey the command if it begins with "Simon says." They are not supposed to move at all if the command does not begin with "Simon says." We may consider two commands to be equal if they have the same effect. Two commands can be "multiplied" by giving the commands in succession. The product is the single command that has the same effect as the two commands combined. The commands that do not begin with "Simon says" are obviously equal. Let *i* stand for any one of them. Let *r* stand for the command, "Simon says 'right face.' " Let *a* stand for the command, "Simon says 'about face.' " Let *l* stand for the command "Simon says 'left face.' " a) Construct the multiplication table for the set of commands whose members are *i*, *r*, *a* and *l*. b) Show that this set is a group with respect to multiplication.

8

Rules of Computation in a Group

Special Rules and General Rules

Every group has its own multiplication table. The table may be thought of as a set of rules that tell us how to multiply members of the group. Among these rules are some special rules that may apply to one group but not to another. For example, the group of symmetries of an equilateral triangle and the group of symmetries of a square both contain members we have called p and q. In each group there is a different special rule for multiplying these members, and the rule is shown in the table. In the multiplication table for the symmetries of an equilateral triangle we are told that $pq = i$, the identity element of the group. In the multiplication table for the symmetries of a square we are told that $pq = r$, which is not the identity element of the group.

The rules for multiplication in a group also include some general rules that are obeyed by all groups. These general rules arise from the four properties that all groups have in common.

The Substitution Rule

Every group is closed with respect to the multiplication operation in the group. That is, if a and b are any two members of the group, the product ab is also a member of the group. The multiplication table for the group tells us which member it is. *This member*,

which is equal to ab, may be substituted for ab in any product in which ab occurs as a factor.

For example, in the group of symmetries of a square, let us find the value of the product $(pq)r$. We see from the multiplication table on page 89 that $pq = r$, so we may substitute r for pq. Thus we get $(pq)r = rr$. We see in the table that $rr = q$, so we may substitute q for rr. Thus we get $(pq)r = rr = q$.

In the chapters before this one, we have already used the substitution rule many times without mentioning it.

Exercises:

1. Use the substitution rule and the multiplication table for the symmetries of an equilateral triangle to find the values of these products:

a) $r(st)$ b) $(iq)t$ c) $(pq)(rs)$

2. Use the substitution rule and the multiplication table for the symmetries of a rectangle to find the values of these products:

a) $(ab)c$ b) $(ca)b$ c) $(aa)(bb)$

3. Use the substitution rule and the multiplication table for the symmetries of a square to find the values of these products:

a) $(pu)v$ b) $(pq)(rs)$ c) $(pu)(qt)$

The Use of Parentheses

The multiplication operation in every group has the associative property. Because of this property, a product of three or more factors written in a definite order has the same value no matter how we use parentheses to associate the factors with each other. For this reason, the rule stated on page 21 for the group of measuring numbers applies to all other groups as well: *When we write the product of three or more members of a group, we may write it*

either without parentheses, or with parentheses around any two or more factors that are written next to each other in the product.

We usually write a product of three or more members of a group without parentheses. To find the value of such a product, we use the rule stated above together with the substitution rule. For example, to find the value of the product *psqus* in the group of symmetries of a square, we may first put parentheses around *ps* and around *qu*, to get $psqus = (ps)(qu)s$. In the multiplication table we see that $ps = u$, and $qu = v$. Substituting u for *ps*, and v for *qu*, we get $psqus = (ps)(qu)s = uvs$. If we put parentheses around *uv*, we have $psqus = (ps)(qu)s = uvs = (uv)s$. In the multiplication table we see that $uv = q$, and that $qs = t$. Making these substitutions, we see that $psqus = (ps)(qu)s = uvs = (uv)s = qs = t$. The parentheses could have been put in in other ways, too, but the final answer obtained in these other ways would also be t.

Exercises:

4. Use the parentheses rule, the substitution rule, and the multiplication table for the symmetries of a square to find the value of the product *risvut*.

5. Use the parentheses rule, the substitution rule, and the multiplication table for the symmetries of an equilateral triangle to find the value of the product *tqrps*.

The Use of *i*

If i is the identity element of a group, and x is any member of the group, then $ix = xi = x$. This means that in any product in which ix is a factor, ix may be replaced by x; and in any product in which xi is a factor, xi may be replaced by x. For example, in the group of symmetries of a square, the product $ipq = (ip)q = pq = r$.

Exercise:

6. In the group of symmetries of a square, find the values of these products:

a) qir; b) tsi; c) $(uu)p$.

One Identity Element

On page 50 we made the statement that a group has only one identity element. We shall now prove this statement.

Let i be an identity element in a group. If there is an element j in the group that is also an identity element, we shall show that j must be the same element as i: If i is an identity element, then multiplying j by i leaves j unchanged. That is, $ij = ji = j$. If j is an identity element, then multiplying i by j leaves i unchanged. That is, $ji = ij = i$. Then j must be equal to i, since they are both equal to ij and ji.

The Use of an Inverse

If x is any member of a group, x^{-1} is the symbol for its inverse. Since $xx^{-1} = x^{-1}x = i$, the identity element of the group, we may substitute i for xx^{-1} or for $x^{-1}x$. For example, if x and y are members of a group, then $xyy^{-1}x^{-1} = x(yy^{-1})x^{-1} = xix^{-1} = (xi)x^{-1} = xx^{-1} = i$.

Exercise:

7. In the group of symmetries of a square, find the values of these products:

a) $pp^{-1}p^{-1}$; b) stq^{-1}; c) $uv^{-1}v$.

The Multiplication Rule

If a and b are members of a group, and the statement $a = b$ is a true statement, then it means that the symbols "a" and "b"

stand for one and the same member, and either symbol may be substituted for the other. If c is also a member of the group, then, in the product ca we may substitute b for a. Then we see that $ca = cb$ is a true statement. We can also get this statement from the statement $a = b$ by multiplying both members of the equation by c, using c as a factor on the left-hand side in each case. Similarly, in the product ac we may substitute b for a. Then we see that $ac = bc$ is a true statement. We can also get this statement from the statement $a = b$ by multiplying both members of the equation by c, using c as a factor on the right-hand side in each case. Thus we get the rule that if we have a true equation concerning members of a group, we can get another true equation by multiplying both sides of the equation by the same member of the group, provided that the multiplication is done either from the left in both cases, or from the right in both cases.

For example, in the group of symmetries of a square, the inverse of p is r. So the statement $p^{-1} = r$ is a true statement. If we multiply from the left by p, we get the true statement $pp^{-1} = pr$, or $i = pr$.

Exercise:

8. Suppose the statement $px = r$ is true in the group of symmetries of a square, where x is an unknown member of the group. Multiply both sides of this equation from the left by r to find out which member of the group x stands for.

One Inverse

We said on page 50 that each member of a group has only one inverse. We shall now prove this statement.

Let a be a member of a group, and suppose that b and c are both inverses of a. Then we shall show that b must be the same element as c: Since b is an inverse of a, then $ab = ba = i$, the identity element of the group. Since c is an inverse of a then

$ac = ca = i$. Now, starting with the true statement $ab = i$, multiply both sides of this equation by c from the left. Then we get $c(ab) = ci$. Substituting c for ci, we see that $c(ab) = c$. However, by the associative property of multiplication in a group, $c(ab) = (ca)b$. One of the equations above tells us that $ca = i$, so we may substitute i for ca. Then we have $c(ab) = (ca)b = ib = b$. Then b and c are equal to each other, since they are both equal to $c(ab)$.

If a is any member of a group, and we find another member b such that $ab = ba = i$, then we know that b is an inverse of a. But since a can have only one inverse, and a^{-1} is the symbol for that inverse, then we may say that $a^{-1} = b$.

The Inverse of a Product

The fact that every member of a group has one and only one inverse makes it easy to find the inverse of a product of members of the group. *To get the inverse of a product, take the inverse of each factor, and multiply them in reverse order.* For example, the inverse of the product ab is the product $b^{-1}a^{-1}$. To prove that $b^{-1}a^{-1}$ is the inverse of ab, all we have to do is show that $(b^{-1}a^{-1})(ab) = i$, and that $(ab)(b^{-1}a^{-1}) = i$.

Exercises:

9. Prove that if a and b are members of a group, whose identity element is i, then a) $(b^{-1}a^{-1})(ab) = i$; b) $(ab)(b^{-1}a^{-1}) = i$.

10. If a, b and c are members of a group, verify that the inverse of abc is $c^{-1}b^{-1}a^{-1}$.

11. If a is a member of a group, prove that the inverse of a^2 is $(a^{-1})^2$.

The Use of Exponents

If a is a member of a group, there are some products, such as aa, aaa, and so on, that we get by repeated multiplication by a. As we did in the group of measuring numbers, we call such products powers of a, and we represent them by the symbols a^2, a^3, and so on. In general, if n is a counting number that is greater than 1, we define a^n to mean the product $aa \ldots a$ in which there are n factors that are all equal to a. We also define a^1 to mean a.

As we did in Chapter 3, we may extend the idea of a power of a to permit the use of any integer, even if it is zero or a negative integer, as an exponent. In the group of measuring numbers, we defined a^0 to mean 1, which is the identity element of the group. Similarly in any group, if a is any member of the group, we define a^0 to mean i, the identity element of the group. In the group of measuring numbers, if a is any member of the group, and $-n$ is any negative integer, we defined a^{-n} to mean $\frac{1}{a^n}$, the multiplicative inverse of a^n. Similarly, if a is a member of any group, we define a^{-n} to mean the inverse of a^n. Since the inverse of a^n is also represented by the symbol $(a^n)^{-1}$, we may say that $(a^n)^{-1} = a^{-n}$.

By our definition, the inverse of a^2 is represented by the symbol a^{-2}. However, a^2 is equal to the product aa, and we already have a rule for writing the inverse of a product, namely, take the inverse of each factor and multiply them in reverse order. By this rule, the inverse of aa is $a^{-1}a^{-1}$, which can also be written as $(a^{-1})^2$. Thus we see that $a^{-2} = (a^{-1})^2$. In general, if $-n$ is a negative integer, a^{-n} stands for the inverse of a^n. But a^n is the product $aa \ldots a$, in which there are n factors that are all equal to a. The inverse of this product is $a^{-1}a^{-1} \ldots a^{-1}$, in which there are n factors that are all equal to a^{-1}, and which may be written briefly as $(a^{-1})^n$. So we see that $a^{-n} = (a^{-1})^n$. In some books this equation is used as the definition of a^{-n}.

It is not difficult to show that if m and n are any two integers, then $a^m a^n = a^{m+n}$. Specific instances of this rule are verified in the exercises.

Exercises:

12. Show that $aa^2 = a^3$; $aa^3 = a^4$; $aa^4 = a^5$.
13. Use the definition of powers of a and the parentheses rule to verify that $a^3a^2 = a^5$.
14. Verify that $a^3a^{-2} = a$. (Hint: Use the fact that $a^{-2} = a^{-1}a^{-1}$.)
15. Show that $a^{-3}a^{-2} = a^{-5}$.
16. Show that $(a^3)^2 = a^6$.
17. Show that $(a^{-3})^2 = a^{-6}$.
18. Show that $(a^3)^{-2} = a^{-6}$.
19. Show that $(a^{-3})^{-2} = a^6$.

The Subgroup Generated by a Member

Let a be a member of a group, and consider the set of all powers of a. The members of this set include a^0, which is equal to i, the identity element of the group. They also include the powers a, a^2, a^3, and so on, and the powers a^{-1}, a^{-2}, a^{-3}, and so on. As we did in Chapter 2, we call this set *the set generated by a*. All of its members are in the group, since the group is closed with respect to multiplication, and the group contains the inverse of each of its members. Any member of this set has the form a^m, where m is an integer. We now show that this set has all the four properties of a group.

I. Closure. Let a^m and a^n be any two members of the set generated by a. Their product $a^m a^n$ is equal to a^{m+n}, which is also

a member of the set. (See exercises 12, 13 and 14 for examples.) So the set is closed with respect to multiplication.

II. Associative Property. Since multiplication has the associative property in the group, and all members of this set are in the group, then multiplication has the associative property in this set.

III. Identity Element. The set contains $a^0 = i$, which is an identity element for the set.

IV. Inverse. The set contains an inverse for every one of its members. The element a^0, which is i, is its own inverse. The inverse of a is a^{-1}, and the inverse of a^{-1} is a; the inverse of a^2 is a^{-2}, and the inverse of a^{-2} is a^2; the inverse of a^3 is a^{-3}, and the inverse of a^{-3} is a^3; and so on.

Thus the set of elements generated by a is a group with respect to multiplication. A group like this, all of whose members are powers of a single member, is called a *cyclic group.*

We now have two groups, the original group of which a is a member, and the cyclic group whose membership is the set generated by a. To avoid mixing them up as we talk about them, let us give each of them a name. Let us call the original group G. Any group, all of whose members are members of G is called a *subgroup* of G. So we shall call the cyclic group whose membership is the set generated by a the *subgroup of G that is generated by a.*

In our description of the subgroup of G that is generated by a, we have listed an infinite number of symbols, each of which stands for a member of the subgroup. This does not mean that there are necessarily infinitely many members in the subgroup. There may be only a finite number of members in it, with many different symbols standing for the same members. For example, consider the group of all symmetries of a square. Let us call this group S, and let us identify within it the subgroup generated by the element called p. The group of symmetries of a square has exactly eight

members, namely, i, p, q, r, s, t, u and v. (See page 88.) Since every power of p must be equal to one of these members, the subgroup generated by p cannot have more than eight members. Let us identify which member each power of p is equal to, and then we shall be able to see how many of them are different. Let us take first the non-negative powers: $p^0 = i$; $p^1 = p$; $p^2 = pp = q$; $p^3 = pp^2 = pq = r$; $p^4 = pp^3 = pr = i$; $p^5 = pp^4 = pi = p$; $p^6 = p^2p^4 = qi = q$; $p^7 = p^3p^4 = ri = r$; $p^8 = p^4p^4 = ii = i$; $p^9 = pp^8 = pi = p$; and so on. Obviously, as we take higher and higher powers of p, the values i, p, q and r are repeated over and over again, and no other values ever arise.

Now let us identify the values of the negative powers of p: p^{-1} = the inverse of $p = r$; p^{-2} = the inverse of p^2 = the inverse of $q = q$; p^{-3} = the inverse of p^3 = the inverse of $r = p$; p^{-4} = the inverse of p^4 = the inverse of $i = i$; p^{-5} = the inverse of p^5 = the inverse of $p = r$; and so on. Obviously, as we continue the series of negative powers of p, the values r, q, p and i are repeated over and over again, and no other values ever arise.

Thus we see that the subgroup of S generated by p has exactly four members, namely, i, p, q and r. The multiplication table for this group is given in the answer to exercise 26 of Chapter 7. (See page 246.)

In the subgroup of S generated by p, there are only four members, but each member may be represented by infinitely many powers of p. The member i is represented by any one of these powers of p: p^0, p^4, p^8, p^{12}, and so on, as well as p^{-4}, p^{-8}, p^{-12}, and so on. The member p is represented by any one of these powers of p: p^1, p^5, p^9, p^{13}, and so on, as well as p^{-3}, p^{-7}, p^{-11}, and so on. The member q is represented by any one of these powers of p: p^2, p^6, p^{10}, p^{14}, and so on, as well as p^{-2}, p^{-6}, p^{-10}, and so on. The member r is represented by any one of these powers of p: p^3, p^7, p^{11}, and so on, as well as p^{-1}, p^{-5}, p^{-9}, and so on.

Notice that it is possible to represent each member of the subgroup S generated by p by a power of p that has a positive expo-

nent. In fact, we have many such powers to choose from. If we choose in each case the power of p that has the lowest positive exponent, then we can list the entire membership of the subgroup as follows:

$$p^1 = p,\ p^2 = q,\ p^3 = r,\ p^4 = i.$$

In general, if G is a finite group, and a is a member of G, the subgroup of G that is generated by a is also a finite group. If this subgroup contains exactly m members, these members are a^1, a^2, . . . , a^m, and $a^m = i$.

Exercises:

20. Let S be the group of symmetries of a square. Use the multiplication table on page 89 to identify the members of the subgroup of S generated by r. What is the power of r with the lowest positive exponent that represents each of these members?

21. Identify the members of the subgroup of S generated by v.

22. Let T be the group of symmetries of an equilateral triangle. Use the multiplication table on page 81 to identify the members of the subgroup of T generated by p.

23. Identify the members of the subgroup of T generated by s.

24. Let A stand for the multiplicative group whose membership consists of the non-zero members of the seven-point clock-face number system. Use the multiplication table on page 241 to identify the members of the subgroup of A that is generated by *2*.

25. Identify the members of the subgroup of A that is generated by *3*.

26. Let G be a finite group whose identity element is i, and let a be a member of G that is not i. Suppose that in the sequence of powers of a with positive exponents the powers a^m and a^n turn

out to be equal, with m less than n. Find a positive power of a that is equal to i.

27. If a group is cyclic, every member of the group is a power of some one member in the group. Prove that a cyclic group is commutative.

The Cancellation Rule

If a, b and c are members of a group, the multiplication rule tells us that if $a = b$, then $ca = cb$. Let us see whether this rule can be reversed. If $ca = cb$, may we remove or "cancel" the c on both sides of the equals sign and conclude that $a = b$?

Before we try to answer this question for a group, let us see what the answer is in some familiar systems that are not groups. Consider first the set of all counting numbers with the operation multiplication. We know that the set of counting numbers is not a group with respect to multiplication, because it does not contain a multiplicative inverse for each of its members. In this set there is a multiplication rule: if $a = b$, then $ca = cb$, and $ac = bc$. There is also a cancellation rule: if $ca = cb$, or if $ac = bc$, then $a = b$. For example, if we know that $2 \times a = 2 \times 3$, then we may cancel the twos and conclude that $a = 3$. In other words, 3 is the only number which, when multiplied by 2 produces a product equal to 6.

The situation is quite different in the set of all non-zero members of the six-point clock-face number system, used with the operation multiplication. This set, like the set of all counting numbers, is not a group with respect to multiplication. (See page 73.) In this set there is a multiplication rule, but there is no cancellation rule. The multiplication table for this set is given in the answer to exercise 14 of Chapter 6. (See page 240.) The table shows that in this system, $2 \times 1 = 2$, and $2 \times 4 = 2$. So $2 \times 1 = 2 \times 4$, since both products are equal to 2. However, we cannot then simply cancel the twos in this equation and conclude that $1 = 4$, because this state-

ment is false. In this system, *1* and *4* are two different numbers. The trouble arises from the fact that neither *1* nor *4* is the only number which, when multiplied by *2*, produces a product equal to *2*.

Thus we see that if a system has a multiplication rule, this fact alone does not guarantee that it also has a cancellation rule. However, we can prove that if a system is a group, then it does have a cancellation rule. Let us suppose that a, b and c are members of a group, and that $ca = cb$. Since the system is a group, it contains an inverse for every one of its members. In particular, it contains an element c^{-1} which is the inverse of c. Using the multiplication rule, we know we may multiply both sides of the equation by c^{-1}, multiplying from the left in both cases. Then we have $c^{-1}(ca) = c^{-1}(cb)$. Since multiplication in a group has the associative property, we may replace $c^{-1}(ca)$ by $(c^{-1}c)a$, and we may replace $c^{-1}(cb)$ by $(c^{-1}c)b$. So we have $(c^{-1}c)a = (c^{-1}c)b$. Since $c^{-1}c$ may be replaced by i, we have $ia = ib$. But $ia = a$, and $ib = b$, so we conclude that $a = b$. We can show by a similar argument that if $ac = bc$, then $a = b$. Thus there is a cancellation rule in a group.

Because of the cancellation rule in a group, we can be sure that if a is *not* equal to b, then ca is *not* equal to cb. This is so, because the cancellation rule shows that $ca = cb$ only if $a = b$. Similarly, if a is *not* equal to b, then ac is *not* equal to bc. For this reason, if three different members of a group, say p, q, and r, are all multiplied from the left by the same member, say p, then the three products pp, pq, and pr are all different. Similarly, if they are multiplied by p from the right, the three products pp, qp, and rp are all different. In fact, if *all* the different members of a group are multiplied by some one fixed member, either all from the left or all from the right, then *all* the products are different. For example, in the group of symmetries of a rectangle, if all the members i, a, b and c are multiplied from the left by a, the products are a, i, c and b respectively, which are all different from each other. (See the table on page 86.)

Exercises:

28. In the set of non-zero members of the four-point clock-face number system, find two different numbers which, when multiplied by *2*, produce the same product. (See the table on page 72.) Does this system have a cancellation rule for multiplication?

29. Prove that in a group, if $ac = bc$, then $a = b$.

30. Prove that if b is a member of a group and $bb = b$ then $b = i$, the identity element of the group.

Equations in a Group

In the system of counting numbers, the equation $2x = 3$ does not have any solutions. There is no counting number that we can multiply by *2* to produce *3* as a product. On the other hand, in the system of measuring numbers, the equation $2x = 3$ does have a solution. In fact $x = \frac{3}{2}$ is a solution, as we can verify by substituting $\frac{3}{2}$ for x in the equation. Then we get $2\left(\frac{3}{2}\right) = 3$, which is a true statement. Moreover, $\frac{3}{2}$ is the only solution of the equation. To see this fact, let us suppose that x stands for some solution of the equation. Then $2x = 3$ is a true statement. The system of measuring numbers contains the multiplicative inverse of every one of its members. In particular, it contains $\frac{1}{2}$, which is the inverse of *2*. Multiplying both sides of the equation by $\frac{1}{2}$, we get $\frac{1}{2}(2x) = \frac{1}{2}(3)$. Because multiplication of measuring numbers has the associative property, we may replace $\frac{1}{2}(2x)$ by $\left(\frac{1}{2} \times 2\right)x$.

So we have $\left(\frac{1}{2} \times 2\right) x = \frac{1}{2}(3)$. We may replace $\frac{1}{2} \times 2$ by 1, and we may replace $\frac{1}{2}(3)$ by $\frac{3}{2}$. Then we have $1x = \frac{3}{2}$. But, since 1 is the multiplicative identity element, $1x = x$. When we replace $1x$ by x, we find that $x = \frac{3}{2}$. This shows that the number $\frac{3}{2}$ is the only solution of the equation $2x = 3$.

By a similar argument we can show that if a and b are members of a group, the equation $ax = b$ has one and only one solution in the group. We can verify that it has a solution by substituting $a^{-1}b$ for x. We can show that it has only one solution by multiplying both sides of the equation by a^{-1}, multiplying from the left in both cases.

Example: In the group of symmetries of a square, solve the equation $px = u$. We see from the table on page 89 that the inverse of p is r. Multiplying from the left by r on both sides of the equation, we get $r(px) = ru$, $(rp)x = ru$, $ix = ru$, and $x = ru = s$. The table shows that ps is indeed equal to u. Because this group is a finite group, whose multiplication table can be written down completely, there is also a shorter way of solving the equation. Simply look for u in the p row of the table. Since it occurs in the s column, then we know that $ps = u$, and $x = u$ is the solution of the equation.

Exercises:

31. If a and b are members of a group, verify that $a^{-1}b$ is a solution of the equation $ax = b$.

32. Show that $a^{-1}b$ is the only solution of $ax = b$.

33. If a and b are members of a group, verify that ba^{-1} is a solution of the equation $xa = b$.

34. Show that ba^{-1} is the only solution of $xa = b$.

35. In the group of symmetries of a square, solve the equation $sx = v$.

36. In the group of symmetries of a square, solve the equation $xs = v$.

37. In the group of symmetries of an equilateral triangle, solve the equation $qx = p$.

38. In the group of symmetries of an equilateral triangle, solve the equation $xr = t$.

39. In the multiplicative group of the non-zero members of the five-point clock-face number system, solve the equations

a) $2x = 3$; b) $3x = 1$; c) $4x = 2$.

(See the table on page 70.)

Computation in a Commutative Group

In a commutative group, if a and b are any members of the group, $ab = ba$. This fact leads to the special rule that *in a commutative group the factors in a product may be written in any order whatever*. We can prove this rule by making repeated use of the fact that multiplication in a commutative group has the commutative property as well as the associative property. For example, to prove that in a commutative group $abab = aabb$, we note that $abab = a(ba)b = a(ab)b = aabb$.

Because the factors of a product in a commutative group may be rearranged in any order, there is a special rule for the use of exponents in a commutative group: *In a commutative group*, $(ab)^n = a^n b^n$. We prove this rule here for the special case where $n = 2$: $(ab)^2 = (ab)(ab) = abab = aabb = (aa)(bb) = a^2b^2$.

Exercises:

40. Prove that in a commutative group $abc = bca$.

41. Prove that in a commutative group $(ab)^3 = a^3b^3$.

42. Prove that in a commutative group $(ab)^{-1} = a^{-1}b^{-1}$.

43. a) Prove that if the square of every member of a group is equal to the identity element then the group is commutative. (Hint: First identify the inverse of each member.) b) Find in Chapter 7 an example of a group with more than two members where the square of every member is equal to the identity element.

The Use of the Sign +

Sometimes the sign + is used for the group operation in a commutative group. We have already seen examples of this in the group of integers with the operation addition, and in the additive groups of clock-face numbers. *The sign + is never used for the group operation in a group that is not commutative.* So, whenever you see the sign + used for the operation in a group, you know that the operation is assumed to have the commutative property.

When the sign + is used for the operation in a group, these other changes in notation are also made: the identity element of the group is called *zero*, and is represented by the symbol *0*; the inverse of a is written as $-a$, instead of a^{-1}; the nth power of a is written as na instead of a^n.

Exercise:

44. Below are some expressions or statements that may occur in a commutative group in which the group operation is written as

multiplication. What expressions or statements would take their place if the group operation were represented by the sign $+$? a) pq; b) $pp^{-1} = i$; c) $ip = p$; d) $pp = p^2$; e) $p(qr) = (pq)r$; f) $pq = qp$; g) $(pq)^2 = p^2q^2$.

9

Recognizing a Group by Its Table

Patterns in the Table

If a group is finite, its multiplication table can be written out completely. Each member of the group is used once as the label of a row, and it is also used once as the label of a column. The row labels, written from top to bottom, and the column labels, written from left to right, are in the same order. The product of every ordered pair of members of the group appears as an entry in the table. If a and b are members of the group, the product ab appears in the a row and the b column, and the product ba appears in the b row and the a column.

Every group satisfies the four group properties of *Closure*, *Associative Property*, *Identity Element*, and *Inverse*. Each of these properties shows itself as a definite pattern in the arrangement of the products in the multiplication table of the group. We shall examine these patterns one at a time.

The *Closure* Pattern

A group consists of a set of members and a multiplication operation, with the four properties listed above.

The *Closure* property of the group is that every product of two members of the set is also a member of the set. Every entry in the multiplication table of the group is a product of two members of

the set. Then the Closure property of a group shows itself in this pattern in its multiplication table: *Every entry in the table is a member of the set whose elements are being multiplied.*

It is obvious that a multiplication table that has this pattern has the Closure property. For this reason let us call this pattern the *Closure pattern.* Then we may say that if multiplication in a set has the Closure property, then its multiplication table has the Closure pattern, and vice versa.

For example, the membership of the group of symmetries of an equilateral triangle consists of the elements i, p, q, r, s and t, defined on page 77. In the multiplication table for this group, printed on page 81, we see that each entry is a member of this set, so the table has the Closure pattern.

Exercises:

1. The membership of the group of symmetries of a rectangle consists of the elements i, a, b and c defined on page 85. Verify that each entry in the multiplication table for this group is a member of this set. (See page 86.)

2. The membership of the group of symmetries of a square consists of the elements i, p, q, r, s, t, u and v defined on page 88. Verify that each entry in the multiplication table for this group is a member of this set. (See page 89.)

The *Identity Element* Pattern

The *Identity Element* property of a group is that for every member a of the group, $ia = a$, and $ai = a$. The entry that is in the i row and the a column of the multiplication table is the product ia. Since this product is equal to a, the entry in the i row and the

a column is a copy of the column label. For this reason, the entire *i* row is a copy of the column labels, column by column. Similarly, the entry that is in the *a* row and the *i* column of the multiplication table is the product *ai*. Since this product is equal to *a*, the entry in the *a* row and the *i* column is a copy of the row label. For this reason, the entire *i* column is a copy of the row labels, row by row. So the Identity Element property of a group shows itself in this pattern in its multiplication table: *There is an element whose row is a copy of the column labels, column by column, and whose column is a copy of the row labels, row by row.*

For example, in the multiplication table on page 81 for the group of symmetries of an equilateral triangle, the column labels from left to right and the row labels from top to bottom are *i*, *p*, *q*, *r*, *s* and *t*. The entries in the *i* row and in the *i* column are also *i*, *p*, *q*, *r*, *s* and *t* in the same order.

It is obvious, too, that if a multiplication table for a set has this pattern, then the set has an identity element for multiplication. In fact, suppose *i* is the name of the element whose row is a copy of the column labels and whose column is a copy of the row labels. Then the entry in the *i* row and *a* column is *a*, which is the column label. But the entry in the *i* row and the *a* column is the product *ia*. So $ia = a$. Also, the entry in the *a* row and the *i* column is *a*, which is the row label. But the entry in the *a* row and the *i* column is the product *ai*. So $ai = a$. Since this is true for every element *a* in the set, the element *i* satisfies the requirements for being an identity element. Because of this fact, let us call this pattern the *Identity Element pattern*. Then we may say that if a set has an identity element for multiplication, then its multiplication table has the Identity Element pattern, and vice versa.

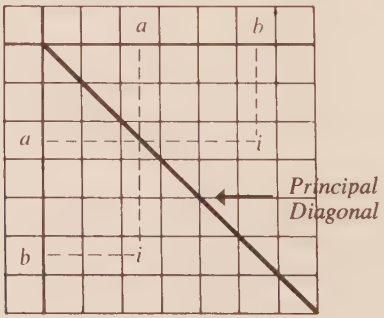
Exercise:

3. In the multiplication table for the group of symmetries of a rectangle, the row labels and the column labels are *i*, *a*, *b*, *c* and *d*,

listed in that order. Verify that the *i* row and the *i* column are copies of these labels listed in the same order.

The *Inverse* Pattern

The *Inverse* property of a group is that for every element *a* in the group there is another element *b* in the group for which the products *ab* and *ba* are both equal to *i*. The product *ab* is in the *a* row, and the product *ba* is in the *a* column. So there is an *i* in the *a* row and there is also an *i* in the *a* column. We can describe more exactly how these two entries of the symbol *i* are placed with respect to each other in the table. The multiplication table is a square array of boxes. Imagine a line drawn from the upper left-hand corner to the lower right-hand corner of the array. This line is called the *principal diagonal* of the array. Each box whose row label is the same as its column label lies on the principal diagonal. The box that is in the *a* row and the *b* column and the box that is in the *b* row and the *a* column are symmetrically placed with respect to the principal diagonal. (That is, if the table were folded in half along the principal diagonal, these two boxes would fall on each other.) Since there is an *i* in each of these boxes, we may say that the *i* in the *a* row and the *i* in the *a* column are symmetrically placed with respect to the principal diagonal, as shown in the diagram below. We see then that the inverse property of a



group shows itself in this pattern in its multiplication table: *If i is the identity element, there is an i in every row and in every column of the table, and all these entries of the symbol i are symmetrically placed with respect to the principal diagonal.*

As an example of this pattern, we show below the location of all entries that are equal to i in the multiplication table for the symmetries of an equilateral triangle. The boxes on the principal diagonal are shaded.

	i	p	q	r	s	t
i	i					
p			i			
q		i				
r				i		
s					i	
t						i

It is easy to see that if a multiplication table for a set has this pattern, then the set contains an inverse for every one of its members. In fact, let a be any member of the set. If the table has the pattern described above, then there is an i in the a row and another one in the a column, and they are symmetrically placed with respect to the principal diagonal. If the i in the a row is in the b column, then the i in the a column is in the b row. The entry i in these two places means that $ab = i$, and $ba = i$, so that b is the inverse of a, and a is the inverse of b.

Because of this fact, let us call the pattern described above the *Inverse pattern.* Then we may say that if a set has an inverse for every one of its members, then its multiplication table has the Inverse pattern, and vice versa.

Exercises:

4. Make a copy of the multiplication table for the group of symmetries of a rectangle, showing only the row and column labels

and the location of all entries equal to i. Shade the boxes that are on the principal diagonal. Verify that an i appears in every row and every column, and that the i entries are symmetrically placed with respect to the principal diagonal. (See page 86.)

5. Follow the directions given in exercise 4, using the multiplication table for the group of symmetries of a square. (See page 89.)

The *Once and Only Once* Pattern

The inverse property of a group also leads to another pattern in its multiplication table. We saw on page 107 that if a and b are any two members of a group, then there is one and only one solution of the equation $ax = b$. This means that the element b appears in the a row once and only once. Since this is true for any two members a and b of the group, we see that each member of the group appears in each row once and only once. Similarly, if a and b are any two members of a group, there is one and only one solution of the equation $xa = b$. This means that the element b appears in the a column once and only once. Since this is true for any two members a and b of the group, we see that each member of the group appears in each column once and only once. Thus we see that the inverse property of a group shows itself in a second pattern in the multiplication table of the group: *Each member of the set whose elements are being multiplied in the table appears in each row and in each column once and only once.*

Exercises:

6. Verify that in the multiplication table for the group of symmetries of an equilateral triangle, each of the elements i, p, q, r, s and t appears in each row and in each column once and only once.

7. Verify that in the multiplication table for the group of symmetries of a rectangle, each of the elements i, a, b and c appears in each row and in each column once and only once.

8. Verify that in the addition table for the additive group of the five-point clock-face number system, each of the numbers *0*, *1*, *2*, *3* and *4* appears in each row and in each column once and only once. (See page 68.)

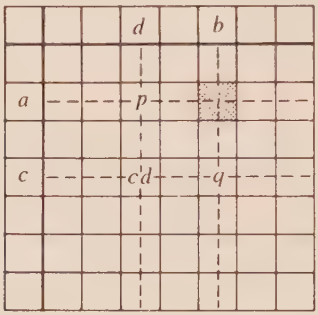
9. Verify that in the multiplication table for the multiplicative group of the non-zero members in the five-point clock-face number system, each of the numbers *1*, *2*, *3* and *4* appears in each row and in each column once and only once. (See page 71.)

The *Rectangle* Pattern

The Associative Property of a group is that if a, b and c are any three members of the group, then $a(bc) = (ab)c$. We know that this property leads to the rule that we may write a product of three or more members of a group either without parentheses, or with parentheses around any two or more factors that are written next to each other in the product. We shall see that this rule leads to a definite pattern in the arrangement of the entries in the multiplication table of a group.

Because of the Inverse pattern, there is an i in every row and every column of the table. Choose any one of these i entries, and suppose it is in the a row and in the b column. Then choose any other row, say the c row, and any other column, say the d column. The a and c rows cross the b and d columns in four places in the table. These four places are at the corners of a rectangle. At one corner, where the a row crosses the b column, is the entry i. There is a definite pattern that relates the entries that are at the other three corners of the rectangle. This is how we find it. The entry in the c row and the d column is the product cd, whatever its name may be. Suppose that the entry in the a row and d column is p.

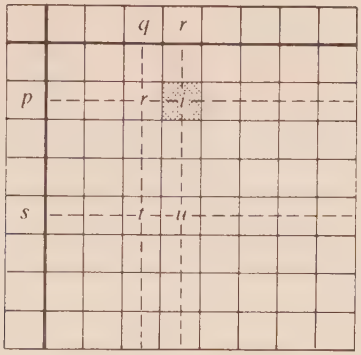
Then $ad = p$. Suppose that the entry in the c row and the b column is q. Then $cb = q$. Since the entry i is in the a row and the b column, it must also appear in the b row and the a column, because of the



The Rectangle Pattern:
$qp=cd$

Inverse pattern we have already observed. Then not only is ab equal to i, but ba is also equal to i. With this information in mind, let us examine the product of q and p. Since $q = cb$, and $p = ad$, then $qp = (cb)(ad)$. By the parentheses rule which arises from the associative property, we may remove and insert parentheses as we wish. So $qp = (cb)(ad) = cbad = c(ba)d$. But $ba = i$. So substituting i for ba, we may continue the chain of equalities in this way: $qp = (cb)(ad) = cbad = c(ba)d = cid = (ci)d = cd$. Thus we see that $qp = cd$. Notice that q, p and cd are the entries in three corners of the rectangle. The entry cd is in the corner diagonally opposite the one where i is. The entry q is in the same column with i. The entry p is in the same row with i. Thus we see that the associative property shows itself in this pattern in the multiplication table relating the entries at the corners of certain rectangles in the table: *In every rectangle in the table that has i at one corner of the rectangle, the entry at the corner that is in the same column with i times the entry at the corner that is in the same row with i is equal to the entry at the corner that is diagonally opposite i.* Let us call this pattern the *Rectangle pattern.*

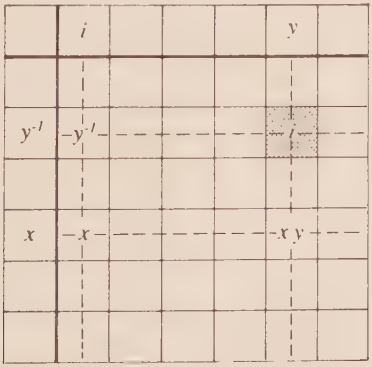
As an example of this pattern, we show below a rectangle taken from the multiplication table for the symmetries of a square. This



$ur=t$

rectangle is formed by the p and s rows and the q and r columns. There is an i at one corner of the rectangle. In the same column with i is the entry u. In the same row with i is the entry r. Diagonally opposite i is the entry t. According to the table on page 89, $ur = t$.

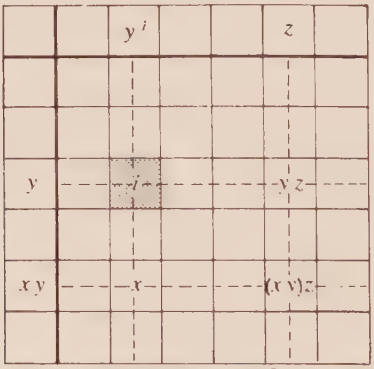
It is easy to see, too, that if the multiplication table for a set has this pattern, and the set also has the Closure property, the Identity Element property, and the Inverse property, then the multiplication operation in the set has the associative property. In fact, let x, y and z be any three members of the set. Since the set contains



$(xy)y^{-1}=x$

an inverse for every one of its members, it must contain an element y^{-1} for which $yy^{-1} = y^{-1}y = i$. Consider the rectangle in the multiplication table formed by the x and y^{-1} rows and the i and y columns. It is shown on page 119. The entry at the corner where the y^{-1} row crosses the y column is i, since $y^{-1}y = i$. The entry at the corner where the x row crosses the y column is xy. The entry at the corner where y^{-1} row crosses the i column is y^{-1}, since $y^{-1}i = y^{-1}$. The entry at the corner where the x row crosses the i column is x, since $xi = x$. Since the multiplication table we are talking about is assumed to have the Rectangle pattern, then $(xy)y^{-1} = x$.

Now consider the rectangle formed by the y and xy rows and the y^{-1} and z columns. This rectangle is shown below. The entry at the corner where the xy row crosses the y^{-1} column is x, because



$x(yz)=(xy)z$

as we have just seen, $(xy)y^{-1} = x$. The entry at the corner where the y row crosses the y^{-1} column is i, because $yy^{-1} = i$. The entry at the corner where the y row crosses the z column is yz. The entry at the corner where the xy row crosses the z column is $(xy)z$. Because we are assuming that the table has the Rectangle pattern, we have $x(yz) = (xy)z$. Thus we see that if a set which has the

Closure property, the Identity Element property, and the Inverse property also displays the rectangle pattern in its table, then the operation in the set has the associative property. Then we may say that if a set which is closed with respect to multiplication, which has an identity element, and which has an inverse for each of its members also has the associative property, then its multiplication table has the Rectangle pattern, and vice versa.

Exercises:

10. The multiplication table for the non-zero members of the three-point clock-face number system is shown in the answer to exercise 13 of Chapter 6. (See page 240.) In this table, *1* is the identity element, and every element has an inverse. There is only one rectangle that has *1* at a corner. Verify that this rectangle shows the Rectangle pattern.

11. The multiplication table for the non-zero members of the five-point clock-face number system is shown on page 71. In this table, *1* is the identity element, and every element has an inverse. The number *1* appears in row *2* and column *3*. There are nine rectangles that have this particular *1* at a corner. Using a separate diagram for each of these rectangles, show the location in the table of the four corners of the rectangle. Verify from the table that each of these rectangles shows the Rectangle pattern.

Five Patterns

We have observed five patterns that can be seen in the multiplication table of a group. We list all the patterns together here for convenience:

The Closure pattern. Every entry in the table is a member of the set whose elements are being multiplied.

The Identity Element pattern. There is an element whose row is a copy of the column labels, column by column, and whose column is a copy of the row labels, row by row.

The Inverse pattern. If i is the identity element, there is an i in every row and in every column of the table, and all these entries of the symbol i are symmetrically placed with respect to the principal diagonal.

The Once and Only Once pattern. Each member of the set whose elements are being multiplied in the table appears in each row and in each column once and only once.

The Rectangle pattern. In every rectangle in the table that has i at one corner of the rectangle, the entry at the corner that is in the same column with i times the entry at the corner that is in the same row with i is equal to the entry at the corner that is diagonally opposite i.

We have seen, too, that if a multiplication table has all five of these patterns in it, then the set to which the multiplication table belongs, taken with the multiplication operation shown in the table, has the properties of *Closure*, *Associative Property*, *Identity Element* and *Inverse*, so it must be a group. Thus we have a way of recognizing a group by examining the patterns in its multiplication table. A table that shows all five patterns is the table of a group. If any one of the five patterns is missing then the table is

I

	i	a
i	i	a
a	a	b

II

	i	a	b
i	i	b	a
a	b	a	b
b	a	a	i

not the table of a group. For example, Table I on page 122 cannot be a group table, because it does not have the closure pattern since $aa = b$, and b is not a member of the set whose elements are being multiplied. Table II cannot be a group table, since it does not have the identity element pattern. That is, there is no element whose row is a copy of the column labels and whose column is a copy of the row labels.

Exercise:

12. Which of the five patterns does Table I on page 122 show? Which patterns does it not show?

The Commutative Pattern

In the multiplication table for a group, the product ab appears in the box that is in the a row and the b column. If the table is folded in half along the principal diagonal, this box falls on the box that is in the b row and the a column. The entry in the latter box is the product ba. In a commutative group, $ab = ba$. So the multiplication table for a commutative group shows this additional pattern: two boxes that are symmetrically placed with respect to the principal diagonal contain the same entries. It is obvious, too, that if a multiplication table shows this pattern, then the multiplication has the commutative property. For this reason, let us call this pattern the *Commutative pattern.* Then we may say that if a group is commutative, its table has the Commutative pattern, and vice versa.

For example, the additive group of the five-point clock-face number system is a commutative group. Its table, shown on page 68, has the Commutative pattern. That is, any two boxes

that are symmetrically placed with respect to the principal diagonal have the same entries.

Exercise:

13. The multiplication table shown below belongs to a commutative group. Some of the entries in the table are missing. Use the Commutative pattern to fill in the missing entries.

	i	a	b	c
i	i	a		
a		b		i
b	b	c	i	
c	c		a	b

10

Transformations

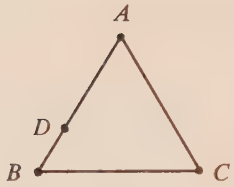
A Second Look at Motion

In Chapter 7 we talked about certain motions called symmetries of plane figures. We said there that two motions are equal if they have the same effect. We shall now examine these motions further to get a better understanding of what is meant by the *effect* of a motion. We shall find that it is possible to produce the effect of a motion without using any motion at all.

A Triangle Is a Set of Points

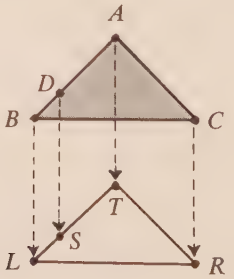
To study the symmetries of an equilateral triangle, we first cut one out of paper and labeled its vertices A, B and C. Strictly speaking, the triangle is not where the paper is. The triangle is the edge of the triangular piece of paper, or its boundary. What is inside this edge, where the paper is, is called a *triangular region.* We had to cut out a triangular region of paper so that we could hold it, write on it, and turn it over. We were not concerned with the region, but only with its boundary. So let us pay no further attention to the region, and look only at its boundary.

The triangle which is the boundary of the triangular region is a set of points. Three of these points are the vertices A, B and C of the triangle. The rest of the points lie between pairs of vertices on the straight lines that join them. Some of the points, like the point D, shown in the diagram, lie between A and B. Some of the points are between B and C, and some are between C and A.



The Position of Each Point

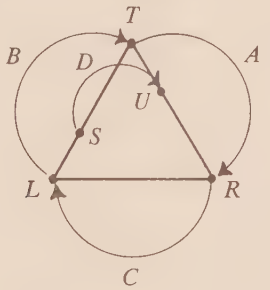
Each symmetry of a triangle brings it from a first "position" to a final "position." Let us see exactly what is meant by the word "position" as we use it here. Before we move the triangle, we have it resting flat on a table. Then each point of the triangle which is the boundary of the paper triangular region is directly over a point on the surface of the table. The set of all these points on the surface of the table is a triangle on this surface. In order to "see" this triangle, let us imagine it drawn with pencil directly on the surface of the table. Let us call its top vertex T, its left-hand vertex L, and its right-hand vertex R. When triangle ABC is put on top of triangle TLR, each point of triangle ABC rests on a point of triangle TLR. The point that it rests on is what we mean by its "position." For example, if we put triangle ABC down with A at the top, B at the left, and C at the right, then A rests on T, B rests on L, and C rests on R. The point D, which is between A and B, will then rest on some point S between T and L. Then the position of A is T, the position of B is L, the position of C is R, and the position of D is S.



We can use each point of triangle *ABC* and its position on triangle *TLR* to form an ordered pair. Thus we get ordered pairs like (*A*, *T*), (*B*, *L*), (*C*, *R*), (*D*, *S*), and so on. In each of these ordered pairs, the first member is a point of triangle *ABC*, and the second member is the position of that point. We call the set of all these ordered pairs the "position" of triangle *ABC*. It tells us where each individual point of triangle *ABC* rests on triangle *TLR*.

The Effect of a Motion

Each symmetry of triangle *ABC* moves it from a first position on triangle *TLR* to a final position on triangle *TLR*. In both positions, triangle *ABC* looks the same to the eye, because in both positions it fits exactly over the fixed triangle *TLR*. But the two positions may be different, with the separate points of triangle *ABC* in different positions in each case. For example, suppose that in the first position of triangle *ABC* the point *A* is resting on *T*, the point *B* is resting on *L*, and the point *C* is resting on *R*. If we use the motion *p* which rotates the triangle *120°* clockwise around its center, then in the final position of triangle *ABC*, point *A* will be resting on *R*, point *B* will be resting on *T*, and point *C* will be resting on *L*. The point *D* will be resting on some point *U* between *T* and *R*. The change that the motion *p* produces in the positions of the separate points of triangle *ABC* is what we mean by its "effect." So the effect of motion *p* is to



The Effect of Motion p

move A from T to R, to move B from L to T, to move C from R to L, to move D from S to U, and so on. We can write down a list of these changes of position as follows:

Old position		New position
T	$\xrightarrow{A}$	R
L	$\xrightarrow{B}$	T
R	$\xrightarrow{C}$	L
S	$\xrightarrow{D}$	U
.	.	.

The three dots show that there are many more points whose changes of position are not listed. We cannot actually show them all because there are infinitely many of them. In this list, the A over the first arrow shows that it is point A that moved from T to R. But motion p moves A from T to R only if A rested on T to begin with. We might have started with triangle ABC in a different position, say with point B on T. Then motion p would have moved B from T to R. That is, motion p moves from T to R whatever point of triangle ABC was resting on T to begin with. Since this point need not be A, it is better not to mention the points of triangle ABC in the list at all. So we write it this way:

Old position		New position
T	$\longrightarrow$	R
L	$\longrightarrow$	T
R	$\longrightarrow$	L
S	$\longrightarrow$	U
.	.	.

This list, which matches each old position of a point of triangle *ABC* with its new position, is a description of the effect of motion *p*. Since two motions that have the same effect are considered to be equal, we can go even further. We can say that *this list is a complete description of motion p*, because any motion whose effect is described by the same list must be equal to *p*. From this point of view, motion *p* concerns triangle *TLR* more than it does triangle *ABC*, because the list mentions the points of triangle *TLR*, but it does not mention the points of triangle *ABC*.

Jumping Fleas

In fact, we can get the list that describes motion *p* without using triangle *ABC* at all. Triangle *ABC* served its purpose by supplying us with points each of which moved from an old position on triangle *TLR* to a new position on triangle *TLR*. To serve the same purpose, let us use an infinite set of imaginary fleas, replacing each point of triangle *ABC* by a flea. Let us imagine that at first each flea is standing on triangle *TLR*, with one and only one flea on each point of the triangle. Then suppose that each flea jumps to a new position on the triangle, choosing precisely that new position to which the motion *p* would have carried the point of triangle *ABC* that the flea has replaced. Each flea, by its jump, matches an old position with a new position on triangle *TLR*. The list of these matched positions would be the same as those produced by motion *p*. Thus we can use the jumping fleas to produce the same effect as that produced by rotating triangle *ABC* 120° clockwise.

There is an advantage to using the imaginary fleas instead of triangle *ABC*. When we move triangle *ABC*, all the points on the triangle move together in such a way that the distances that separate them do not change. Then, for example, two points that are a half inch apart in their old positions on triangle *TLR* would

also be a half inch apart in their new positions on triangle *TLR*. However, the jumping fleas need not move together in this way. If they do move together, they can produce the same effects as the motions of triangle *ABC*. But they can also produce other different effects by moving in such a way that the distances between the new positions of some fleas will be different from the distances between the old positions of the same fleas. So by using the imaginary jumping fleas we can produce more effects than we can by using triangle *ABC*.

However, we shall not allow the fleas to jump as they please. To be sure that we can reverse the jumps of the fleas we make these restrictions: First, no two fleas are allowed to jump to the same new position, and second, after the fleas jump, each point of triangle *TLR* has a flea on it. Then, if the fleas all reverse their jumps, the reverse jumps match each point on triangle *TLR* with one and only one point on it, just as the original jumps did.

The Effect without the Motion

We replaced triangle *ABC* by a set of jumping fleas, because the jumping fleas could produce the same effects as a moving triangle, and many more effects as well. But since we are interested only in the effects of the jumps, we don't even have to use the fleas. The effect produced by the jumping fleas is to match each point on triangle *TLR*, used as an old position, with another point on triangle *TLR*, used as a new position. We can make such a matching by simply preparing a list of ordered pairs of points of triangle *TLR*, in which the first member of each ordered pair serves as an old position, and the second member serves as the new position that is matched with it. We may draw an arrow pointing to the right from the first member to the new member in each ordered pair to help remind us that the old positions are written on the left and the new positions are written on the right. In this notation, $T \rightarrow R$ is another way of writing the ordered

pair (T, R). When we use the jumping fleas, each point of triangle TLR serves as the old position for one and only one flea. So we shall require that each point of triangle TLR be used as a first member in one and only one ordered pair. We had also made some restrictions on the way in which the fleas may jump. We make the same restrictions on the ordered pairs: No point of the triangle may appear as second member in two different ordered pairs. (This is like saying that no two fleas are allowed to jump to the same new position.) And every point appears as second member in at least one ordered pair. (This is like saying that after the fleas jump, each point of the triangle has a flea on it.) A set of ordered pairs of the points of triangle TLR that meets all these requirements is called a *transformation* of the points of the triangle. There are many different transformations of the points of triangle TLR. Six of them are the effects of the six motions of triangle ABC that we called symmetries. These six transformations preserve distances between points. There are other transformations of triangle TLR that do not preserve distances between points.

A Transformation of a Set

A transformation can be defined for any set at all: A transformation of a set is a set of ordered pairs of elements of the set, with these properties: 1) Every element of the set is used as first member in one and only one ordered pair. 2) Every element of the set is used as second member in one and only one ordered pair. The set of ordered pairs gives us the effect of a motion without using any motion. However, it is often helpful, when we think of a transformation of a set to imagine it as being produced by jumping fleas. We think of the first member of each ordered pair as the old position of a flea, and the second member of that ordered pair as the new position to which the flea has jumped.

Transformations of a Line

Transformations of sets of points play an important part in the study of geometry. The effects of the symmetries of plane figures are examples of such transformations. As additional examples we shall describe some transformations of the points on a straight line.

On page 37, where we described the system of integers, we attached each integer to a point on the number line. There are many points on the number line that do not have integers attached to them. However, if we use a larger number system called the system of *real numbers*, it is possible to attach a number to *every* point on the number line. The system of real numbers includes the number zero, the measuring numbers (which are all to the right of zero), the negatives of the measuring numbers (which are all to the left of zero), and some other numbers known as *irrational* numbers. It is convenient to use these numbers as the names of the points to which they are attached.

Let us imagine that there is a flea on each point of the number line. If each flea jumps *5* units to the right, the jumps define a transformation of the points of the line. Let us call this transformation *t*. The transformation *t* contains an ordered pair for each flea. The first member of the ordered pair is the flea's old position. The second member of the ordered pair is the flea's new position. The flea that jumped from the point *2* landed on the point *7*. So the ordered pair (*2*, *7*), which we may also write as $2 \rightarrow 7$, is a member of the transformation *t*. Similarly, the ordered pairs $3 \rightarrow 8$, $4 \rightarrow 9$, $\frac{1}{2} \rightarrow 5\frac{1}{2}$, and $0 \rightarrow 5$, are also members of *t*.



Some Jumps in the Transformation $t : x \longrightarrow x+5$

In general, if x is any real number, the ordered pair $x \to x + 5$ is a member of t.

Exercises:

1. In the transformation t described above, what is the ordered pair whose first member is -3?

2. Let s be a transformation of the points of the number line in which each flea jumps 2 units to the left. What is the ordered pair in s whose first member is 6? whose first member is 10? whose first member is x?

3. The reverse of transformation t is a transformation in which each flea jumps 5 units to the left. Let us call this transformation t^{-1}. What is the ordered pair in t^{-1} whose first member is 7? whose first member is 8? whose first member is 9? whose first member is $x + 5$?

4. Denote by ts the result of having each flea make two jumps one right after the other, first the jump required by transformation t, and then the jump required by transformation s. What single jump by each flea would have the same effect? What is the ordered pair in ts whose first member is 12? whose first member is x?

5. Let w be a transformation of the points of the number line in which each flea jumps to a new position that is on the same side of zero as his old position was, but is twice as far from zero as his old position was. What is the ordered pair in w whose first member is 3? whose first member is -2? whose first member is x?

Transformations of a Finite Set

A set for which a transformation is defined need not be a set of points. It may be any kind of set at all, and it may be either a

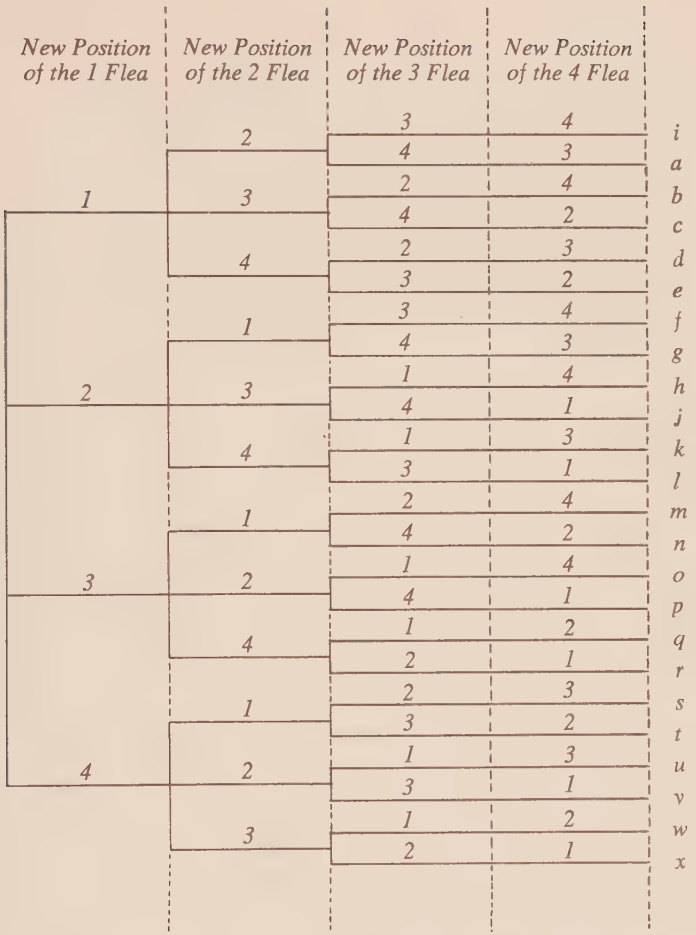
finite set or an infinite set. Transformations of finite sets play an important part in the study of algebra.

If a set is finite, it is easy to identify all the possible transformations of the set. As an example, let us identify all the possible transformations of a set that has exactly four elements, represented by the four numbers *1*, *2*, *3* and *4*. We shall find it helpful to use imaginary fleas again. Assume that there is a flea on each member of the set, and that each flea jumps from his old position to a new position in the set. Let us call the flea whose old position is *1* the *1* flea, call the flea whose old position is *2* the *2* flea, and so on. To find all the possible choices that the fleas might make for their jumps, we consider one flea at a time. As we discover these possible choices, we shall record them in a special kind of diagram known as a *tree diagram.*

To prepare for the tree diagram, take a large blank sheet of paper and draw a vertical line near the left-hand edge of the sheet. This line is the trunk of the tree. Later we shall put some branches on the trunk, then we shall put some smaller branches on each branch, then some still smaller branches on each small branch, and so on. Using dotted lines, divide the sheet into four vertical columns. Label the first one "new position of the *1* flea," label the second column "new position of the *2* flea," and so on.

Let us consider first the possible jumps of the *1* flea. He is free to jump to any one of the four elements, *1*, *2*, *3* or *4*. So any one of these four elements may be his new position. To show this fact, draw four horizontal lines in the first column, and write above them *1*, *2*, *3* and *4* respectively, as shown in the diagram on page 135. Leave lots of space above and below each of these lines, and attach them to the vertical line at the left. These four lines are the branches of the tree. At the right-hand end of each of these branches draw a short vertical line.

The short vertical lines are the places where we shall attach small branches to each large branch of the tree. These small branches will represent the possible jumps of the *2* flea. The first branch in the first column stands for the cases where the *1* flea chose *1* as his new position. Since two fleas may not have the same



new position, and the *1* position is already reserved for the *1* flea, the *2* flea has only three possible choices, *2*, *3* or *4* for his new position. To show these possible choices, draw three horizontal lines in the second column, attached as branches to the *1* branch that is in the first column. Label them *2*, *3* and *4* respectively.

The second branch in the first column stands for the cases where the *1* flea chose *2* as his new position. In these cases, since the *2* position is already reserved for the *1* flea, the *2* flea may

choose only *1*, *3* or *4* for his new position. To show these possible choices, attach three branches to the *2* branch that is in the first column, and label them *1*, *3* and *4* respectively. Similarly, attach three branches labeled *1*, *2* and *4* to the *3* branch that is in the first column, and attach three branches labeled *1*, *2* and *3* to the *4* branch that is in the first column.

Now draw a short vertical line at the right-hand end of each branch that is in the second column. We shall use these vertical lines to attach small branches to each branch that is in the second column. These small branches will represent the possible jumps of the *3* flea.

The first branch in the second column stands for the cases where the new position of the *1* flea is *1*, and the new position of the *2* flea is *2*. Since the *1* and *2* positions are already reserved for the *1* flea and the *2* flea, there are only two places left, namely *3* and *4*, that the *3* flea may choose for his new position. To show this fact, draw two horizontal lines in the third column, attached as branches to the first branch in the second column, and label them *3* and *4* respectively. Similarly, attach two branches to every other branch that is in the second column. In each case the two branches drawn in the third column represent the two places that are not already reserved for the *1* flea and the *2* flea.

The first branch in the third column stands for the case where the new position of the *1* flea is *1*, the new position of the *2* flea is *2*, and the new position of the *3* flea is *3*. Since the *1*, *2* and *3* positions are already reserved for the *1* flea, the *2* flea and the *3* flea, there is only one place, the *4* place, left open for the *4* flea. To show this fact, draw one horizontal line in the fourth column, attached to the first branch in the third column, and label it *4*. Similarly, attach one branch to every other branch that is in the third column. In each case the branch drawn in the fourth column represents the one place that is not already reserved for the *1* flea, the *2* flea, and the *3* flea.

We can now read from the tree diagram all possible combinations of new positions for the four fleas. To get a combination,

start with a branch in the first column, and then move to the right to a branch that is attached to it, then continue to the right to a branch that is attached to that one, and so on. Each combination ends with a branch in the fourth column, and for each branch in the fourth column, there is a combination that ends with it. So there are as many combinations of new positions as there are branches in the fourth column. Each combination of new positions of the four fleas defines a transformation of the set whose members are *1*, *2*, *3* and *4*. There are *24* such transformations.

Let us examine the combinations of new positions of the fleas one by one, taking them in the order in which they appear in the tree diagram, starting at the top. The first combination of new positions is shown by taking the first branch in each column. For this combination, the new position of the *1* flea is *1*, the new position of the *2* flea is *2*, the new position of the *3* flea is *3*, and the new position of the *4* flea is *4*. Let us denote by the letter *i* the transformation that is defined by this combination of new positions. The second combination of new positions is shown by taking the top branch in the first and second columns, and the second branch in the third and fourth columns. For this combination, the new position of the *1* flea is *1*, the new position of the *2* flea is *2*, the new position of the *3* flea is *4*, and the new position of the *4* flea is *3*. Let us denote by *a* the transformation that is defined by this combination of new positions. Let us call the other transformations, in the order in which they appear in the tree diagram, *b*, *c*, *d* and so on, respectively, except that we skip the letter *i* which has already been used to name the first transformation. The ordered pairs (or the flea jumps) that make up each of the twenty-four transformations are shown below.

i	*a*	*b*	*c*
$1 \to 1$	$1 \to 1$	$1 \to 1$	$1 \to 1$
$2 \to 2$	$2 \to 2$	$2 \to 3$	$2 \to 3$
$3 \to 3$	$3 \to 4$	$3 \to 2$	$3 \to 4$
$4 \to 4$	$4 \to 3$	$4 \to 4$	$4 \to 2$

d	e	f	g
$1 \to 1$	$1 \to 1$	$1 \to 2$	$1 \to 2$
$2 \to 4$	$2 \to 4$	$2 \to 1$	$2 \to 1$
$3 \to 2$	$3 \to 3$	$3 \to 3$	$3 \to 4$
$4 \to 3$	$4 \to 2$	$4 \to 4$	$4 \to 3$

h	j	k	l
$1 \to 2$	$1 \to 2$	$1 \to 2$	$1 \to 2$
$2 \to 3$	$2 \to 3$	$2 \to 4$	$2 \to 4$
$3 \to 1$	$3 \to 4$	$3 \to 1$	$3 \to 3$
$4 \to 4$	$4 \to 1$	$4 \to 3$	$4 \to 1$

m	n	o	p
$1 \to 3$	$1 \to 3$	$1 \to 3$	$1 \to 3$
$2 \to 1$	$2 \to 1$	$2 \to 2$	$2 \to 2$
$3 \to 2$	$3 \to 4$	$3 \to 1$	$3 \to 4$
$4 \to 4$	$4 \to 2$	$4 \to 4$	$4 \to 1$

q	r	s	t
$1 \to 3$	$1 \to 3$	$1 \to 4$	$1 \to 4$
$2 \to 4$	$2 \to 4$	$2 \to 1$	$2 \to 1$
$3 \to 1$	$3 \to 2$	$3 \to 2$	$3 \to 3$
$4 \to 2$	$4 \to 1$	$4 \to 3$	$4 \to 2$

u	v	w	x
$1 \to 4$	$1 \to 4$	$1 \to 4$	$1 \to 4$
$2 \to 2$	$2 \to 2$	$2 \to 3$	$2 \to 3$
$3 \to 1$	$3 \to 3$	$3 \to 1$	$3 \to 2$
$4 \to 3$	$4 \to 1$	$4 \to 2$	$4 \to 1$

If a set has only one element represented by the number *1*, only one transformation of this set is possible: $1 \to 1$. This transformation is usually called *i*.

Exercises:

6. Consider a set that has exactly two elements, represented by the numbers *1* and *2*. Use imaginary jumping fleas to define all possible transformations of this set. Construct a tree diagram to show all possible combinations of new positions of the fleas. Call the transformations *i* and *a* in the order in which they appear in the tree diagram. List the ordered pairs that make up each transformation.

7. Consider a set that has exactly three elements, represented by the numbers *1*, *2* and *3*. Use imaginary jumping fleas to define all possible transformations of this set. Construct a tree diagram to show all possible combinations of new positions of the fleas. Call the transformations *i*, *a*, *b*, *c*, *d* and *e* in the order in which they appear in the tree diagram. List the ordered pairs that make up each transformation.

8. Consider the transformations of the set of four elements *1*, *2*, *3* and *4*. Denote by *ek* the result of having each flea make two jumps one right after the other, first the jump required by transformation *e*, and then the jump required by transformation *k*. Show the effect of the two jumps as a set of ordered pairs. What transformation, requiring only a single jump for each flea, would have the same effect?

Counting the Transformations

The use of a tree diagram to show the transformations of a finite set makes it easy to count the transformations. The counting is based on this simple rule: If the branches on a tree have the same number of twigs on each branch, then the total number of twigs is equal to the number of branches times the number of twigs on each branch. We apply this rule to a tree diagram col-

umn by column, going from left to right, to calculate the number of branches in each column. The number of branches in the last column is equal to the number of transformations, as we saw in the paragraph before this one.

For example, consider the tree diagram for the transformations of the set whose members are *1*, *2*, *3* and *4*. The first column has *4* branches, one for each new position the *1* flea might choose. Each branch has *3* twigs on it, one for each new position the *2* flea might choose after one position has already been reserved for the *1* flea. So the total number of twigs is 4×3. This is the number of branches in the second column. Each of these branches has *2* twigs on it, one for each new position the *3* flea might choose after two positions have already been reserved for the *1* flea and the *2* flea. So the total number of twigs is $(4 \times 3) \times 2$, or $4 \times 3 \times 2$. This is the number of branches in the third column. Each of these branches has *1* twig on it, one for each new position the *4* flea might choose after three positions have already been reserved for the *1* flea, the *2* flea and the *3* flea. So the total number of twigs is $(4 \times 3 \times 2) \times 1$, or $4 \times 3 \times 2 \times 1$. This is the number of branches in the last column. So this is the number of possible transformations of a set of *4* things.

By a similar argument we see that the number of possible transformations of a set of *3* things is $3 \times 2 \times 1$, and the number of possible transformations of a set of *2* things is 2×1.

Exercise:

9. How many possible transformations are there of a set of *5* things? of a set of 6 things? of a set of *7* things?

Multiplying Transformations

Let us take any fixed set of things, and consider all the possible transformations of this set. We can introduce a multiplication

operation for these transformations in a very simple way: As in exercises 4 and 8, picture each transformation as the effect produced when a flea on each member of the set jumps to a new position. To multiply two transformations s and t, let each flea make two jumps one right after the other, first the jump required by transformation s, and then the jump required by transformation t. The product st is the transformation that has the same effect as these two jumps.

If the set is finite, it is easy to multiply two transformations of the set by using the list of ordered pairs in each transformation. For example, let us find the product qu of the transformations q and u of the set whose members are *1*, *2*, *3* and *4*. First write down the ordered pairs that make up transformation q. (See page 138.) These ordered pairs describe the first jump made by each flea as required by transformation q. Now, to show the second jump made by each flea, as required by transformation u, use each new position that resulted from the first jump as the old position from which the second jump is made. Write the new positions that result from the second jump in a third column on the right. In the second jump, the flea that starts from *1* jumps to *4*, the flea that starts from *2* jumps to *2*, the flea that starts from *3* jumps to *1*, and the flea that starts from *4* jumps to *3*. (See the ordered pairs in transformation u, listed on page 138.) Then the three columns that describe the two jumps look like this:

$$\begin{array}{ccc} q & & u \\ 1 & \rightarrow 3 & \rightarrow 1 \\ 2 & \rightarrow 4 & \rightarrow 3 \\ 3 & \rightarrow 1 & \rightarrow 4 \\ 4 & \rightarrow 2 & \rightarrow 2 \end{array}$$

The combined effect of the two jumps is to carry each flea from the position shown in the first column to the position shown in the last column. (See the top of page 142.)

qu

$1 \rightarrow 1$
$2 \rightarrow 3$
$3 \rightarrow 4$
$4 \rightarrow 2$

But we see on page 137 that the transformation that has the same effect is transformation c. So $qu = c$.

Exercises:

10. Use the three column arrangement to find the following products of transformations of the set whose members are *1*, *2*, *3* and *4*: *uq*, *ia*, *ib*, *aa*, *ab*.

11. Construct the multiplication table for the set of all transformations of the four-member set whose members are *1*, *2*, *3* and *4*.

12. In exercise 7 we found all the possible transformations of the three-member set whose members are *1*, *2* and *3*. Construct the multiplication table for the set of all these transformations.

13. In exercise 6 we found all the possible transformations of the two-member set whose members are *1* and *2*. Construct the multiplication table for the set of all these transformations.

Groups of Transformations

For any fixed set of things, the set of all possible transformations of that set is a group with respect to multiplication of transformations. To prove this fact, all that we have to do is show that it has all four properties of a group:

1. Closure. From our definition of multiplication, it is clear that the product of two transformations of a fixed set is also a transformation of that set. So the set of all transformations of the fixed set is closed with respect to multiplication.

II. Associative Property. Since multiplying transformations is like combining the motions of fleas, an argument like that given on page 42 shows that multiplication of transformations, like multiplication of motions, has the associative property.

III. Identity Element. Among the transformations there is one in which the new position of each flea is the same as its old position. Let us call this transformation i. When a flea makes the jump required by transformation i, it is as if he didn't jump at all. When this jump which is no jump is combined with any other jump, it is as if only the other jump had been made. So combining i with any transformation a of the fixed set produces the same effect as that transformation alone. That is, $ia = ai = a$.

IV. Inverse Element. Let a be any transformation of the fixed set. If we interchange the first and second members in each ordered pair in a, we get another transformation. Let us call it b. The flea jumps described by b are the reverse of the flea jumps described by a. Then the product ab is the transformation that results from having each flea jump as required by a, and then jump right back to his starting position again. The result is that his first position and his final position are the same. This means that $ab = i$. Similarly, $ba = i$, so b is the inverse of a.

For example, the transformation r of the set whose members are *1*, *2*, *3* and *4* consists of the following ordered pairs:

$$1 \rightarrow 3$$
$$2 \rightarrow 4$$
$$3 \rightarrow 2$$
$$4 \rightarrow 1$$

If we interchange the first and second member of each ordered pair, we get these ordered pairs:

$$3 \rightarrow 1$$
$$4 \rightarrow 2$$
$$2 \rightarrow 3$$
$$1 \rightarrow 4$$

These are the same ordered pairs that are in the transformation w:

$$1 \rightarrow 4$$
$$2 \rightarrow 3$$
$$3 \rightarrow 1$$
$$4 \rightarrow 2$$

So r and w are inverses of each other.

The group of all transformations of a fixed set is called the *symmetric* group of the set. If the fixed set is finite and has exactly n things in it, we call its symmetric group S_n. (Read this as "S sub-n".) Thus the group of all transformations of the set whose members are *1*, *2*, *3* and *4* is called S_4. The group of all transformations of the set whose members are *1*, *2* and *3* is called S_3. The group of all transformations of the set whose members are *1* and *2* is called S_2. The group of all transformations of the set whose only member is *1* is called S_1. The multiplication tables for S_1, S_2, S_3, and S_4 are shown below.

	i
i	i

Multiplication Table for the Symmetric Group S_1

	i	a
i	i	a
a	a	i

Multiplication Table for the Symmetric Group S_2

	i	a	b	c	d	e
i	i	a	b	c	d	e
a	a	i	c	b	e	d
b	b	d	i	e	a	c
c	c	e	a	d	i	b
d	d	b	e	i	c	a
e	e	c	d	a	b	i

Multiplication Table for the Symmetric Group S_3

	a	b	c	d	e	f	g	h	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
a	a	b	c	d	e	f	g	h	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
i	i	c	b	e	d	g	f	j	h	l	k	n	m	p	o	r	q	t	s	v	u	x	w
d	d	i	e	a	c	h	k	f	l	g	j	o	q	m	r	n	p	u	w	s	x	t	v
e	e	a	d	i	b	j	l	g	k	f	h	p	r	n	q	m	o	v	x	t	w	s	u
b	b	e	i	c	a	k	h	l	f	j	g	q	o	r	m	p	n	w	u	x	s	v	t
c	c	d	a	b	i	l	j	k	g	h	f	r	p	q	n	o	m	x	v	w	t	u	s
g	g	m	n	s	t	i	a	o	p	u	v	b	c	h	j	w	x	d	e	k	l	q	r
f	f	n	m	t	s	a	i	p	o	v	u	c	b	j	h	x	w	e	d	l	k	r	q
k	k	o	q	u	w	b	d	m	r	s	x	i	e	f	l	t	v	a	c	g	j	n	p
l	l	p	r	v	x	c	e	n	q	t	w	a	d	g	k	s	u	i	b	f	h	m	o
h	h	q	o	w	u	d	b	r	m	x	s	e	i	l	f	v	t	c	a	j	g	p	n
j	j	r	p	x	v	e	c	q	n	w	t	d	a	k	g	u	s	b	i	h	f	o	m
s	s	f	t	g	n	o	u	i	v	a	p	h	w	b	x	c	j	k	q	d	r	e	l
t	t	g	s	f	m	p	v	a	u	i	o	j	x	c	w	b	h	l	r	e	q	d	k
u	u	h	w	k	q	m	s	b	x	d	r	f	t	i	v	e	l	g	n	a	p	c	j
v	v	j	x	l	r	n	t	c	w	e	q	g	s	a	u	d	k	f	m	i	o	b	h
w	w	k	u	h	o	r	x	d	s	b	m	l	v	e	t	i	f	j	p	c	n	a	g
x	x	l	v	j	p	q	w	e	t	c	n	k	u	d	s	a	g	h	o	b	m	i	f
m	m	t	f	n	g	u	o	v	i	p	a	w	h	x	b	j	c	q	k	r	d	l	e
n	n	s	g	m	f	v	p	u	a	o	i	x	j	w	c	h	b	r	l	q	e	k	d
o	o	w	h	q	k	s	m	x	b	r	d	t	f	v	i	l	e	n	g	p	a	j	c
p	p	x	j	r	l	t	n	w	c	q	e	s	g	u	a	k	d	m	f	o	i	h	b
q	q	u	k	o	h	x	r	s	d	m	b	v	l	t	e	f	i	p	j	n	c	g	a
r	r	v	l	p	j	w	q	t	e	n	c	u	k	s	d	g	a	o	h	m	b	f	i

Multiplication Table for the Symmetric Group S_4

Exercises:

14. Which of the groups S_1, S_2, S_3 or S_4 is commutative?

15. The transformation q in S_4 consists of these ordered pairs:

$$1 \rightarrow 3$$
$$2 \rightarrow 4$$
$$3 \rightarrow 1$$
$$4 \rightarrow 2$$

a) What are the ordered pairs that make up the inverse of q?
b) What transformation is the inverse of q?

16. What are the members of the subgroup of S_4 that are generated by a? by d? by j? (See page 100.)

17. In S_4, the multiplication table shows that $k^2 = x$, $k^3 = n$, and $k^4 = i$. Using this information, and without using the multiplication table or the ordered pairs in the transformation n, find the values of n^2, n^3, and n^4.

11

Groups That Are Alike

Tables That Look Alike

In Chapter 5, which deals with the turns of a wheel, we let *i*, *a*, *b* and *c* stand for turns of *0°*, *90°*, *180°* and *270°* respectively. We found that the set whose members are *i*, *a*, *b* and *c* is a group with respect to the operation "multiplication" of turns. Let us call this group *B*. The multiplication table for the group *B* appears on page 60.

In Chapter 6, which deals with clock-face numbers, we let *0*, *1*, *2* and *3* stand for the members of the four-point clock-face number system. We found that the set whose members are these four points *0*, *1*, *2* and *3* is a group with respect to the operation "addition" of clock-face numbers. Let us call this group *C*. The addition table for the group *C* appears on page 239 in the answer to exercise 2.

The tables for the groups *B* and *C* are reproduced side by side below so that we may compare them with each other. A glance at the tables shows that they look alike: The arrangement of the

○	*i*	*a*	*b*	*c*
i	*i*	*a*	*b*	*c*
a	*a*	*b*	*c*	*i*
b	*b*	*c*	*i*	*a*
c	*c*	*i*	*a*	*b*

Table for the Group B

+	*0*	*1*	*2*	*3*
0	*0*	*1*	*2*	*3*
1	*1*	*2*	*3*	*0*
2	*2*	*3*	*0*	*1*
3	*3*	*0*	*1*	*2*

Table for the Group C

symbols *i*, *a*, *b* and *c* in the table for the group *B* is the same as the arrangement of the symbols *0*, *1*, *2* and *3* in the table for the group *C*. That is, wherever *i* appears in one table, *0* appears in the other table; wherever *a* appears in one table, *1* appears in the other table; wherever *b* appears in one table, *2* appears in the other table, and wherever *c* appears in one table, *3* appears in the other table. Because of this fact, each of these tables can easily be changed into the other one by some substitutions. If we start with the table for group *B*, replace the symbol $\circ$ by $+$, and replace every *i* by *0*, every *a* by *1*, every *b* by *2*, and every *c* by *3*, then it becomes the table for group *C*. Or if we start with the table for group *C*, replace the symbol $+$ by $\circ$, and replace every *0* by *i*, every *1* by *a*, every *2* by *b*, and every *3* by *c*, then it becomes the table for group *B*.

The substitutions that change the table for group *B* into the table for group *C* are listed below:

$$\circ \rightarrow +$$
$$i \rightarrow 0$$
$$a \rightarrow 1$$
$$b \rightarrow 2$$
$$c \rightarrow 3$$

In this list there are two columns. The left-hand column contains all the symbols that appear in the table for group *B*. The right-hand column contains all the symbols that appear in the table for group *C*. Each symbol on the left is matched with a symbol on the right and vice-versa. To change the table for group *B* into the table for group *C* we replace each symbol in the table for group *B* by the symbol that it is matched with in this list. Each arrow in the list points from a symbol that is replaced to the symbol that replaces it.

The substitutions that change the table for group *C* into the table for group *B* are the reverse of those listed above.

One Table in Two Languages

The way in which the table for group B can be changed into the table for group C and vice versa suggests that they are not really two different tables but only one table written in two different languages. We may think of the symbols $\circ$, i, a, b and c that appear in the table for group B as the words of a language. Let us call it the B language. The table for group B is a short way of writing in the B language sixteen statements of the form $x \circ y = z$, such as $i \circ i = i$, $i \circ a = a$, and so on. Similarly, we may think of the symbols $+$, *0*, *1*, *2* and *3* that appear in the table for group C as the words of a language. Let us call it the C language. The table for group C is a short way of writing in the C language sixteen statements of the form $x + y = z$, such as $0 + 0 = 0$, $0 + 1 = 1$, and so on. The list of matched symbols in the preceding paragraph is like a dictionary for translating from the B language into the C language. It matches each word in the B language with a word in the C language. When we replace each symbol in the table for group B by the symbol that it is matched with in the dictionary, we are translating the table from the B language to the C language. The fact that this substitution changes the table for group B into the table for group C shows that the table for group C says in the C language the same things that the table for group B says in the B language. So the two tables are really the same table written in two different languages.

If the tables for two groups are really the same table written in two different languages, we say that the two groups are *isomorphic*, or have the same structure, or that each group is isomorphic to the other. If two groups are isomorphic, the dictionary used to translate from the language of one group into the language of the other group is called an *isomorphism* from the first group to the second group. The reverse dictionary used to translate from the language of the second group into the language of the

first group, is an isomorphism from the second group to the first group. In each of these two dictionaries, the symbol for the operation in one group is matched with the symbol for the operation in the other group. The rest of the dictionary matches each member of each group with one and only one member of the other group. This is obviously possible only if the two groups have the same number of members.

If two groups have the same number of members, there are usually many different ways of matching each member of one group with one and only one member of the other group. Each of these matchings may be used as part of a dictionary for translating one of the group tables into another language. But the translated table may or may not turn out to be the same as the table for the other group. *The dictionary is an isomorphism from one group to the other if and only if the translated table is the same as the table for the other group.* For example, the dictionary on page 148 is an isomorphism from the group B to the group C. However, here is another dictionary that matches $\circ$ with $+$, and matches the members of B with the members of C:

$$\circ \rightarrow +$$
$$i \rightarrow 1$$
$$a \rightarrow 0$$
$$b \rightarrow 2$$
$$c \rightarrow 3$$

If we use this dictionary for translating the table for group B into another language, we get the following table:

+	1	0	2	3
1	1	0	2	3
0	0	2	3	1
2	2	3	1	0
3	3	1	0	2

To compare this table with the table for Group C, we first rearrange the columns and rows of the table so that the column labels and the row labels appear in the order *0*, *1*, *2*, *3*. Rearranging the rows and columns of the table will change the positions of the sixteen statements that are recorded in the table, but will not change the statements themselves. So the rearranged table may be thought of as being the same as the original table, since it contains the same sixteen statements.

To get the column labels in the order *0*, *1*, *2*, *3*, we interchange the *1* column and the *0* column. Then the table looks like this:

+	*0*	*1*	*2*	*3*
1	*0*	*1*	*2*	*3*
0	*2*	*0*	*3*	*1*
2	*3*	*2*	*1*	*0*
3	*1*	*3*	*0*	*2*

To get the row labels in the order *0*, *1*, *2*, *3*, we interchange the *1* row and the *0* row. Then the table looks like this:

+	*0*	*1*	*2*	*3*
0	*2*	*0*	*3*	*1*
1	*0*	*1*	*2*	*3*
2	*3*	*2*	*1*	*0*
3	*1*	*3*	*0*	*2*

Comparing this table with the table for group C, we see that they are not the same. This table and the table for group C give different sums. For example, this table says $0 + 0 = 2$. But the table for group C says $0 + 0 = 0$. Since this translation of the table for group B did not produce the table for group C, the dictionary we used for making the translation is not an isomorphism from B to C. Nevertheless, the group B and the group C are isomorphic, because although this particular dictionary is not an

isomorphism from B to C, there is another dictionary (the one on page 148) that is.

To show that two groups are isomorphic, all we have to do is produce *one* dictionary that is an isomorphism from one of the groups to the other, that is, that translates the table for one group into the table for the other group. To show that two groups are *not* isomorphic, we have to show that *no* such dictionary can be produced.

Example: As before, let us use the name C for the additive group of the four-point clock-face number system. Let us use the name D for the multiplicative group of the non-zero members of the five-point clock-face number system. (See page 71.) We shall show that the groups C and D are isomorphic.

The tables for the groups C and D are printed below.

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table for the Group C

$\times$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

Table for the Group D

Let us translate the table for group C into another language with the help of the following dictionary:

$$+ \rightarrow \times$$
$$0 \rightarrow 1$$
$$1 \rightarrow 2$$
$$2 \rightarrow 4$$
$$3 \rightarrow 3$$

To carry out the translation, replace the $+$ in the table by $\times$, replace each 0 by 1, each 1 by 2, each 2 by 4, and each 3 by 3. The translated table is shown at the top of page 153.

×	1	2	4	3
1	1	2	4	3
2	2	4	3	1
4	4	3	1	2
3	3	1	2	4

Translated table for the Group C

To be able to compare this table with the table for group *D*, we first rearrange its columns and rows so that the column labels and the row labels will have the order *1*, *2*, *3*, *4*. If we interchange the *3* column and the *4* column, the table looks like this:

×	1	2	3	4
1	1	2	3	4
2	2	4	1	3
4	4	3	2	1
3	3	1	4	2

Translated Table for Group C
with Columns Rearranged

If we then interchange the *3* row and the *4* row, the table looks like this:

×	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

Translated Table for Group C
with Columns and Rows Rearranged

But this table is exactly the same as the table for group *D*. Since we have produced a dictionary that translates the table for

group C into the table for group D, the groups C and D are isomorphic.

Example: Let S be the group of symmetries of a square. Its multiplication table is on page 89. Let T be the group of symmetries of an equilateral triangle. Its multiplication table is on page 81. The group S has eight members, and the group T has six members. Since they do not have equal numbers of members, it is impossible to produce a dictionary which matches each member of each group with one and only one member of the other group. Therefore the groups S and T are not isomorphic.

Exercises:

1. In Chapter 5, exercise 11, page 61, we constructed the table for a group whose members are a turn of $0°$ and a turn of $180°$. In Chapter 6, exercise 4, page 67, we constructed the table for the additive group of the two-point clock-face number system. Show that these two groups are isomorphic by constructing a dictionary that translates the table for one of these groups into the table for the other.

2. The multiplication table for the symmetric group S_2 is printed on page 144. Show that the group whose members are a turn of $0°$ and a turn of $180°$ is isomorphic to S_2.

3. In Chapter 5, exercise 12, page 61, we constructed the table for a group whose members are a turn of $0°$, a turn of $120°$, and a turn of $240°$. In Chapter 6, exercise 3, page 66, we constructed the table for the additive group of the three-point clock-face number system. Show that these two groups are isomorphic.

4. The multiplication table for T, the group of symmetries of an equilateral triangle, is printed on page 81. The multiplication table for the symmetric group S_3 is printed on page 144. Show that the groups T and S_3 are isomorphic.

5. In Chapter 7, exercise 35, page 92, we constructed the table for the group whose members are the commands in the game of "Simon Says." In Chapter 5, page 60, we constructed the table for the group whose members are turns of *0°*, *90°*, *180°* and *270°* respectively (the group we called *B* on page 147). Show that these two groups are isomorphic.

6. In Chapter 7, exercise 27, page 90, we constructed a four-member subgroup of the group of symmetries of a square. Show that this subgroup is isomorphic to the group of symmetries of a rectangle, whose multiplication table is given on page 86.

7. The multiplication tables for the groups S_3 and S_4 are on pages 144 and 145. Show that S_3 is isomorphic to a subgroup of S_4.

8. In Chapter 10, exercise 16, page 146, we found the members of the subgroup of S_4 generated by j. Use the table for S_4 to construct the multiplication table for this subgroup. Show that this subgroup is isomorphic to the additive group of the four-point clock-face number system.

9. Use the table for S_4 to construct the table for the subgroup of S_4 that is generated by m. Show that this subgroup is isomorphic to the additive group of the three-point clock-face number system.

Multiplication Facts

A statement that the product of two members of a group is equal to some third member of the group is a *multiplication statement*. Some multiplication statements are true and some are false. For example, in the group *B* whose table is printed on page 147, the statement $a \circ b = c$ is true, but the statement $a \circ b = b$ is false. Let us call a multiplication statement that is true for the members of a group a *multiplication fact* of the group. Then the statement $a \circ b = c$ is a multiplication fact of the group *B*, but the statement $a \circ b = b$ is not a multiplication fact of the group *B*.

The multiplication table for a group is a collection of all the multiplication facts of the group, recorded in a short and convenient way. Each entry in the table represents a multiplication fact. For example, in the table for group B, the c that is entered in the b row and the a column represents the multiplication fact $b \circ a = c$. If we keep in mind that a multiplication table is a collection of all the multiplication facts of the group, we can find a second simple and useful way of recognizing when a dictionary that matches the membership of two groups is an isomorphism from one group to the other.

In the preceding exercises we recognized an isomorphism by applying this test: A dictionary is an isomorphism from one group to another group if it has three properties: 1) It matches the operation in one group with the operation in the other group. 2) It matches each member of each group with one and only one member of the other group. 3) If the table of the first group is translated by means of the dictionary, it becomes the table of the second group.

Since the table of a group is a collection of all the multiplication facts of the group, property 3) can be restated in this form: 3′) Every multiplication fact of the first group, translated by means of the dictionary, becomes a multiplication fact of the second group. So we have a second way of recognizing an isomorphism between two groups: A dictionary is an isomorphism between two groups if it has properties 1), 2) and 3′).

Example: We know from Chapter 4 that the set of all integers is a group with respect to the operation $+$. We found in exercise 3, Chapter 4, page 52, that the set of all even integers is also a group with respect to the operation $+$. We shall show that the group of integers and the group of even integers are isomorphic.

Every even integer is two times some integer. For example, $4 = 2 \times 2$, $6 = 2 \times 3$, etc. Thus we may say that the set of all even integers is the set of all integers of the form $2n$, where n is an integer. To show that the group of integers and the group of

even integers are isomorphic, let us first set up a dictionary that matches their operations and that matches their memberships with each other. We use the following dictionary, in which each integer n is matched with the even integer $2n$:

$$
\begin{array}{c}
+ \rightarrow + \\
0 \rightarrow 2\,(0) \\
1 \rightarrow 2\,(1) \\
2 \rightarrow 2\,(2) \\
3 \rightarrow 2\,(3) \\
\cdot \qquad \cdot \qquad \cdot \\
-1 \rightarrow 2\,(-1) \\
-2 \rightarrow 2\,(-2) \\
\cdot \qquad \cdot \qquad \cdot
\end{array}
$$

This dictionary obviously has property 1): It matches the operation in one group with the operation in the other group. It also obviously has property 2): It matches each member of each group with one and only one member of the other group. Now to show that the dictionary is an isomorphism, we show that it also has property 3′): Every multiplication fact of the first group, translated by means of the dictionary, becomes a multiplication fact of the second group. (In this case, since the group operation is called $+$, the "multiplication" facts will be addition facts.)

Let a and b be any two integers, and let c be their sum. Then $a + b = c$ is an addition fact of the group of integers. The dictionary above matches each integer with two times itself. So it matches a with $2a$, b with $2b$, and c with $2c$. To translate the statement $a + b = c$ by means of the dictionary, we replace a by $2a$, b by $2b$, and c by $2c$. Then the translated statement is $2a + 2b = 2c$. Since $a + b = c$, we may replace c by $a + b$. Then the translated statement looks like this: $2a + 2b = 2(a + b)$. This statement is true because the system of integers has the following property: If a, b, and c are any three integers, then $c(a + b) = ca + cb$. (This property is usually described by saying that "multiplication is distributive with respect to addition.") Moreover, the

numbers $2a$, $2b$, and $2(a + b)$ are all even integers. So the statement $2a + 2b = 2(a + b)$ is an addition fact of the group of even integers. Thus every addition fact of the group of integers, translated by means of the dictionary above, becomes an addition fact of the group of even integers. Since the dictionary has properties 1), 2) and 3′), it is an isomorphism from the group of integers to the group of even integers, and the two groups are isomorphic.

Exercises:

10. Consider the set of all integers of the form $3n$, where n is an integer. Its members are $0, 3, 6, 9, \ldots$, and $-3, -6, -9, \ldots$. Show that this set is a group with respect to addition, and that this group is isomorphic to the group of all integers.

11. Consider the set of all integers of the form $5n$, where n is an integer. Its members are $0, 5, 10, 15, \ldots$, and $-5, -10, -15 \ldots$. Show that this set is a group with respect to addition, and that this group is isomorphic to the group of all integers.

12. Consider the set of all integers of the form kn, where k is a fixed integer, and n is any integer. Its members are $0, k, 2k, 3k, \ldots$, and $-k, -2k, -3k, \ldots$. Show that this set is a group with respect to addition, and that this group is isomorphic to the group of all integers.

Cyclic Groups

A cyclic group is one in which all the members are powers of a single member. (See page 101.) In any group, the subgroup that is generated by a single member is a cyclic group. For example, we found in Chapter 10, exercise 16, page 146, that in S_4, the element a generates the cyclic group with two members, a, and $a^2 = i$; the element d generates the cyclic group with three mem-

bers, d, $d^2 = c$, and $d^3 = i$; the element j generates the cyclic group with four members, j, $j^2 = q$, $j^3 = s$, and $j^4 = i$.

The multiplication table for the cyclic subgroup of S_4 generated by j looks like this:

o	i	j	q	s
i	i	j	q	s
j	j	q	s	i
q	q	s	i	j
s	s	i	j	q

Table for the Subgroup of S_4 Generated by j

However, since each member of the group is a power of j, it may be written as that power of j. Then the product of any two members of the group can be calculated with the help of the rules for using exponents (see page 99), and the fact that $j^4 = i =$ the identity element. For example, the product $j^3j^2 = j^5 = j^4j = ij = j$. When each member of the group except i is written as a power of j, the multiplication table for the group looks like this:

o	i	j	j^2	j^3
i	i	j	j^2	j^3
j	j	j^2	j^3	
j^2	j^2	j^3	i	j
j^3	j^3	i	j	j^2

Table for the Cyclic Group Generated by j

In S_4, the subgroup generated by the element r is also a cyclic group with four members. When each member of this group except i is written as a power of r, the multiplication table for the group takes this form:

o	i	r	r^2	r^3
i	i	r	r^2	r^3
r	r	r^2	r^3	i
r^2	r^2	r^3	i	r
r^3	r^3	i	r	r^2

Table for the Cyclic Group Generated by r

A glance at the tables for the cyclic group generated by r and the cyclic group generated by j shows that they look alike. In fact, we may change one of the tables into the other by translating it with the help of the following dictionary:

$$\circ \rightarrow \circ$$
$$i \rightarrow i$$
$$j \rightarrow r$$
$$j^2 \rightarrow r^2$$
$$j^3 \rightarrow r^3$$

This shows that the four-member cyclic group generated by j is isomorphic to the four-member cyclic group generated by r.

A similar argument shows that any two finite cyclic groups that have the same number of members are isomorphic. Then any cyclic group can serve as a model of the structure of any other cyclic group that has the same number of members. In order to have a set of models of all possible finite cyclic groups, we construct them as follows: For every positive integer n, let C_n be the set whose members are $a^0, a^1, a^2, a^3, \ldots, a^{n-1}$. Define multiplication in C_n by means of these two rules: 1) For any two nonnegative integers h and k, $a^h a^k = a^{h+k}$. 2) $a^n = a^0$. Then it is easy to see that C_n is a group with respect to this multiplication operation. Every cyclic group of order n, that is, that has exactly n members, is isomorphic to C_n.

Exercises:

13. Consider C_5, whose members are a^0, a^1, a^2, a^3, a^4. Use the two rules given above for multiplication in C_5 to show that a^0 is the identity element of C_5. In this group, what is the inverse of a^2?

14. Construct the multiplication table for the model cyclic group C_5.

15. The element d in S_4 generates a cyclic subgroup with three members. Construct a dictionary that is an isomorphism from this group to C_3.

16. Prove that the additive group of the five-point clock-face number system is isomorphic to C_5.

17. What generalization is suggested by exercise 16?

The Symmetric Group of a Finite Set

The symmetric group of a finite set is the group of all transformations of the set. (See page 144.) In Chapter 10, when we constructed the symmetric group of a set of four elements, we first numbered the elements *1*, *2*, *3* and *4*, and then represented each element by its number. Then each transformation of the original set was also a transformation of the set whose members are the numbers *1*, *2*, *3* and *4*. For this reason, the group of transformations of the original set is the same as the group of transformations of the set of numbers *1*, *2*, *3*, *4*. Since this is true for any set of four elements, we see that all sets of four elements have the same symmetric group S_4. Similarly, if n is any positive integer, all sets of n elements have the same symmetric group, S_n.

Models for All Finite Groups

The symmetric groups S_1, S_2, S_3, and so on, are especially important because they and their subgroups provide us with models of all possible finite groups. It can be shown, in fact, that if a finite group has exactly n numbers, then it is isomorphic to a subgroup of the symmetric group S_n. This fact is known as *Cayley's theorem*, after the mathematician who first proved it.

We have already seen an example of Cayley's theorem in exercise 8 on page 155. There we saw that the additive group of the

four-point clock-face number system, a group with four members in it, is isomorphic to a subgroup of S_4.

As another example of Cayley's theorem, we shall show that the three-member group described in exercise 12, Chapter 5, page 61, is isomorphic to a subgroup of S_3. The members of the group are i, p and q, and the multiplication table for the group is shown on page 238. The members of S_3 are the transformations called i, a, b, c, d and e, shown in the answer to exercise 7, Chapter 10, page 253. The multiplication table for S_3 is shown on page 144.

We shall match each member of the set whose members are i, p and q with a transformation of the set in the following way: Match i with the transformation that matches each member with i times that member. Match p with the transformation that matches each member with p times that member. Match q with the transformation that matches each member with q times that member.

Then i is matched with the following transformation:

$$\begin{cases} i \to ii = i \\ p \to ip = p \\ q \to iq = q, \end{cases} \quad \text{or} \quad \begin{cases} i \to i \\ p \to p \\ q \to q. \end{cases}$$

If the elements i, p and q are numbered *1*, *2*, and *3* respectively, this is the transformation

$$\begin{aligned} 1 &\to 1 \\ 2 &\to 2 \\ 3 &\to 3, \end{aligned}$$

which is the member of S_3 that is called i.

The element p is matched with the following transformation:

$$\begin{cases} i \to pi = p \\ p \to pp = q \\ q \to pq = i, \end{cases} \quad \text{or} \quad \begin{cases} i \to p \\ p \to q \\ q \to i. \end{cases}$$

Expressed in terms of the numbers attached to the elements, this is the transformation

$$\begin{aligned} 1 &\to 2 \\ 2 &\to 3 \\ 3 &\to 1, \end{aligned}$$

which is the member of S_3 that is called c.

The element q is matched with the following transformation:

$$\begin{cases} i \to qi = q \\ p \to qp = i \\ q \to qq = p, \end{cases} \qquad \text{or} \qquad \begin{cases} i \to q \\ p \to i \\ q \to p. \end{cases}$$

Expressed in terms of the numbers attached to the elements, this is the transformation

$$\begin{aligned} 1 &\to 3 \\ 2 &\to 1 \\ 3 &\to 2, \end{aligned}$$

which is the member of S_3 that is called d.

The set whose members are i, c and d is a subgroup of S_3 with the following multiplication table:

○	i	c	d
i	i	c	d
c	c	d	i
d	d	i	c

We have matched i, p and q with members of this subgroup as follows:

$$\begin{aligned} i &\to i \\ p &\to c \\ q &\to d \end{aligned}$$

If we also match the group operations, we have a dictionary for translating from the language ($\circ$, i, p, q) to the language ($\circ$, i, c, d). It is easy to see that this dictionary translates the multiplication table of the group whose members are i, p and q into the multiplication table of the group whose members are i, c and d. Thus these two groups are isomorphic.

A similar method can be used to show that any group with n members is isomorphic to a subgroup of S_n. To construct the isomorphism, match each member x of the n-member group with the transformation that matches each member of the group with x times that member.

Exercise:

18. The group of symmetries of a rectangle has four members. According to Cayley's theorem, it is isomorphic to a subgroup of S_4. Use the method described above to verify this fact.

Isomorphism of a Group to Itself

An isomorphism is a dictionary which matches the operation of one group with the operation of a second group, which matches each member of each group with one and only one member of the other group, and which translates the table of the first group into the table of the second group. In all the isomorphisms we have examined so far, we have been dealing with two *different* groups which have the same structure. It is also possible to have an isomorphism in which the second group is the *same group* as the first group. Such an isomorphism of a group to itself is called an *automorphism* of the group. An automorphism of a group is a dictionary that has these properties: 1) It matches the operation of the group with itself. 2) Whether read from left to right or from right to left, it matches each member of the group with one and

only one member of the same group. 3) If the table of the group is translated by means of the dictionary, the translated table turns out to be the same as the original table.

Example: Consider the group of turns of *0°*, *120°*, and *240°*, called *i*, *p* and *q* respectively. Let us examine the following dictionary to see if it is an automorphism of the group:

$$\circ \rightarrow \circ$$
$$i \rightarrow i$$
$$p \rightarrow q$$
$$q \rightarrow p$$

This dictionary is an automorphism of the group if it has properties 1), 2) and 3) listed above. It obviously has property 1), since it matches the operation $\circ$ with the operation $\circ$. It obviously has property 2), since each member of the group appears once and only once in the left-hand column and once and only once in the right-hand column of the dictionary. Now let us see if it also has property 3).

The multiplication table of the group is shown below:

$\circ$	i	p	q
i	i	p	q
p	p	q	i
q	q	i	p

Multiplication Table of the Group

Using the dictionary, we translate the table into another language by replacing $\circ$ by $\circ$, i by i, p by q, and q by p. The translated table looks like this:

$\circ$	i	q	p
i	i	q	p
q	q	p	i
p	p	i	q

Translated Table

To be able to compare this table with the original table, we have to rearrange its rows and columns to put the row labels and column labels in the order i, p, q. If we interchange the q column and the p column, the table looks like this:

∘	i	p	q
i	i	p	q
q	q	i	p
p	p	q	i

Translated Table with Columns Rearranged

If we then interchange the q row and the p row, the table looks like this:

∘	i	p	q
i	i	p	q
p	p	q	i
q	q	i	p

Translated Table with Columns and Rows Rearranged

This last table is exactly the same as the original table. So the dictionary does have property 3), and it is an automorphism of the group.

Example: Consider again the same group, whose members are i, p and q, and whose multiplication table is shown on page 165. Let us examine this dictionary to see if it is an automorphism:

$$\begin{aligned} \circ &\rightarrow \circ \\ i &\rightarrow p \\ p &\rightarrow q \\ q &\rightarrow i \end{aligned}$$

This table obviously has properties 1) and 2) of an automorphism. To see if it has property 3), we use the dictionary to translate the group table into another language. We replace $\circ$ by $\circ$, i by p, p by q, and q by i, wherever they occur in the table. The translated table is shown at the top of page 167.

o	p	q	i
p	p	q	i
q	q	i	p
i	i	p	q

Translated Table

If we rearrange the columns of this table, to put the column labels in the order i, p, q, then the table looks like this:

o	i	p	q
p	i	p	q
q	p	q	i
i	q	i	p

Translated Table with Columns Rearranged

If we then rearrange the rows of the table, to put the row labels in the order i, p, q, the table looks like this:

o	i	p	q
i	q	i	p
p	i	p	q
q	p	q	i

Translated Table with Columns and Rows Rearranged

This last table is not the same as the original table. So the dictionary does not have property 3), and it is not an automorphism of the group.

There is also a shorter way of recognizing when a dictionary with properties 1) and 2) is not an automorphism. A dictionary that is an automorphism of a group must translate every multiplication fact of the group into a multiplication fact of the group. So a dictionary is not an automorphism of a group if it translates at least one multiplication fact into a statement that is not a multiplication fact. In the example just considered, the statement $ip = p$ is a multiplication fact of the group. The dictionary used in the example replaces i by p, and replaces p by q, so it translates this statement into the statement $pq = q$, which is not a multi-

plication fact of the group. This shows immediately that the dictionary is not an automorphism of the group.

Exercises:

19. Consider the group B whose members i, a, b and c are turns of $0°$, $90°$, $180°$ and $270°$ respectively. The multiplication table for the group is shown on page 147. Consider this dictionary:

$$\circ \rightarrow \circ$$
$$i \rightarrow i$$
$$a \rightarrow c$$
$$b \rightarrow b$$
$$c \rightarrow a$$

a) Using this dictionary, translate the table of the group into a new language.
b) Rearrange the columns of the translated table to put the column labels in the order i, a, b, c.
c) Using the answer to b), rearrange the rows of the table to put the row labels in the order i, a, b, c.
d) Is this dictionary an automorphism of the group?

20. Start again with the multiplication table for group B, and consider this dictionary:

$$\circ \rightarrow \circ$$
$$i \rightarrow i$$
$$a \rightarrow b$$
$$b \rightarrow c$$
$$c \rightarrow a$$

Using this dictionary, answer questions a), b), c) and d) listed in exercise 19 above.

21. Consider the group of symmetries of a rectangle, whose multiplication table is given on page 86. Consider this dictionary:

$$\circ \rightarrow \circ$$
$$i \rightarrow i$$
$$a \rightarrow a$$
$$b \rightarrow c$$
$$c \rightarrow b$$

Using this group and this dictionary, answer questions a), b), c) and d) listed in exercise 19 above.

22. Start again with the multiplication table for the group of symmetries of a rectangle, and consider this dictionary:

$$\circ \rightarrow \circ$$
$$i \rightarrow i$$
$$a \rightarrow b$$
$$b \rightarrow a$$
$$c \rightarrow c$$

Using this group and this dictionary, answer questions a), b), c) and d) listed in exercise 19 above.

23. Start again with the multiplication table for the group of symmetries of a rectangle, and consider this dictionary:

$$\circ \rightarrow \circ$$
$$i \rightarrow i$$
$$a \rightarrow b$$
$$b \rightarrow c$$
$$c \rightarrow a$$

Using this group and this dictionary answer questions a), b), c) and d) listed in exercise 19 above.

An Automorphism Is a Transformation

The first line of a dictionary which is an automorphism of a group matches the operation in the group with itself. This means

that when the automorphism is used to translate the group table into another language, the group operation is replaced by itself. In other words, the group operation is not changed at all. For this reason it is not necessary to say anything about the operation in the dictionary at all. So it is customary to list in an automorphism of a group only the substitutions that it makes for the members of the group. This changes slightly the meaning of the word *automorphism* so that instead of having properties 1), 2) and 3) listed on page 164, it has only properties 2) and 3). From now on we shall use the word automorphism in this changed sense.

An automorphism of a group matches each member of the group, written in the left-hand column, with one and only one member of the group, written in the right-hand column. In Chapter 10 we called such a matching of the members of a set a *transformation* of the set. So every automorphism of a group is a transformation of the membership of the group. However, the example given on page 166 shows that not every transformation of the membership of a group is an automorphism of the group. This suggests the problem of finding out which transformations of the membership of a group are automorphisms of the group, and which ones are not. As a step toward solving this problem, we shall first observe some additional properties that every automorphism of a group has.

The Element Matched with *i*

In each of the automorphisms we examined in the preceding pages, the identity element *i* was matched with itself. This is not an accident. In fact, we can prove that in an automorphism of a group, the identity element *i must* be matched with itself. To prove this, consider any automorphism of a group, and let *e* stand for the element in the right-hand column that is matched with the *i* in the left-hand column. Let *x* be any element in the right-hand column. (It may be any element at all in the group, since every

element in the group appears in the right-hand column.) It is matched with some element y in the left-hand column. Then these two lines appear in the automorphism:

$$i \rightarrow e$$
$$y \rightarrow x$$

Since i is the identity element of the group, the statement $iy = y$ and the statement $yi = y$ are multiplication facts of the group. When these statements are translated with the help of the automorphism, each i is replaced by e, and each y is replaced by x. Then the translated statements are these: $ex = x$, and $xe = x$. Because an automorphism is an isomorphism, these translations of multiplication facts are also multiplication facts. That is, they are true statements. But if it is true that $ex = x$, and $xe = x$ for all members x of the group, then e must be the identity element of the group. That is, $e = i$. In other words, the element that is matched with i must be i itself.

The element i appears only once in the right-hand column of an automorphism of a group. Since it already appears once where it is matched with the i that is in the left-hand column, it cannot appear again, matched with some element in the left-hand column that is different from i. So for every member of the group that is not i, the member that is matched with it is also not i.

The Elements Matched with Powers of a

We shall prove next that if a is an element of a group, and an automorphism of the group matches a in the left-hand column with x in the right-hand column, then it matches a^2 with x^2, a^3 with x^3, a^4 with x^4, and so on. That is, if an automorphism of a group matches a with x, then it matches a^n with x^n for every positive integer n.

Suppose that $a^2 = b$, and that the automorphism matches b with y. Then the two lines that follow appear in the automorphism:

$$a \rightarrow x$$
$$b \rightarrow y$$

Since $a^2 = b$, the statement $aa = b$ is a multiplication fact of the group. To translate this statement with the help of the automorphism, we replace a by x and we replace b by y. Then the translated statement is $xx = y$, or $x^2 = y$. Because an automorphism is an isomorphism, this translation of the multiplication fact is also a multiplication fact. That is, it is a true statement. Since $a^2 = b$, and $x^2 = y$, then a^2 may be substituted for b, and x^2 may be substituted for y in the line of the automorphism which says $b \rightarrow y$. When these substitutions are made, the line becomes $a^2 \rightarrow x^2$. That is, the automorphism matches a^2 with x^2.

Suppose that $a^3 = c$, and that the automorphism matches c with z. Then the line $c \rightarrow z$ appears in the automorphism. We observe that $ab = aa^2 = a^3 = c$. So the statement $ab = c$ is a multiplication fact of the group. To translate this statement with the help of the automorphism, we replace a by x, b by y, and c by z. The translated statement is $xy = z$, and it is a multiplication fact of the group. Since $y = x^2$, we may replace y by x^2 in this multiplication fact. Then it becomes $xx^2 = z$ or $x^3 = z$. Since $a^3 = c$, and $x^3 = z$, then a^3 may be substituted for c, and x^3 may be substituted for z in the line of the automorphism which says $c \rightarrow z$. When these substitutions are made, the line becomes $a^3 \rightarrow x^3$. That is, the automorphism matches a^3 with x^3. The argument is easily extended to show that the automorphism matches a^4 with x^4, and so on.

The Order of Matched Elements

We saw on page 103 that every element a of a finite group generates a cyclic subgroup whose members are $a, a^2, a^3, \ldots, a^m$, where m is the least positive integer for which $a^m = i$, the identity element of the group. The number m is obviously the number of elements in the subgroup generated by a, or the order of the sub-

group generated by a. It is also called the *order of a*. Thus the order of a is *2*, if a generates a subgroup with *2* members; the order of a is *3* if a generates a subgroup with *3* members; and so on.

Consider any finite group, and let a be an element in it whose order is *4*. Suppose that an automorphism of the group matches a with b. We shall show that the order of b is also *4*. The cyclic group generated by a has the four members a, a^2, a^3, and $a^4 = i$. Since the automorphism matches a with b, it also matches a^2 with b^2, a^3 with b^3, and a^4 or i with b^4. Therefore $b^4 = i$, because, as we saw on page 170, every automorphism of a group matches i with i. Since a^4 is the lowest power of a that is equal to i, we know that a, a^2 and a^3 are not equal to i. So the elements they are matched with are not equal to i. That is, b, b^2 and b^3 are not equal to i. So b^4 is the lowest power of b that is equal to i. This means that the order of b is *4*.

A similar argument shows that in general, if an automorphism of a finite group matches a with b, and the order of a is m, then the order of b is m. Since the only element whose order is *1* is i, this rule, applied to the case where $m = 1$, becomes the special rule we have already observed that an automorphism of a group matches i with i.

Finding All the Automorphisms of a Group

We are now ready to identify easily all the automorphisms of a finite group. As an example, we shall find the automorphisms of the group we called B on page 147. The multiplication table for this group is reproduced below.

	i	a	b	c
i	i	a	b	c
a	a	b	c	i
b	b	c	i	a
c	c	i	a	b

Since this group has *4* members, the number of possible transformations of the set of its members is $4 \times 3 \times 2 \times 1 = 24$. Some of these transformations are automorphisms of the group, and some are not. However, we don't have to examine all *24* transformations to see which ones are automorphisms of the group. Because of the result proved in the last paragraph, we have to examine only those transformations which match each element with an element that has the same order.

We therefore begin by finding the order of each element in the group. The order of i is *1*. We see from the table that $a^2 = b$, $a^3 = aa^2 = ab = c$, and $a^4 = aa^3 = ac = i$. So the order of a is *4*. We see next that $b^2 = i$, so the order of b is *2*. Finally, we see that $c^2 = b$, $c^3 = cc^2 = cb = a$, and $c^4 = cc^3 = ca = i$. So the order of c is *4*.

Since i is the only element of order *1*, an automorphism of the group must match i with i. Since b is the only element of order *2*, an automorphism of the group must match b with b. Since a and c are the only elements of order *4*, an automorphism of the group either matches a with a and c with c, or it matches a with c and c with a. Thus there are only two transformations that could possibly be automorphisms of this group. They are shown below:

$$\begin{array}{ll} i \to i & i \to i \\ a \to a & a \to c \\ b \to b & b \to b \\ c \to c & c \to a \end{array}$$

The first transformation, which matches each element of the group with itself, is the identity transformation. It is obviously an automorphism of the group, because when we use it as a dictionary for translating the multiplication table of the group, the table is not changed at all, and so the translated table is the same as the original table. The second transformation is also an automorphism of the group, as we found in exercise 20. Thus this group has exactly two automorphisms.

Exercises:

24. Does every group have at least one automorphism?

25. Find all the automorphisms of the group of symmetries of a rectangle. (The multiplication table of the group is on page 86.)

26. Find all the automorphisms of the group whose members are turns of *0°*, *120°*, and *240°* respectively. (The multiplication table of the group is on page 165.)

27. Find all the automorphisms of the group of symmetries of an equilateral triangle. (The multiplication table of the group is on page 81.)

The Product of Two Automorphisms

In the next few paragraphs we shall use the capital letters P, Q and I to stand for transformations of a group G. Any two transformations P and Q of the group G can be multiplied, as in Chapter 10, to produce a product PQ. The product PQ is the single transformation that has the same effect as the transformation P followed by the transformation Q. We shall show that if P and Q are automorphisms of the group G, then PQ is also an automorphism of the group G.

Consider any multiplication fact $ab = c$ of the group G. Suppose that the transformation P matches a with r, b with s, and c with t, so that these three ordered pairs are in P:

$$a \rightarrow r$$
$$b \rightarrow s$$
$$c \rightarrow t$$

Suppose that the transformation Q matches r with x, s with y, and t with z, so that these three ordered pairs are in Q:

$$r \rightarrow x$$
$$s \rightarrow y$$
$$t \rightarrow z$$

Then the transformation PQ matches a with x, b with y, and c with z, so that these three ordered pairs are in PQ:

$$a \rightarrow x$$
$$b \rightarrow y$$
$$c \rightarrow z$$

If we use the transformation P as a dictionary to translate the statement $ab = c$ into another language, the translated statement is $rs = t$. Because the statement $ab = c$ is a multiplication fact of the group G, and the transformation P is an automorphism of the group G, the translated statement $rs = t$ is also a multiplication fact of the group G. If we use the transformation Q as a dictionary to translate the statement $rs = t$ into another language, the translated statement is $xy = z$. Because the statement $rs = t$ is a multiplication fact of the group G, and the transformation Q is an automorphism of the group G, the translated statement $xy = z$ is also a multiplication fact of the group G. If we use the transformation PQ as a dictionary to translate the multiplication fact $ab = c$ into another language, the translation is the statement $xy = z$, which, as we have just seen, is also a multiplication fact. Thus, the transformation PQ translates every multiplication fact of the group G into a multiplication fact of the group G. Therefore PQ is an automorphism of the group G.

The Identity Automorphism

Every group G has an identity transformation which matches each member of G with itself. Let us call this identity transformation I. The argument given on page 174 shows that the identity transformation I is an automorphism of the group G.

The Inverse of an Automorphism

For every transformation P of a group G, there is an inverse transformation P^{-1} with the property that $PP^{-1} = P^{-1}P = I$. P may be written down as a two-column dictionary in which each member of G appears once and only once in each column. To obtain P^{-1} from P, all we have to do is interchange the two columns. (See page 143.) We shall show that if P is an automorphism of G, then P^{-1} is also an automorphism of G.

Let us consider as an example the group of symmetries of a rectangle, and let P be the transformation shown below:

$$\begin{array}{c} P: \\ i \to i \\ a \to b \\ b \to c \\ c \to a \end{array}$$

Call the multiplication table of the group "table I." If we use P as a dictionary to translate the table into another language, a second table is produced. Let us call it "table II." If we arrange the rows and columns of table II to put the row labels and the column labels in the order i, a, b, c, a third table is produced. Let us call it "table III." The three tables are shown below:

	i	a	b	c
i	i	a	b	c
a	a	i	c	b
b	b	c	i	a
c	c	b	a	i

Table I

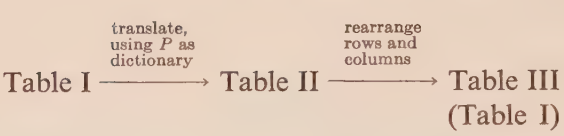
	i	b	c	a
i	i	b	c	a
b	b	i	a	c
c	c	a	i	b
a	a	c	b	i

Table II

	i	a	b	c
i	i	a	b	c
a	a	i	c	b
b	b	c	i	a
c	c	b	a	i

Table III

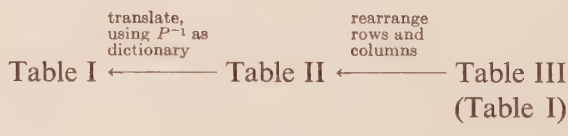
Since table III is the same as table I, the transformation P is an automorphism of the group. The way in which these three tables are related to each other is shown in the next diagram.



To get the transformation P^{-1}, we interchange the two columns in P. The result is shown below:

$$
\begin{array}{c}
P^{-1}: \\
i \rightarrow i \\
b \rightarrow a \\
c \rightarrow b \\
a \rightarrow c
\end{array}
$$

To find out if P^{-1} is an automorphism, we use it as a dictionary to translate the table of the group. Table III is the table of the group, so we may start with table III. However, before translating, let us first rearrange the rows and columns in the table to restore the order that they had in table II. This will produce table II. Since we have obtained table II by merely rearranging the rows and columns of the group table, it is another way of writing the group table. Now let us translate the group table by translating table II, using P^{-1} as a dictionary. The result will be table I, since P^{-1} merely reverses the translation which produced table II from table I. Since the translated table is the same as the original group table, the transformation P^{-1} is an automorphism of the group. The argument we have just given is summarized in the diagram below.



Exactly the same argument shows that the inverse of a given automorphism of any group is also an automorphism.

The Group of Automorphisms of a Group

With the help of the information uncovered in the last few paragraphs we can show that the set of all automorphisms of a group G is a group with respect to the operation multiplication of transformations:

I. Closure. The product of two automorphisms of a group is an automorphism of the group. So the set of all automorphisms of the group is closed with respect to multiplication.

II. Associative Property. We saw in Chapter 10 that multiplication of transformations of a set has the associative property.

III. Identity Element. The identity transformation of the membership of the group G is an automorphism of the group G. It is an identity element for the set of all automorphisms of G because every automorphism of G is a transformation of the membership of G.

IV. Inverse. The inverse of any automorphism of G is also an automorphism of G.

Thus the set of all automorphisms of a group G has all the properties of a group with respect to the operation multiplication of transformations.

If G is a finite group with n members, the group of all transformations of the membership of G is the symmetric group S_n. The group of all automorphisms of G is a subgroup of S_n.

In Chapter 7 we became acquainted with the symmetries of a plane figure. These are the transformations of the figure that leave unchanged its geometric structure consisting of distances between points. In the paragraphs above we have become acquainted with the automorphisms of a group. These are the transformations of

the group that leave unchanged its group structure consisting of multiplication facts. Thus the automorphisms of a group play the same role for a group that the symmetries of a plane figure play for that figure. The automorphisms of a group are the "symmetries" of the group, and the group of automorphisms of a group is the group of "symmetries" of the group.

Exercises:

28. Using the results of exercise 25 (see answer section, page 259), construct the multiplication table of the group of automorphisms of the group of symmetries of a rectangle.

29. Using the results of exercise 26 (see answer section, page 260), construct the multiplication table of the group of automorphisms of the group of turns whose members are turns of $0°$, $120°$ and $240°$ respectively.

30. Using the results of exercise 27 (see answer section, page 260), construct the multiplication table of the group of automorphisms of the group of symmetries of an equilateral triangle.

Abstract Groups

The groups that we have examined have had many different kinds of members. We have seen groups whose members are numbers, groups whose members are turns of a wheel, groups whose members are symmetries of a plane figure, groups whose members are transformations of a set, and groups whose members are automorphisms of a group. In each case we represented the members of the group by some symbols, and if the groups were finite, we arranged the symbols in a multiplication table. Once we had written out the multiplication table of a group, we could study the properties of the group by looking at its table without paying any attention to what the symbols in the table stand for.

Thus in the group of turns of *0°*, *90°*, *180°*, and *270°*, we represented these turns by the symbols *i*, *a*, *b* and *c* respectively. After we wrote out the multiplication table for the group, we could find out the order of each element of the group without keeping in mind that it was a turn of a certain number of degrees. All we had to do was look up certain products in the table. In this way, using the table alone, we could find that the order of *i* is *1*, the order of *a* is *4*, the order of *b* is *2*, and the order of *c* is *4*.

Thus it is possible to disregard the original meaning of the members of a group, and think of the group as merely a set of symbols that can be multiplied and whose products are listed in a multiplication table. The only requirement is that the table have the properties of a group table. (See Chapter 9.) A set of symbols related by such a table is called an *abstract group*. A set of specific things such as numbers, turns, or transformations, which has a multiplication operation defined in it, and which has the same multiplication table as the abstract group is called a concrete example of the abstract group. For example, the set of turns of *0°*, *90°*, *180°* and *270°* is a concrete example of the abstract group whose table is shown on page 60.

We saw on page 147 that the additive group of the four-point clock-face number system is isomorphic to the group of turns of *0°*, *90°*, *180°* and *270°*. This means that if we represented the clock-face numbers by the symbols *1*, *a*, *b* and *c* instead of the symbols *0*, *1*, *2* and *3*, and represented the addition operation by the symbol ∘ instead of the symbol +, the table for the additive group of the four-point clock-face number system would be the same as the table on page 60. Thus the additive group of the four-point clock-face number system and the group of turns of *0°*, *90°*, *180°* and *270°* are both concrete examples of the same abstract group. This shows that an abstract group can have many different concrete examples. For this reason it is worth while studying abstract groups. Then, each time we discover a property of an abstract group, we know that it is a property of all the concrete examples of that abstract group.

All the Group Structures of Order 1 to 4

To construct an abstract group with n members, all we have to do is take n different symbols and put them into a multiplication table that has n rows and n columns and has all the properties of a group table. If we do this in all possible ways, we can find out how many different group structures there are with order n. We shall do this now for the orders *1*, *2*, *3* and *4*.

Every group must have an identity element. Let us use the symbol i to stand for it. Then as we introduce more members in a group, we shall use the symbols a, b, c, etc.

If a group has only one member, this member is i, and it must have the following multiplication table:

	i
i	i

Table for Abstract Group of Order 1

If a group has only two members, i and a, its multiplication table has two rows and two columns, with column labels i and a and with row labels i and a, as shown below:

	i	a
i		
a		

We know from our study of the properties of a group table that the i row must be a copy of the column labels, and the i column must be a copy of the row labels. When we fill in the i row and the i column according to this rule, the table looks like this:

	i	a
i	i	a
a	a	

To complete the table, we have to put a symbol into the box in the lower right-hand corner. Because of the rule that each member of a group must appear once and only once in each row and in each column of the table, the symbol that must be in this box is *i*. Thus a group of order two, whose members are *i* and *a*, must have the following multiplication table:

	i	*a*
i	*i*	*a*
a	*a*	*i*

Table for Abstract Group of Order 2

Every group of order two must be isomorphic to this one.

If a group has only three members, *i*, *a* and *b*, its multiplication table has three rows and three columns, with column labels *i*, *a* and *b*, and with row labels *i*, *a* and *b*. The *i* row must be a copy of the column labels, and the *i* column must be a copy of the row labels, as shown below:

	i	*a*	*b*
i	*i*	*a*	*b*
a	*a*		
b	*b*		

Now consider the box in the *a* row and the *b* column. The symbol in this box cannot be *a*, because there already is an *a* in the same row. It cannot be *b* because there already is a *b* in the same column. Then it must be *i*. When we make this entry, the table looks like this:

	i	*a*	*b*
i	*i*	*a*	*b*
a	*a*		*i*
b	*b*		

The *a* row now has *a* and *i* in it. Then *b*, the only symbol not yet used in the *a* row, must be put into the empty box in that row. The *b* column now has *b* and *i* in it. Then *a*, the only symbol not yet used in the *b* column, must be put into the empty box in that column. When we make these entries, the table looks like this:

	i	*a*	*b*
i	*i*	*a*	*b*
a	*a*	*b*	*i*
b	*b*		*a*

Since the *b* row already contains *a* and *b*, we must put *i* into the empty box in that row. Then a group of order three, whose members are *i*, *a* and *b*, must have the following multiplication table:

	i	*a*	*b*
i	*i*	*a*	*b*
a	*a*	*b*	*i*
b	*b*	*i*	*a*

Table for Abstract Group of Order 3

Every group of order three must be isomorphic to this one.

If a group has only four members, *i*, *a*, *b* and *c*, its multiplication table has four rows and four columns, with column labels *i*, *a*, *b* and *c*, and with row labels *i*, *a*, *b* and *c*. The *i* row must be a copy of the column labels, and the *i* column must be a copy of the row labels, as shown below:

	i	*a*	*b*	*c*
i	*i*	*a*	*b*	*c*
a	*a*			
b	*b*			
c	*c*			

Now consider the box in the *a* row and the *b* column. It cannot contain *a*, since there already is an *a* in that row. It cannot con-

tain b, since there already is a b in that column. So the entry in the a row must be either i or c. These two possibilities are shown below:

	i	a	b	c
i	i	a	b	c
a	a		i	
b	b			
c	c			

Table I

	i	a	b	c
i	i	a	b	c
a	a		c	
b	b			
c	c			

Table II

To complete the a row in table I above, we must put in b and c. The c may not go into the c column, because there already is a c there. So it must go into the a column, and the b goes into the c column. Then table I looks like this:

	i	a	b	c
i	i	a	b	c
a	a	c	i	b
b	b			
c	c			

Table I

There is only one way to complete this table so that each element i, a, b and c appears once and only once in each row and in each column. The completed table is shown below:

	i	a	b	c
i	i	a	b	c
a	a	c	i	b
b	b	i	c	a
c	c	b	a	i

Table I Completed

To complete the *a* row in table II above, we must put in *b* and *i*. There are two ways of doing this, as shown below:

	i	*a*	*b*	*c*
i	*i*	*a*	*b*	*c*
a	*a*	*b*	*c*	*i*
b	*b*			
c	*c*			

Table IIA

	i	*a*	*b*	*c*
i	*i*	*a*	*b*	*c*
a	*a*	*i*	*c*	*b*
b	*b*			
c	*c*			

Table IIB

There is only one way to complete table IIA so that each element *i*, *a*, *b* and *c* appears once and only once in each row and in each column. The completed table looks like this:

	i	*a*	*b*	*c*
i	*i*	*a*	*b*	*c*
a	*a*	*b*	*c*	*i*
b	*b*	*c*	*i*	*a*
c	*c*	*i*	*a*	*b*

Table IIA Completed

There is only one way to complete the *a* column in table IIB, if we obey the rule that each element appears only once in a row or column:

	i	*a*	*b*	*c*
i	*i*	*a*	*b*	*c*
a	*a*	*i*	*c*	*b*
b	*b*	*c*		
c	*c*	*b*		

Table IIB

There are two ways of completing table IIB, so that each element appears once and only once in each row and in each column:

	i	a	b	c
i	i	a	b	c
a	a	i	c	b
b	b	c	a	i
c	c	b	i	a

Table IIB1

	i	a	b	c
i	i	a	b	c
a	a	i	c	b
b	b	c	i	a
c	c	b	a	i

Table IIB2

Thus a group of order four, whose members are i, a, b and c, must have one of the four multiplication tables I, IIA, IIB1, and IIB2. However, this need not mean that there are four different group structures of order 4. It is possible that some of these tables may represent the same structure. We shall look into this possibility now.

Let us translate table I into another language with the help of the following dictionary:

$$a \rightarrow a$$
$$b \rightarrow c$$
$$c \rightarrow b$$

That is, replace a by a, b by c, and c by b. Then the translated table looks like this:

	i	a	c	b
i	i	a	c	b
a	a	b	i	c
c	c	i	b	a
b	b	c	a	i

Table I Translated

If we rearrange the columns to give the column labels the order i, a, b, c, the table takes the form shown at the top of page 188.

	i	a	b	c
i	i	a	b	c
a	a	b	c	i
c	c	i	a	b
b	b	c	i	a

Table I, Translated, and with Columns Rearranged

If we then rearrange the rows to give the row labels the order i, a, b, c, the table looks like this:

	i	a	b	c
i	i	a	b	c
a	a	b	c	i
b	b	c	i	a
c	c	i	a	b

Table I, Translated, and with Rows and Columns Rearranged

But this is exactly what table IIA looks like. Thus table IIA represents the same structure as table I, except that it is written in a different language.

Similarly it can be shown that table IIB1 represents the same structure as table I.

We must check, too, to see whether table IIB2 also represents the same structure as table I. In table IIB2, $a^2 = i$, $b^2 = i$, and $c^2 = i$. That is, the elements a, b and c all have order *2*. However, in table I, the order of a is *4*. Since there is no element of order *4* in table IIB2, there is no isomorphism that can change table I into table IIB2. Therefore tables I and IIB2 represent different structures.

Thus a group of order four, whose members are i, a, b and c, must have the structure represented by table I, or the structure represented by table IIB2. In the exercises below, we shall see that these tables are actually group tables. Every group of order four is isomorphic either to the abstract group whose multiplication table is table I, or to the abstract group whose multiplication table is table IIB2.

Exercises:

31. In the cyclic group C_2, whose members are a^0 and a^1, let i stand for a^0. Then compare the multiplication table for C_2 with the table for the abstract group of order two. (See page 183.)

32. In the cyclic group C_3 whose members are a^0, a^1 and a^2, let i stand for a^0, and let b stand for the element a^2. Then compare the multiplication table for C_3 with the table for the abstract group of order three. (See page 184.)

33. Show that table IIB1 represents the same structure as table I.

34. The multiplication table for the symmetries of a rectangle appears on page 86. Compare this table with table IIB2 on page 187.

35. In the cyclic group C_4, whose members are a^0, a^1, a^2 and a^3, let i stand for a^0, let c stand for a^2, and let b stand for a^3. Then rearrange the rows and columns of the table for C_4 to give the row labels and the column labels the order i, a ,b, c. Compare the table with table I.

It is obvious that every group of order one is isomorphic to the cyclic group C_1.

Exercise 31 shows that every group of order two is isomorphic to the cyclic group C_2. Exercise 32 shows that every group of order three is isomorphic to the cyclic group C_3. Exercises 34 and 35 show that every group of order four is isomorphic to either the cyclic group C_4 or the group of symmetries of a rectangle.

Exercise:

36. Prove that every group of order *1*, *2*, *3* or *4* is commutative.

12

Groups Within a Group

Subgroups of a Group

If G is a group, any group whose members are all members of G is called a subgroup of G. In this chapter we shall learn how to identify and count all the subgroups of a finite group. We shall also learn some properties of the subgroups of a group.

Any *set* whose members are all members of a group G is called a *subset* of G. A subset of G is a subgroup of G only if it has the four properties of a group. For convenience in talking about the subsets of a group, we shall assign names to them. If a set is finite, we shall sometimes identify it by writing a list of its members inside a pair of braces. Thus, we may write $A = \{i, p, q\}$. To read this statement, say "A is the set whose members are i, p and q."

Example: If G is a group, then G itself is a subgroup of G, because all its members are in G. It is the largest subgroup that G may have.

Example: If G is a group, it contains an identity element i. The set whose only member is i is a subgroup of G. It is the smallest subgroup that G may have.

The subgroup of G that contains all the members of G and the subgroup of G that contains only the identity element i are called the *trivial* subgroups of G. Some groups have other subgroups besides the trivial subgroups. These other subgroups are called *non-trivial* subgroups. A non-trivial subgroup of G contains other elements besides i, but does not contain all the elements of G.

Example: Let T be the group of symmetries of an equilateral triangle. Then $T = \{i, p, q, r, s, t\}$, with the multiplication table shown on page 81. In Chapter 7, exercise 8, page 82, we saw that $\{i, p, q\}$ is a subgroup of T.

Example: Let S be the group of symmetries of a square. Then $S = \{i, p, q, r, s, t, u, v\}$, with the multiplication table shown on page 89. In Chapter 7, exercise 27, page 90, we saw that $\{i, q, s, t\}$ is a subgroup of S.

Example: If G is a finite group, and a is a member of G, and m is the least positive integer for which $a^m = i$, then $\{i, a, a^2, \ldots, a^{m-1}\}$ is the subgroup of G generated by a.

A Test for Subgroups of a Finite Group

Let H be a subset of a group G. H is a subgroup of G if it has the four group properties, *Closure*, *Associative Property*, *Identity Element*, and *Inverse*. However, to prove that H is a subgroup of G, it is not necessary to test it for all four properties. It is not necessary to test H for the associative property, because, since multiplication is associative in all of G, it is certainly associative in H, which is part of G. This leaves only three properties for which H has to be tested to prove that it is a subgroup of G.

If G is a finite group, we can reduce even further the number of properties for which H must be tested. It is sufficient then to test H only for the property of *Closure*. This is so, because, if H is closed with respect to multiplication, then if a is a member of H, a^2, a^3, a^4, and so on, are also members of H. Then H contains the subgroup generated by a. But the subgroup generated by a contains the identity element i, and also the inverse of a. Thus, i and the inverse of a are also in H. This means that if a subset of a finite group has the property of *Closure*, it automatically also has the properties *Identity Element* and *Inverse*. Therefore it is not

necessary to test the subset separately for these two properties, if it has already been tested for the property of *Closure*. Thus we have a shortcut for recognizing subgroups of a finite group: *A subset of a finite group is a subgroup of the group if it has the property of Closure.*

Finding All the Subgroups of a Finite Group

There is a simple procedure that you can follow to find all the subgroups of a finite group G. Begin with the smallest set that is a subgroup of G. This is the set $\{i\}$. Then, for any element a in G that is different from i, enlarge the set by putting a into it. Then the enlarged set is $\{i, a\}$. If this set is not closed with respect to multiplication, enlarge it further by putting in a^2. Then the enlarged set is $\{i, a, a^2\}$. If this set is not closed with respect to multiplication, enlarge it further by putting in a^3. Continue putting in more and more powers of a in this way, until the enlarged set is closed with respect to multiplication. Then it will be the subgroup generated by a. Repeat this procedure with each element of G that is different from i. Thus you obtain some subgroups that are larger than the trivial subgroup i.

Now follow a similar procedure for finding some still larger subgroups. Take any subgroup you already have, and enlarge its membership by putting in an element that is not already there. Form all possible products of members of the set. If any elements obtained as products are not already in the set, enlarge the set again by putting them in. Then again take all possible products of members of the set, and put into the set any products that are not already there. Repeat this over and over again, enlarging the set step by step until you get a set that already contains all possible products of its members. Then this set has the property of closure, and is a subgroup of G.

Continue in this way, enlarging each subgroup to form all possible larger subgroups. The process of enlarging a subgroup stops

whenever the enlarged subgroup is G, which is the largest possible subgroup of G.

Example: Let A be the group of turns that are multiples of $120°$. $A = \{i, p, q\}$, with the multiplication table shown on page 238. The smallest subgroup of A is $\{i\}$. Let us call it I. To enlarge I, we must put in either p or q. If we put in p, we must also put in $p^2 = q$. Then we get the subgroup $\{i, p, q\}$, which is A. If we put in q, we must also put in $q^2 = p$. Then the enlarged group again is $\{i, p, q\}$, which is A. Since A is the largest possible subgroup of A, there are no other subgroups of A. Thus we see that A has only two subgroups, I and A.

Example: Let S = the group of all symmetries of a square. $S = \{i, p, q, r, s, t, u, v\}$, with the multiplication table shown on page 89. Let $I = \{i\}$. It is the smallest subgroup of S. To enlarge I, we must put in another element and all the powers of that element until we reach the first power that is equal to i. If we put in p, we also put in $p^2 = q$ and $p^3 = pp^2 = pq = r$. We don't have to put in $p^4 = pp^3 = pr = i$, since i is already there. The enlarged set is $\{i, p, q, r\}$, which is closed with respect to multiplication, so it is a subgroup of S. Let us call this subgroup A. Start again with I and put in q. The enlarged set is $\{i, q\}$, which is closed with respect to multiplication, so it, too, is a subgroup of S. Let us call this subgroup B. Start again with I and put in r, $r^2 = q$, and $r^3 = p$. Then the enlarged set is the group A which we already have. There are four more subgroups that we can get by enlarging I. Let us assign names to them as follows: $C = \{i, s\}$ = the subgroup generated by s. $D = \{i, t\}$ = the subgroup generated by t. $E = \{i, u\}$ = the subgroup generated by u. $F = \{i, v\}$ = the subgroup generated by v.

Now we proceed to enlarge the subgroups A, B, C, D, E and F.

To enlarge $A = \{i, p, q, r\}$, we must first put in an element that is not already in it. We may put in either s or t or u or v. If we enlarge A by putting in s, we must also put in any new elements

we get by multiplying p, q, or r by s. Since $ps = u$, $qs = t$, and $rs = v$, we must put in u, t, and v. Then the enlarged set that is a subgroup is $\{i, p, q, r, s, t, u, v\}$, which is S. If we enlarge A by putting in t, we must also put in s, since $qt = s$. But if we put in s, then, as we have already seen, we must also put in u and v. So, once again, the enlarged set that is a subgroup is S. If we enlarge A by putting in u, we must also put in t, since $pu = t$. But, if we put in t, the set is not closed, as we have seen, until we also put in s and v. For a third time, the enlarged set that is a subgroup is S. If we enlarge A by putting in v, we must also put in s, since $pv = s$. But, if we put in s, the set is not closed, as we have seen, until we also put in t and u. Then, for a fourth time, the enlarged set that is a subgroup is S.

To enlarge $B = \{i, q\}$, we must first put in an element that is not already in it. We may put in either p or r or s or t or u or v. If we enlarge B by putting in p, we must also put in r, since $pq = r$. Then the enlarged set is $\{i, p, q, r\}$, which is the subgroup A that we already have. If we enlarge B by putting in r, we must also put in p, since $qr = p$. The enlarged set is again $\{i, p, q, r\}$, which is the subgroup A. If we enlarge B by putting in s, we must also put in t, since $qs = t$. The enlarged set $\{i, q, s, t\}$ is closed with respect to multiplication, so it is a subgroup of S. Let us call it G. If we enlarge B by putting in t we must also put in s, since $qt = s$. Thus we get again the subgroup G. If we enlarge B by putting in u, we must also put in v, since $qu = v$. The enlarged set $\{i, q, u, v\}$ is closed with respect to multiplication, so it is a subgroup of S. Let us call it H. If we enlarge B by putting in v, we must also put in u, since $qv = u$. Thus we get again the subgroup H.

To enlarge $C = \{i, s\}$, we begin by putting in either p or q or r or t or u or v. If we enlarge C by putting in p, then we must also put in q, since $p^2 = q$. Then the set contains all the members of A as well as s. As we saw above, we will also have to put in r, t, u and v in order to get the property of closure. But with all these elements in it, the enlarged set will be the subgroup S. If we enlarge C by putting in q, we must also put in t, since $qs = t$. This gives

us again the subgroup G. If we enlarge C by putting in r, then we must also put in q, since $r^2 = q$. Then we must also put in p, since $rq = p$. Then the set contains all the members of A as well as s, and it won't be closed until it is enlarged some more to become S. If we enlarge C by putting in t, we must also put in q, since $st = q$. This gives us again the subgroup G. If we enlarge C by putting in u, we must also put in r, since $su = r$. But, as we saw above, with r in it, the set will not be closed until it contains all the members of S. If we enlarge C by putting in v, we must also put in p, since $sv = p$. But, as we saw above, with p in it, the set will not be closed until it contains all the members of S. Thus, by enlarging C, we get the subgroups S or G.

Similarly, we can show that enlarging D produces either the subgroup S or the subgroup G; enlarging E produces either the subgroup S or the subgroup H; and enlarging F produces either the subgroup S or the subgroup H. (See exercise 1 below.) Thus, by enlarging the subgroups A, B, C, D, E and F, we get the subgroups G, H and S.

Next we have to enlarge the subgroups G and H. When we do so, we find that they are not closed until they contain all the members of S. (See exercise 1 below.) We cannot enlarge S, because S is the largest subgroup of S. Since we cannot produce any more subgroups by enlarging those we already have, then there aren't any other subgroups of S. The complete list of subgroups of S is shown below:

$$
\begin{aligned}
I &= \{i\} \\
A &= \{i, p, q, r\} \\
B &= \{i, q\} \\
C &= \{i, s\} \\
D &= \{i, t\} \\
E &= \{i, u\} \\
F &= \{i, v\} \\
G &= \{i, q, s, t\} \\
H &= \{i, q, u, v\} \\
S &= \{i, p, q, r, s, t, u, v\}
\end{aligned}
$$

Exercises:

1. In the example above, verify that a) enlargement of the subgroup D produces either the subgroup S or the subgroup G; b) enlargement of the subgroup E produces either the subgroup S or the subgroup H; c) enlargement of the subgroup F produces either the subgroup S or the subgroup H; d) the only subgroup of S that is larger than G is S; e) the only subgroup of S that is larger than H is S.

2. Let B = the group of all turns that are multiples of $90°$. $B = \{i, a, b, c\}$, with the multiplication table shown on page 147. Find all the subgroups of B.

3. Let R = the group of symmetries of a rectangle. $R = \{i, a, b, c\}$, with the multiplication table shown on page 86. Find all the subgroups of R.

4. Let T = the group of symmetries of an equilateral triangle. $T = \{i, p, q, r, s, t\}$, with the multiplication table shown on page 81. Find all the subgroups of T.

Conjugates

If you multiply a measuring number r by 3, and then multiply the product by $\frac{1}{3}$, which is the multiplicative inverse of 3, then the final product is the original number r: $\frac{1}{3} \times (3 \times r) = \left(\frac{1}{3} \times 3\right) \times r = 1 \times r = r$. In general, if you multiply a measuring number r by a measuring number t, and then multiply the product by t^{-1}, which is the multiplicative inverse of t, then the final product is the original number r: $t^{-1} \times (t \times r) = (t^{-1} \times t) \times r = 1 \times r = r$. This is true even if you multiply by t on the left, and multiply by t^{-1} on the right, so that the product is written at first as $t \times r \times t^{-1}$. Since

multiplication of measuring numbers is commutative, $r \times t^{-1} = t^{-1} \times r$. Then $t \times r \times t^{-1} = t \times t^{-1} \times r = (t \times t^{-1}) \times r = 1 \times r = r$. The situation is quite different in a group which is not commutative. There a product trt^{-1} need not be equal to r. For example, in the group of symmetries of an equilateral triangle, $t^{-1} = t$, so $trt^{-1} = trt = (tr)t = pt = s$. In this case, the product trt^{-1} is equal to s, not r. The element s obtained in this way from r by multiplying on the left by t and on the right by t^{-1} is called the *conjugate of r with respect to t.*

Exercises:

5. In the group of symmetries of an equilateral triangle, find the conjugate of s with respect to p.

6. In the group of symmetries of an equilateral triangle, find the conjugate of q with respect to p.

7. In the group of symmetries of a rectangle, find the conjugate of a with respect to b.

8. Prove that if a and b are elements of a commutative group, then $aba^{-1} = b$.

Matching Each Element with Its Conjugate

In the group of symmetries of an equilateral triangle, let us find the conjugate of each element with respect to p. We note first that since $pq = i$, then $p^{-1} = q$. Using the multiplication table on page 81, we find that

$$\begin{aligned} pip^{-1} &= (pi)q = pq = i; \\ ppp^{-1} &= p(pp^{-1}) = pi = p; \\ pqp^{-1} &= (pq)q = iq = q; \\ prp^{-1} &= (pr)q = tq = s; \\ psp^{-1} &= (ps)q = rq = t; \\ ptp^{-1} &= (pt)q = sq = r. \end{aligned}$$

If we match each element with its conjugate with respect to p, we get the following set of ordered pairs:

$$i \rightarrow i$$
$$p \rightarrow p$$
$$q \rightarrow q$$
$$r \rightarrow s$$
$$s \rightarrow t$$
$$t \rightarrow r$$

This set of ordered pairs is the automorphism D shown in the answer to exercise 27 in Chapter 11. This is an example of a general rule: If a is an element of a group G, the set of ordered pairs in which each element b of G is matched with its conjugate with respect to a is an automorphism of G. It is called the *inner automorphism* of G determined by a.

To prove this rule, we shall show that 1) every member of G appears once and only once in the left-hand column of such a set of ordered pairs; 2) every member of G appears once and only once in the right-hand column; 3) if the set of ordered pairs is used as a dictionary, it translates every multiplication fact of G into a multiplication fact.

1) The set of ordered pairs is constructed by putting each element b into the left-hand column and then matching it with its conjugate aba^{-1}. So, obviously, each element b of G appears in the left-hand column once and only once.

2) Let y be any element of G. Then $a^{-1}ya$ is also an element of G. The conjugate of $a^{-1}ya$ with respect to a is $a(a^{-1}ya)a^{-1} = (aa^{-1})y(aa^{-1}) = iyi = y$. Thus y appears in the right-hand column of the set of ordered pairs as the conjugate of $a^{-1}ya$. Moreover, $a^{-1}ya$ is the only member of G which has y as its conjugate with respect to a. In fact, if c is an element which has y as its conjugate with respect to a, then $aca^{-1} = y$. Multiplying both sides of this equation by a^{-1} on the left and by a on the right, we find that $a^{-1}(aca^{-1})a = a^{-1}ya$. But $a^{-1}(aca^{-1})a = (a^{-1}a)c(a^{-1}a) = ici = c$, so we have $c = a^{-1}ya$. Thus y appears in the right-hand column only once.

3) Suppose that $xy = z$ is a multiplication fact of the group G. If we use as a dictionary the set of ordered pairs in which each element is matched with its conjugate with respect to a, then we translate this fact by replacing x by axa^{-1}, y by aya^{-1}, and z by aza^{-1}. Then the translated statement is this one: $(axa^{-1})(aya^{-1}) = aza^{-1}$. To show that this statement is true, begin with the true statement $xy = z$, and multiply both sides of the equation by a on the left and by a^{-1} on the right. Then we get the true statement $axya^{-1} = aza^{-1}$. In this statement, replace y by iy. Then we get the true statement $axiya^{-1} = aza^{-1}$. Now replace i by $a^{-1}a$. Then we get the true statement $axa^{-1}aya^{-1} = aza^{-1}$. Then the statement $(axa^{-1})(aya^{-1}) = aza^{-1}$ is true, and is therefore a multiplication fact of G. Thus, the set of ordered pairs, used as a dictionary, translates every multiplication fact of G into a multiplication fact.

Because of 1) and 2), the set of ordered pairs is a transformation of G. Because of 3), it is an automorphism of G. This completes the proof of the rule.

The rule we have just proved gives us a quick way of identifying some of the automorphisms of a group, namely those that are inner automorphisms. Some groups also have automorphisms that are not inner automorphisms. An automorphism that is not an inner automorphism is called an *outer automorphism.*

Example: In any group G, the inner automorphism determined by i matches each element a with iai^{-1}. But $i^{-1} = i$, so $iai^{-1} = iai = a$. Thus the inner automorphism determined by i matches each element of G with itself. The automorphism of G which matches each element with itself is the identity element in the group of all automorphisms of G. (See page 176.) So it is called the *identity automorphism.*

Exercise:

9. Prove that if G is a commutative group and a is any element of G, then the inner automorphism determined by a is the identity automorphism.

Example: The multiplication table for the group of symmetries of a square is shown on page 89. We shall find all the automorphisms of this group, and identify those that are inner automorphisms.

The order of the group is 8, so the number of transformations of the group is $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40{,}320$. Our problem is to find out which of these *40,320* transformations are automorphisms of the group, and to distinguish the inner automorphisms from the outer automorphisms. However, to solve this problem, we do not have to list all these thousands of transformations and test them one at a time. Our knowledge about groups gives us some shortcuts that will take us very quickly to our goal.

We begin by identifying the order of each member of the group. The order of i is *1*. We see from the table that $p^2 = q$, $p^3 = pp^2 = pq = r$, and $p^4 = pp^3 = pr = i$. So the order of p is *4*. Similarly, since $r^2 = q$, $r^3 = rr^2 = rq = p$, and $r^4 = rr^3 = rp = i$, the order of r is *4*. The elements q, s, t, u and v are all of order *2*, since $q^2 = i$, $s^2 = i$, $t^2 = i$, $u^2 = i$, and $v^2 = i$.

We know that an automorphism of a group must match each element of the group with an element that has the same order. So the only transformations that might possibly be automorphisms of the group are those that match i with i, that match p and r with p and r in some order, and that match q, s, t, u and v with q, s, t, u and v in some order.

There is only one way to match i with i, using the ordered pair $i \rightarrow i$.

There are two ways of matching p and r with p and r. We may use the ordered pairs $p \rightarrow p$ and $r \rightarrow r$, or the ordered pairs $p \rightarrow r$ and $r \rightarrow p$. In each case we can find out what element q must be matched with by using the fact that an automorphism must translate every multiplication fact of the group into a multiplication fact. Let us suppose that an automorphism contains the ordered pairs $p \rightarrow p$, $r \rightarrow r$, and $q \rightarrow x$. In the multiplication table of the group we see that $pp = q$. The automorphism will translate this

multiplication fact into the statement $pp = x$. This will be a multiplication fact only if $x = q$.

Similarly, suppose that an automorphism contains the ordered pairs $p \rightarrow r$, $r \rightarrow p$, and $q \rightarrow x$. This automorphism will translate the multiplication fact $pp = q$ into the statement $rr = x$. We see from the table that this statement is a multiplication fact only if $x = q$. So any automorphism of the group must match q with itself.

Thus we see that a transformation of the group is an automorphism only if it matches the elements i, p, q and r in one of these two ways:

$i \rightarrow i$	or	$i \rightarrow i$
$p \rightarrow p$		$p \rightarrow r$
$q \rightarrow q$		$q \rightarrow q$
$r \rightarrow r$		$r \rightarrow p$

We examine next how s may be matched in these automorphisms. There are four ways in which s may be matched, using $s \rightarrow s$, or $s \rightarrow t$, or $s \rightarrow u$, or $s \rightarrow v$. In each case we shall find out what element t must be matched with. Suppose an automorphism contains the ordered pairs $q \rightarrow q$, $s \rightarrow s$, and $t \rightarrow x$. Then it will translate the multiplication fact $sq = t$ into the statement $sq = x$. This statement is a multiplication fact only if $x = t$. So if an automorphism contains the ordered pair $s \rightarrow s$, it also contains $t \rightarrow t$.

Suppose an automorphism contains the ordered pairs $q \rightarrow q$, $s \rightarrow t$, and $t \rightarrow x$. Then it will translate the multiplication fact $sq = t$ into the statement $tq = x$. This statement is a multiplication fact only if $x = s$, since $tq = s$. So if an automorphism contains the ordered pair $s \rightarrow t$, it also contains $t \rightarrow s$.

Suppose an automorphism contains the ordered pairs $q \rightarrow q$, $s \rightarrow u$, and $t \rightarrow x$. Then it will translate the multiplication fact $sq = t$ into the statement $uq = x$. This statement is a multiplication fact only if $x = v$, since $uq = v$. So, if an automorphism contains the ordered pair $s \rightarrow u$, it also contains $t \rightarrow v$.

Suppose an automorphism contains the ordered pairs $q \to q$, $s \to v$, and $t \to x$. Then it will translate the multiplication fact $sq = t$ into the statement $vq = x$. This statement is a multiplication fact only if $x = u$, since $vq = u$. So, if an automorphism contains the ordered pair $s \to v$, it also contains $t \to u$.

Thus there are only four ways in which an automorphism of the group may match s and t with elements of the group:

$s \to s$	or	$s \to t$	or	$s \to u$	or	$s \to v$
$t \to t$		$t \to s$		$t \to v$		$t \to u$

If we combine these with the two ways in which an automorphism of the group may match i, p, q and r with elements of the group, we get eight ways in which an automorphism of the group may match i, p, q, r, s and t with elements of the group:

1.	2.	3.	4.
$i \to i$	$i \to i$	$i \to i$	$i \to i$
$p \to p$	$p \to p$	$p \to p$	$p \to p$
$q \to q$	$q \to q$	$q \to q$	$q \to q$
$r \to r$	$r \to r$	$r \to r$	$r \to r$
$s \to s$	$s \to t$	$s \to u$	$s \to v$
$t \to t$	$t \to s$	$t \to v$	$t \to u$
5.	**6.**	**7.**	**8.**
$i \to i$	$i \to i$	$i \to i$	$i \to i$
$p \to r$	$p \to r$	$p \to r$	$p \to r$
$q \to q$	$q \to q$	$q \to q$	$q \to q$
$r \to p$	$r \to p$	$r \to p$	$r \to p$
$s \to s$	$s \to t$	$s \to u$	$s \to v$
$t \to t$	$t \to s$	$t \to v$	$t \to u$

We shall find that in each case there is only one way in which we may match u and v to complete the transformation of the group.

If an automorphism contains the ordered pairs in list 1 and also the ordered pair $u \to y$, then it translates the multiplication fact $ps = u$ into the statement $ps = y$. This statement is a multiplica-

tion fact only if $y = u$. Therefore an automorphism that begins with list 1 can be completed only by the ordered pairs $u \to u$ and $v \to v$.

If an automorphism contains the ordered pairs in list 2 and also the ordered pair $u \to y$, then it translates the multiplication fact $ps = u$ into the statement $pt = y$. This statement is a multiplication fact only if $y = v$, since $pt = v$. Therefore an automorphism that begins with list 2 can be completed only by the ordered pairs $u \to v$ and $v \to u$.

If an automorphism contains the ordered pairs in list 3 and also the ordered pair $u \to y$, then it translates the multiplication fact $ps = u$ into the statement $pu = y$. This statement is a multiplication fact only if $y = t$, since $pu = t$. Therefore an automorphism that begins with list 3 can be completed only by the ordered pairs $u \to t$ and $v \to s$.

In the same way we can show (see exercise 10) that an automorphism that begins with list 4 can be completed only by the ordered pairs $u \to s$ and $v \to t$; an automorphism that begins with list 5 can be completed only by the ordered pairs $u \to v$ and $v \to u$; an automorphism that begins with list 6 can be completed only by the ordered pairs $u \to u$ and $v \to v$; an automorphism that begins with list 7 can be completed only by the ordered pairs $u \to s$ and $v \to t$; an automorphism that begins with list 8 can be completed only by the ordered pairs $u \to t$ and $v \to s$.

Exercise:

10. Verify that there is only one way of completing each of the lists 4, 5, 6, 7 and 8 to form a transformation that might be an automorphism.

Thus we see that there are only eight transformations of the group that may be automorphisms of the group. We assign names to these transformations as shown below. We shall show that each of these transformations actually is an automorphism of the group.

I:	K:	L:	M:
$i \to i$	$i \to i$	$i \to i$	$i \to i$
$p \to p$	$p \to p$	$p \to p$	$p \to p$
$q \to q$	$q \to q$	$q \to q$	$q \to q$
$r \to r$	$r \to r$	$r \to r$	$r \to r$
$s \to s$	$s \to t$	$s \to u$	$s \to v$
$t \to t$	$t \to s$	$t \to v$	$t \to u$
$u \to u$	$u \to v$	$u \to t$	$u \to s$
$v \to v$	$v \to u$	$v \to s$	$v \to t$
N:	P:	Q:	R:
$i \to i$	$i \to i$	$i \to i$	$i \to i$
$p \to r$	$p \to r$	$p \to r$	$p \to r$
$q \to q$	$q \to q$	$q \to q$	$q \to q$
$r \to p$	$r \to p$	$r \to p$	$r \to p$
$s \to s$	$s \to t$	$s \to u$	$s \to v$
$t \to t$	$t \to s$	$t \to v$	$t \to u$
$u \to v$	$u \to u$	$u \to s$	$u \to t$
$v \to u$	$v \to v$	$v \to t$	$v \to s$

To show that these transformations are automorphisms of the group, we shall first identify which of them are inner automorphisms of the group. We already know that the inner automorphism determined by i is the transformation I. The inner automorphism determined by p is shown below, where we take into account the fact that $p^{-1} = r$.

$$
\begin{aligned}
i &\to pir = pr = i \\
p &\to ppr = qr = p \\
q &\to pqr = rr = q \\
r &\to prr = ir = r \\
s &\to psr = ur = t \\
t &\to ptr = vr = s \\
u &\to pur = tr = v \\
v &\to pvr = sr = u
\end{aligned}
$$

This automorphism is the transformation K shown on page 204. By writing out in the same way the inner automorphisms determined by q, r, s, t, u and v, we find that they are the transformations I, K, N, N, P and P respectively. Thus the transformations I, K, N and P are the only inner automorphisms of the group.

Exercise:

11. Verify that the inner automorphisms determined by q, r, s, t, u and v in the group of symmetries of a square are the transformations I, K, N, N, P and P respectively.

It remains for us to show that the transformations L, M, Q and R are also automorphisms. We can show that L is an automorphism of the group by employing the method described on page 165. (See exercise 12 below.)

Exercise:

12. Use the transformation L as a dictionary for translating into another language the multiplication table for the group of symmetries of a square. Rearrange the columns and rows of the translated table to give the column labels and the row labels the order i, p, q, r, s, t, u, v. Verify that the translated table, rearranged in this way, is the same as the original table.

We have proved in exercise 12 that the transformation L is an automorphism of the group. We also know that K, N and P are automorphisms, since they are inner automorphisms. We can identify some more automorphisms by using the rule that the product of two automorphisms is an automorphism. For example, in the left-hand table below, columns 1 and 2 show the ordered pairs of transformation L, and columns 2 and 3 show the ordered pairs of transformation K. Then columns 1 and 3 show the ordered pairs of transformation LK.

L:	K:	LK:
$i \to i$	$\to i$	$i \to i$
$p \to p$	$\to p$	$p \to p$
$q \to q$	$\to q$	$q \to q$
$r \to r$	$\to r$	$r \to r$
$s \to u$	$\to v$	$s \to v$
$t \to v$	$\to u$	$t \to u$
$u \to t$	$\to s$	$u \to s$
$v \to s$	$\to t$	$v \to t$

Comparing the right-hand table with the tables on page 204, we see that $LK = M$. We can show in the same way that $LN = R$, and $LP = Q$. (See exercise 13 below.) Thus we see that M, R and Q are automorphisms of the group. The group of symmetries of a square therefore has exactly eight automorphisms as follows:

inner automorphisms: I, K, N and P
outer automorphisms: L, M, R and Q

Exercises:

13. If L, N, P, Q and R are the transformations shown on page 204, verify that $LN = R$, and $LP = Q$.

14. The automorphisms of the group of symmetries of a rectangle are given in the answer to exercise 25, Chapter 11. Which of them are inner automorphisms?

15. The automorphisms of the group of turns that are multiples of $120°$ are given in the answer to exercise 26, Chapter 11. Which of them are inner automorphisms?

16. The automorphisms of the group of symmetries of an equilateral triangle are given in the answer to exercise 27, Chapter 11. Which of them are inner automorphisms?

The Center of a Group

If a and b are members of a group, we say that *a commutes with b* if $ab = ba$. For example, every element a of a group commutes with its inverse a^{-1}, since $aa^{-1} = a^{-1}a$. There are some elements of a group that commute with every member of the group. For example, the identity element i commutes with every member of the group, since $ia = ai$ for every member a of the group. In the group of symmetries of a square, the element q also commutes with every member of the group. There is a simple way of recognizing in the multiplication table of a group an element that commutes with every member of the group: *An element a commutes with every member of the group if the a row, read from left to right, is the same as the a column, read from top to bottom.* The set of all those members of a group that commute with every member of the group is called the *center* of the group.

It is easy to see that an element belongs to the center of a group if and only if the inner automorphism determined by that element is the identity automorphism. (See exercises 17 and 18 below.)

Exercises:

17. Prove that if an element a belongs to the center of a group G, then the inner automorphism of G determined by a is the identity automorphism.

18. Prove that if the inner automorphism of a group G determined by an element a is the identity automorphism, then a belongs to the center of G.

19. What elements belong to the center of the group of symmetries of a square?

20. What elements belong to the center of the group of symmetries of an equilateral triangle?

21. What elements belong to the center of the group of symmetries of a rectangle?

22. What elements belong to the center of any commutative group?

Conjugates of a Subgroup

If R is a subset of a group G, and a is a member of G, the inner automorphism of G determined by a matches every member x of R with its conjugate axa^{-1}. The set of all these conjugates of the members of R is called the conjugate of R with respect to a, and is represented by the symbol aRa^{-1}. For example, consider the group S of all symmetries of a square, let $R = \{p, q\}$, and consider the inner automorphism determined by s. This inner automorphism matches p with sps^{-1}, which is equal to r, and it matches q with sqs^{-1}, which is equal to q. So the conjugate of R with respect to s is the set $\{r, q\}$, and we write $sRs^{-1} = \{r, q\}$.

As another example, let us find a conjugate of a subset that is a *subgroup* of a group. Consider the group T of all symmetries of an equilateral triangle, and the subgroup $B = \{i, r\}$. Let us find the conjugate pBp^{-1} of B: $pBp^{-1} = \{pip^{-1}, prp^{-1}\} = \{i, s\}$. Notice that this set is also a subgroup of T. This is an example of a general rule: A conjugate of a subgroup of a group G is also a subgroup of G. This rule is proved by the results of exercises 23, 24 and 25.

Exercises:

23. If S is a subgroup of a group G, and a is a member of G, prove that aSa^{-1} contains the identity element i of G.

24. If S is a subgroup of G, and a is a member of G, prove that aSa^{-1} contains the inverse of every one of its members.

25. If S is a subgroup of G, and a is a member of G, prove that aSa^{-1} contains the product of any two of its members.

26. Using the result of exercise 25, prove that the subgroup S is isomorphic to the subgroup aSa^{-1}.

27. In the group of symmetries of an equilateral triangle, the subgroup $C = \{i, s\}$. Find all the conjugates of C. That is, find iCi^{-1}, pCp^{-1}, qCq^{-1}, rCr^{-1}, sCs^{-1}, and tCt^{-1}.

28. In the group of symmetries of an equilateral triangle, the subgroup $A = \{i, p, q\}$. Find all the conjugates of A. That is, find iAi^{-1}, pAp^{-1}, qAq^{-1}, rAr^{-1}, sAs^{-1}, and tAt^{-1}.

29. In the group of symmetries of a square, the subgroup $A = \{i, p, q, r\}$. Find all the conjugates of A.

30. In the group of symmetries of a square, the subgroup $G = \{i, q, s, t\}$. Find all the conjugates of G.

31. In the group of symmetries of a square, the subgroup $B = \{i, q\}$. Find all the conjugates of B.

32. In the group of symmetries of a square, the subgroup $F = \{i, v\}$. Find all the conjugates of F.

33. In the group of symmetries of a square, the subgroup $H = \{i, q, u, v\}$. Find all the conjugates of H.

34. In the group of symmetries of a square, the subgroup $C = \{i, s\}$. Find all the conjugates of C.

35. In the group of symmetries of a square, the subgroup $D = \{i, t\}$. Find all the conjugates of D.

36. In the group of symmetries of a square, the subgroup $E = \{i, u\}$. Find all the conjugates of E.

Self-Conjugate Subgroups

In the group of symmetries of an equilateral triangle, the subgroup C which is equal to $\{i, s\}$ has three different conjugates, as we found in exercise 27:

$$iCi^{-1} = sCs^{-1} = \{i, s\} = C;$$
$$pCp^{-1} = rCr^{-1} = \{i, t\} = D;$$
$$qCq^{-1} = tCt^{-1} = \{i, r\} = B.$$

However, the subgroup A which is equal to $\{i, p, q\}$ has only one conjugate, because every conjugate of A is A itself. (See the answer to exercise 28.) If every conjugate of a subgroup is the subgroup itself, the subgroup is called a *normal subgroup.* Thus the subgroup A is a normal subgroup of the group of symmetries of an equilateral triangle, but the subgroup C is not a normal subgroup.

If G is a group, and $I = \{i\}$, I, the smallest subgroup of G, is obviously a normal subgroup, because for every element a of G, $aia^{-1} = i$. G, the largest subgroup of G is also a normal subgroup of G.

Exercises:

37. The group T of symmetries of an equilateral triangle has six subgroups, I, A, B, C, D and T, listed in the answer to exercise 4, page 261. Which of these subgroups are normal subgroups?

38. The group S of symmetries of a square has ten subgroups, I, A, B, C, D, E, F, G, H and S, listed on page 195. Which of these subgroups are normal subgroups?

39. Prove that in a commutative group every subgroup is a normal subgroup.

The Order of a Subgroup

The order of the group S of all symmetries of a square is 8. The subgroups of S, as we can see from the list on page 195, have orders 1, 2, 4 and 8. Notice that the order of each subgroup of S is a divisor of the order of S. This is an example of a general rule

known as Lagrange's theorem: *The order of each subgroup of a finite group is a divisor of the order of the group.* To prove this theorem we first introduce the idea of a coset of a subgroup.

Left Cosets of a Subgroup

Let S be a finite group, let G be a subgroup of S, and let a be any member of S. We shall use the symbol aG to stand for the set of elements we get by multiplying each member of G on the left by a. Obviously, one of these elements is a itself, since the identity element i is in G, and $ai = a$. The set aG is called a *left coset* of G. To get acquainted with the properties of the left cosets of a subgroup, we shall first examine a familiar example.

	i	p	q	r	s	t	u	v
i	i	p	q	r	s	t	u	v
p	p	q	r	i	u	v	t	s
q	q	r	i	p	t	s	v	u
r	r	i	p	q	v	u	s	t
s	s	v	t	u	i	q	r	p
t	t	u	s	v	q	i	p	r
u	u	s	v	t	p	r	i	q
v	v	t	u	s	r	p	q	i

The Subgroup $G = \{i, q, s, t\}$

The Left Coset $pG = \{p, r, u, v\}$

The Subgroup G and its Left Cosets

Consider the group S of symmetries of a square, and the subgroup $G = \{i, q, s, t\}$. The multiplication table for S is shown above. We have shaded those columns in the table whose column labels are members of the subgroup G. The left coset $pG = \{pi, pq, ps, pt\}$. Each of these products that is a member of pG is in the p row and in a column whose column label is a member of G. But these are the columns we have shaded. *So the*

left coset pG is the set of all those entries in the p row that are in shaded boxes. Similarly, the set of entries in the shaded boxes in any row is the left coset obtained by multiplying the members of G on the left by the label of that row. From this fact we see immediately that the number of members there are in a left coset of G is equal to the order of G.

We observe next that each member of the group S is in some left coset of G. In fact, the complete membership of S appears in the i column.

Now let us examine the cosets more closely. Let us look first at the cosets iG, qG, sG, and tG, obtained by multiplying the members of G on the left by an element that is a member of G. We see from the table that $iG = \{i, q, s, t\}$, $qG = \{q, i, t, s\}$, $sG = \{s, t, i, q\}$, and $tG = \{t, s, q, i\}$. Each of these sets has the same members as G, so it is equal to G. This is not surprising, of course, since we multiplied the members of G by a member of G, and since G is a subgroup of S, it has the property of closure. Thus we see that G itself is a left coset of G.

Now let us pick any member of S, and examine all the left cosets of G that contain that member. For example, let us look at all the left cosets of G in which p is a member. We see from the table that p occurs in each of the left cosets pG, rG, uG and vG, and only in these. Notice that pG, rG, uG and vG all have the same members, p, r, u and v. Thus all the left cosets that have the member p in common have all their members in common, so they are the same sets. We would have arrived at a similar result had we chosen first any other member of S instead of p: Any left cosets of G that have a single member in common have all their members in common, and therefore are the same sets. So if two left cosets of G are different, they have no members in common at all.

In this case the 8 members of S are divided into just two different left cosets of G. They are $\{i, q, s, t\}$ and $\{p, r, u, v\}$. The number of members in each left coset of G is 4, the order of the subgroup G.

So the number of members in both together is 2×4. Thus, since we have $8 = 2 \times 4$, the order of G, which is 4, is a divisor of the order of S, which is 8.

We can now show that if S is any finite group, and G is any subgroup of S, the left cosets of G have properties like those we have seen above. Take the multiplication table for S and shade those columns whose labels are members of G. Then, if a is any member of S, the left coset aG is the set of all those entries in the a row that are in shaded boxes. Those left cosets that are in rows that are labeled by members of G are all equal to G. That is, G itself is a left coset of G. The number of members in each left coset of G is equal to the number of shaded columns in the table, and this is equal to the number of members of G. Thus the order of the subgroup G is equal to the number of members in each of its left cosets.

Every member of S is in some left coset of G, since the i column contains all the members of S.

Now pick any member a of S, and any two left cosets bG and cG in which a is a member. We shall show that every member of bG is a member of cG, and vice versa. Since a is a member of bG, a is the result of multiplying some member g of G on the left by b. That is, $a = bg$. Since a is a member of cG, a is the result of multiplying some member h of G on the left by c. That is, $a = ch$. Thus we have $bg = ch$. Multiplying both sides of this equation on the right by g^{-1}, we get $b = chg^{-1}$. Now consider any member bk of bG, where k is some member of G. Since $b = chg^{-1}$, $bk = chg^{-1}k = c(hg^{-1}k)$. The elements h, g^{-1} and k are all elements of G, so their product is also a member of G, since G is a subgroup of S. Thus we have shown that bk equals c times a member of G. But every product we get by multiplying a member of G on the left by c is a member of the left coset cG. So bk is a member of cG. Thus every member of bG is a member of cG. We can prove in the same way that every member of cG is a member of bG. Then bG and cG have the same members. Thus any left cosets of G

that have a single member in common have all their members in common, and therefore are the same sets. So left cosets of G that are different have no common members at all.

Suppose that the order of the group S is n, the order of the subgroup G is x, and the number of different left cosets of G is y. Then the n members of S are divided among y different cosets each of which contains x members. That is, $n = xy$, and so x is a divisor of n. This completes the proof of Lagrange's theorem.

Exercises:

40. If the order of a group is 15, what are the possible orders of its subgroups?

41. If the order of a group is 5, what are the possible orders of its subgroups?

42. Prove that if the order of a group G is a prime number p, then the only subgroups it has are the trivial subgroups G and $I = \{i\}$.

43. Prove that if the order of a group G is a prime number p, then the group is a cyclic group.

44. Prove that every group whose order is a prime number is commutative.

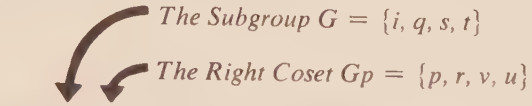
45. Prove that every group whose order is less than 6 is commutative.

Right Cosets of a Subgroup

Let G be a subgroup of a group S, and let a be a member of S. We use the symbol Ga to stand for the set of elements we get by multiplying each member of G on the right by a. The set Ga is called a *right coset* of G. Right cosets have properties very much like those of left cosets. In the multiplication table for S, if we shade each row whose label is a member of G, then the right

coset Ga is the set of all those entries in the a column that are in shaded boxes. The number of members in each right coset of G is equal to the order of G. Each member of S is in some right coset of G. Any right cosets of G that have a single member in common have all their members in common.

We show below the multiplication table for the group S of symmetries of a square, with shading in the rows whose labels are members of the subgroup $G = \{i, q, s, t\}$. The right coset Gi is in the shaded boxes in the i column, the right coset Gp is in the shaded boxes in the p column, and so on. The subgroup G has two different right cosets, $\{i, q, s, t\}$ and $\{p, r, v, u\}$.



	i	p	q	r	s	t	u	v
i	i	p	q	r	s	t	u	v
p	p	q	r	i	u	v	t	s
q	q	r	i	p	t	s	v	u
r	r	i	p	q	v	u	s	t
s	s	v	t	u	i	q	r	p
t	t	u	s	v	q	i	p	r
u	u	s	v	t	p	r	i	q
v	v	t	u	s	r	p	q	i

The Subgroup G and its Right Cosets

The set C, which is equal to $\{i, s\}$, is also a subgroup of S. The first table on page 216 shows the left cosets of C. The second table shows the right cosets of C.

The left cosets of C are $iC = sC = \{i, s\}$; $pC = uC = \{p, u\}$; $qC = tC = \{q, t\}$; and $rC = vC = \{r, v\}$. The right cosets of C are $Ci = Cs = \{i, s\}$; $Cp = Cv = \{p, v\}$; $Cq = Ct = \{q, t\}$; and $Cr = Cu = \{r, u\}$. Notice that the set $\{p, u\}$, which is a left coset

	i	p	q	r	s	t	u	v
i	i	p	q	r	s	t	u	v
p	p	q	r	i	u	v	t	s
q	q	r	i	p	t	s	v	u
r	r	i	p	q	v	u	s	t
s	s	v	t	u	i	q	r	p
t	t	u	s	v	q	i	p	r
u	u	s	v	t	p	r	i	q
v	v	t	u	s	r	p	q	i

The Subgroup $C = \{i, s\}$

The Left Coset $pC = \{p, u\}$

The Subgroup C and its Left Cosets

The Subgroup $C = \{i, s\}$

The Right Coset $Cp = \{p, v\}$

	i	p	q	r	s	t	u	v
i	i	p	q	r	s	t	u	v
p	p	q	r	i	u	v	t	s
q	q	r	i	p	t	s	v	u
r	r	i	p	q	v	u	s	t
s	s	v	t	u	i	q	r	p
t	t	u	s	v	q	i	p	r
u	u	s	v	t	p	r	i	q
v	v	t	u	s	r	p	q	i

The Subgroup C and its Right Cosets

of C, is not a right coset of C, and the set $\{p, v\}$, which is a right coset of C, is not a left coset of C. This shows that a left coset of a subgroup need not be a right coset of that subgroup, and vice versa. There are some special subgroups, however, for which every left coset is a right coset, and every right coset is a left coset. For

example, in the group of symmetries of a square, the left cosets of the subgroup G are:

$$iG = qG = sG = tG = \{i, q, s, t\}; \quad \text{and}$$
$$pG = rG = uG = vG = \{p, r, u, v\}.$$

The right cosets of the subgroup G are:

$$Gi = Gq = Gs = Gt = \{i, q, s, t\}; \quad \text{and}$$
$$Gp = Gr = Gu = Gv = \{p, r, u, v\}.$$

So every left coset of G is a right coset of G, and every right coset of G is a left coset of G. We shall explore the meaning of this special case in the next chapter.

13

Collapsing a Group

The Cosets of a Normal Subgroup

At the end of Chapter 12 we saw that there are some subgroups of a group that have the special property that every left coset of the subgroup is also a right coset of the subgroup, and vice versa. To explore the meaning of this special case, we shall examine an example.

Let S be the group of symmetries of a square, and consider the subgroup $G = \{i, q, s, t\}$. We have already seen that every left coset of G is a right coset of G, and vice versa. Let us choose any member of S, for example, p, and examine the left coset pG and the right coset Gp. As we have seen, they have the same members, p, r, u and v. Take any member s of G, and multiply it on the left by p. The product ps is a member of pG. In fact, from the multiplication table for S, we see that $ps = u$. Because the left coset pG is equal to the right coset Gp, this member u can also be obtained by multiplying a member of G by p on the right. In fact, $u = tp$. Therefore $ps = tp$. Multiplying both sides of this equation on the right by p^{-1}, we find that $psp^{-1} = t$. Repeating this procedure with every member of G, we find these conjugates of the members of G: $pip^{-1} = i$, $pqp^{-1} = q$, $psp^{-1} = t$, and $ptp^{-1} = s$. The set of all these conjugates of members of G is G itself. That is, the conjugate pGp^{-1} of G is G. By a similar argument, in which we begin by choosing any other element of S instead of p, we can show that *every* conjugate of G is G. Therefore G is a normal subgroup of S.

Exactly the same kind of argument can be developed for any subgroup that has the special property that every one of its left

cosets is a right coset and vice versa. So we have the following important conclusion: If every left coset of a subgroup is a right coset and vice versa, then the subgroup is a normal subgroup.

This statement can also be reversed: If a subgroup is normal, then every left coset of the subgroup is a right coset, and vice versa. (See exercise 1 below.)

In exercise 37 of Chapter 12 we found that the subgroup $B = \{i, q\}$ in the group of symmetries of a square is a normal subgroup. Then every left coset of B must be a right coset and vice versa.

On page 216 we saw that the set $\{p, u\}$ is a left coset of $C = \{i, s\}$, but it is not a right coset of C. Then we can conclude immediately that C is not a normal subgroup of the group of symmetries of a square.

Exercises:

1. Let S be a group, and let G be a normal subgroup of S. If a is any member of S, then $aGa^{-1} = G$. Prove that $aG = Ga$. (Hint: Show that every member of aG is a member of Ga, and vice versa.)

2. In the group S of symmetries of a square, the subgroup $A = \{i, p, q, r\}$ is normal. (See exercise 38 in Chapter 12.) Find all the left cosets of A and all the right cosets of A. Verify that every left coset of A is a right coset of A, and vice versa.

3. In the group S of symmetries of a square, the subgroup $B = \{i, q\}$ is normal. Find all the left cosets of B and all the right cosets of B. Verify that every left coset of B is a right coset of B, and vice versa.

4. In the group S of symmetries of a square, the subgroup $H = \{i, q, u, v\}$ is normal. Find all the left cosets of H and all the right cosets of H. Verify that every left coset of H is a right coset of H, and vice versa.

5. In the group T of symmetries of an equilateral triangle, the subgroup $A = \{i, p, q\}$ is normal. Find all the left cosets of A and all the right cosets of A. Verify that every left coset of A is a right coset of A, and vice versa.

Multiplying a Member of One Coset by a Member of Another

In each of the examples below we shall take two left cosets of a given subgroup. We shall pick any member of the first coset and any member of the second coset and then multiply them in that order. Then we shall find the left coset to which the product belongs. We shall do this for every product that can be obtained in this way. Then we shall compare the results we get in the two examples.

Example 1: Consider the subgroup $C = \{i, s\}$ in the group of symmetries of a square. Take the left cosets pC and rC. $pC = \{p, u\}$, and $rC = \{r, v\}$. If we multiply a member of pC times a member of rC, there are four different products that we may get: pr, pv, ur and uv. In the multiplication table for S we see that $pr = i$, $pv = s$, $ur = t$, and $uv = q$. So the possible products are i, s, t and q. The products i and s belong to the left coset $\{i, s\}$. The products q and t belong to the left coset $\{q, t\}$. *In this case, the products do not all belong to the same left coset of the subgroup.*

Example 2: Consider the subgroup $B = \{i, q\}$ in the group of symmetries of a square. Take the left cosets pB and sB. $pB = \{p, r\}$, and $sB = \{s, t\}$. If we multiply a member of pB times a member of sB, there are four different products that we may get: ps, pt, rs and rt. In the multiplication table for S we see that $ps = u$, $pt = v$, $rs = v$, and $rt = u$. So the possible products are u and v. The products u and v belong to the left coset $\{u, v\}$.

In this case, the products all belong to the same left coset of the subgroup.

The difference in the results obtained in these two examples is related to the fact that C is not a normal subgroup of S, while B is a normal subgroup of S. In fact we shall prove the following result of which example 2 is an illustration: Let S be any group and let G be a normal subgroup of S. Take any two left cosets aG and bG. *Then all products obtained by multiplying a member of aG times a member of bG belong to the same left coset of G.* To prove this statement, let us recall first that there is only one left coset of G that contains any particular element of S. So there is only one left coset of G that contains ab. Every member of aG is a product ag, where g is a member of G. Similarly, every member of bG is a product bh, where h is a member of G. The product of ag times bh is $agbh$. The product gb is a member of the right coset Gb. But, since G is a normal subgroup of S, the right coset Gb has the same members as the left coset bG. Thus gb is a member of bG. So there is a member k of G for which the product bk is equal to gb. Then $agbh = a(gb)h = a(bk)h = (ab)(kh)$. Since k and h are both members of G, their product kh is also a member of G. Then $(ab)(kh)$ is a member of the left coset abG that contains ab. Thus every product of a member of aG times a member of bG is a member of abG.

Exercises:

6. The subgroup $B = \{i, q\}$ is a normal subgroup of the group S of symmetries of a square. In the group S, $ur = t$. Verify that the product of any member of uB times any member of rB is a member of tB.

7. The subgroup $H = \{i, q, u, v\}$ is a normal subgroup of the group S of symmetries of a square. In the group S, $us = p$. Verify that the product of any member of uH times any member of sH is a member of pH.

8. The subgroup $A = \{i, p, q\}$ is a normal subgroup of the group T of symmetries of an equilateral triangle. In the group T, $pr = t$. Verify that the product of any member of pA times any member of rA is a member of tA.

Cosets in a Commutative Group

In a commutative group, every subgroup is a normal subgroup. So, in a commutative group, *every* subgroup has the two properties we observed in the preceding paragraphs: 1) Every left coset of a subgroup is a right coset, and vice versa; 2) All products obtained by multiplying a member of one left coset of a subgroup by a member of another left coset of that subgroup belong to a single left coset of that subgroup.

For example, consider the group of symmetries of a rectangle, $R = \{i, a, b, c\}$. We can see from its multiplication table on page 86 that R is a commutative group. Let $A = \{i, a\}$. A is a subgroup of R. Since R is commutative, A is a normal subgroup of R. So A has properties 1) and 2), listed above. (See exercise 9 on page 223.)

In a commutative group we sometimes use the symbol $+$ for the operation in the group. In such cases, addition takes the place of multiplication in all statements about members of the group, subgroups, cosets, and so on. For example, in the six-point clock-face number system, the set $\{0, 1, 2, 3, 4, 5\}$ is a group with respect to the operation $+$, with the following group table:

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Let us call this group M. 0 is the identity element in M. Let $A = \{0, 3\}$. A is obviously a subgroup of M. Since M is a commutative group, A is a normal subgroup of M. Then A has properties 1) and 2) listed above. The left and right cosets of A can be read from the table. For example, the left coset $2 + A = \{2+0, 2+3\} = \{2, 5\}$. The right coset $A + 2 = \{0 + 2, 3 + 2\} = \{2, 5\}$. (See exercise 10 below.)

Another example of a commutative group is the set of all integers, used with the operation $+$. In this group, the set of all even integers, $\{0, 2, 4, \ldots, -2, -4, \ldots\}$, is a subgroup, and therefore it is a normal subgroup with properties 1) and 2) listed above. Similarly, the set of all multiples of 3, $\{0, 3, 6, \ldots, -3, -6, \ldots\}$ is a normal subgroup of the group of integers, and has properties 1) and 2). (See exercises 11 and 12 below.)

Exercises:

9. Consider the subgroup $A = \{i, a\}$ in the group R of symmetries of a rectangle. Find all the left cosets of A and all the right cosets of A. Verify that every left coset of A is a right coset of A, and vice versa. In the group R, $ab = c$. Verify that the product of any member of aA times any member of bA is a member of cA.

10. Consider the subgroup $A = \{0, 3\}$ in the additive group M of the six-point clock-face number system. Find all the left cosets of A and all the right cosets of A. Verify that every left coset of A is a right coset of A, and vice versa. In the group M, $2 + 3 = 5$. Verify that the sum of any member of $2 + A$ plus any member of $3 + A$ is a member of $5 + A$.

11. Let $E = \{0, 2, 4, \ldots, -2, -4, \ldots\}$, the subgroup of all even integers in the group of integers. a) By listing a few typical members inside a pair of braces, as we did for E above, identify each of the following left cosets of E: $0 + E$, $1 + E$, $2 + E$, $3 + E$. b) Identify each of the following right cosets of E: $E + 0$, $E + 1$,

$E + 2$, $E + 3$. c) How many different left cosets does E have? d) How many different right cosets does E have? e) Verify that every left coset of E is a right coset of E, and vice versa. f) In the system of integers, $1 + 2 = 3$. Verify that the sum of any member of $1 + E$ plus any member of $2 + E$ is a member of $3 + E$.

12. Let $T = \{0, 3, 6, \ldots, -3, -6, \ldots\}$, the subgroup of all multiples of 3 in the group of integers. a) Identify each of the following left cosets of T: $0 + T$, $1 + T$, $2 + T$, $3 + T$, $4 + T$. b) Identify each of the following right cosets of T: $T + 0$, $T + 1$, $T + 2$, $T + 3$, $T + 4$. c) How many different left cosets does T have? d) How many different right cosets does T have? e) Verify that every left coset of T is a right coset of T, and vice versa. f) In the system of integers, $1 + 2 = 3$. Verify that the sum of any member of $1 + T$ plus any member of $2 + T$ is a member of $3 + T$.

Products of Cosets of a Normal Subgroup

Let S be a group, and let G be a normal subgroup of S. Since every left coset of G is a right coset of G, and vice versa, we may drop the words "left" and "right" when we talk about them, and simply refer to each as a coset of G. Consider the set of all cosets of G, and choose an ordered pair of cosets, say aG and bG. We saw on page 221 that all products obtained by multiplying a member of aG times a member of bG belong to a *single* coset of G. Let us call this single coset the *product* of aG and bG. Since a is a member of aG, and b is a member of bG, the product of aG and bG is the coset which contains ab, namely abG. Thus we have the following rule for multiplying cosets of a normal subgroup G: $(aG)(bG) = abG$.

For example, consider the normal subgroup $B = \{i, q\}$ in the group of symmetries of a square. The cosets pB and rB are both equal to $\{p, r\}$, so we have a choice of two different ways of writing the coset $\{p, r\}$, namely pB or rB. The cosets sB and tB are both equal to $\{s, t\}$, so we have a choice of two different ways

of writing the coset $\{s, t\}$, namely sB or tB. Then we have a choice of four different ways of writing the product of the coset $\{p, r\}$ and the coset $\{s, t\}$: $(pB)(sB) = psB$, or $(pB)(tB) = ptB$, or $(rB)(sB) = rsB$, or $(rB)(tB) = rtB$. It makes no difference which way the product is written, because the products psB, ptB, rsB, and rtB are all the same, since ps, pt, rs, and rt all belong to the same coset.

If a subgroup is not normal, we cannot use the same procedure to define a multiplication operation for its left cosets, because in that case, as we saw in example 1 on page 220, the products obtained by taking a member of one left coset times a member of another left coset do not all belong to a single left coset.

In the group of symmetries of a square, let us assign some new names to each coset of the normal subgroup B according to the following scheme: If a member of the coset is represented by a certain lower case letter, then use the corresponding capital letter as a name for that coset. There are four cosets of B, and by this scheme, each has two new names as follows:

$\{i, q\}$ is called I or Q;
$\{p, r\}$ is called P or R;
$\{s, t\}$ is called S or T;
$\{u, v\}$ is called U or V.

If we want to emphasize that the coset $\{p, r\}$ is the one that contains p, then we shall call it P. If we want to emphasize that it is the one that contains r, then we shall call it R. Otherwise we are free to use either one of its two names. A similar scheme for naming cosets of a normal subgroup is convenient whenever the elements of the group are represented by lower case letters.

In the group of symmetries of a square, the set of all cosets of the normal subgroup B is $\{I, P, S, U\}$, where each coset is now represented by just one of its new names. We have just defined a multiplication operation in this set. Using this operation, we can identify the product for any ordered pair of members of the set.

For example, using the old names, we saw on page 225, that $(pB)(sB) = psB$. Since $ps = u$, we may write $(pB)(sB) = uB$. P is the new name for pB, S is the new name for sB, and U is the new name for uB. Using the new names, we see that $PS = U$.

Exercise:

13. Construct the multiplication table for the set of cosets $\{I, P, S, U\}$ of the normal subgroup B in the group of symmetries of a square.

A Group Whose Members Are Cosets

In exercise 13 above we constructed a multiplication table for the set $\{I, P, S, U\}$, each of whose members is a coset of the normal subgroup B in the group of symmetries of a square. (See answer section, page 265.) Let us translate this table into another language by using the following dictionary:

$$I \rightarrow i$$
$$P \rightarrow a$$
$$S \rightarrow b$$
$$U \rightarrow c$$

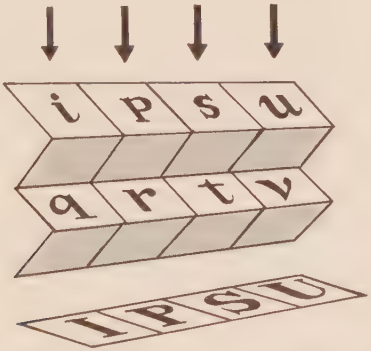
The translated table looks like this:.

	i	a	b	c
i	i	a	b	c
a	a	i	c	b
b	b	c	i	a
c	c	b	a	i

If we compare this table with the table for the group of symmetries of a rectangle, we see that they are the same. This proves that the

set of cosets $\{I, P, S, U\}$ is a group with respect to the operation "multiplication of cosets." In fact, the dictionary that we used is an isomorphism between the group of cosets of B and the group of symmetries of a rectangle.

What we have just observed is an example of a general rule: If S is any group, and G is a normal subgroup of S, then the set of cosets of G is a group with respect to the operation "multiplication of cosets." Since we form this group by *dividing* S into cosets, each of which has the same number of members as the subgroup G, it is called a *quotient group*, and is represented by the symbol S/G. If G has more than one member in it, forming the quotient group S/G is like collapsing the group S to produce a smaller group. In the diagram below we picture the process of collapsing the group S of symmetries of a square as a process of squeezing it like an accordion. The "squeezing" presses together the members of each coset of B, so that each coset becomes a single member in the quotient group S/B.



The proof that the set of cosets of a normal subgroup is a group is carried out in exercises 14, 15 and 16.

Exercises:

For exercises 14, 15 and 16, let S be any group, and let G be a normal subgroup of G. For any member of S, represented by a

lower-case letter, let the corresponding capital letter stand for the coset of G that contains that member. Then, by the definition of multiplication of cosets, if $ab = c$, then $AB = C$. Since the product of two cosets of G is a coset of G, the set of all cosets of G is closed with respect to multiplication.

14. Prove that if a, b and c are members of S, then $(AB)C = A(BC)$.

15. I is the coset of G that contains the identity element i of S. Prove that if A is any coset of G, $IA = AI = A$.

16. If A is any coset of G, there is a coset B with the property $AB = BA = I$.

17. If the order of a group S is n, and the order of a normal subgroup G in S is m, what is the order of the quotient group S/G?

18. In the group S of symmetries of a square, the subgroup $A = \{i, p, q, r\}$ is normal. There are two cosets of A, which we name as follows: $I = \{i, p, q, r\}$, and $S = \{s, t, u, v\}$. Construct the multiplication table for the quotient group S/A.

19. Let M be the additive group of the six-point clock-face number system, and let A be the subgroup $\{0, 3\}$. There are three cosets of A which we name as follows, using boldface numerals: $\mathbf{0} = \{0, 3\}$, $\mathbf{1} = \{1, 4\}$, and $\mathbf{2} = \{2, 5\}$. Construct the addition table for the quotient group M/A.

20. Let Z be the additive group of all integers, and let T be the subgroup of all integers that are multiples of 3. There are three cosets of T which we name as follows:

$$\mathbf{0} = \{0, 3, 6, \ldots, -3, -6, \ldots\}$$
$$\mathbf{1} = \{1, 4, 7, \ldots, -2, -5, \ldots\}$$
$$\mathbf{2} = \{2, 5, 8, \ldots, -1, -4, \ldots\}$$

Construct the addition table for the quotient group Z/T.

21. a) Use the multiplication table for S_4, printed on page 145, to construct the multiplication table for the subset $Y = \{i, g, q, x\}$. Verify from the table that Y is a subgroup of S_4. b) Use the table

for S_4 to identify the left cosets and the right cosets of Y. Verify from your result that Y is a normal subgroup of S_4. c) Assign names to the cosets of Y as follows:

$$I = \{i, g, q, x\}; \quad A = \{a, f, r, w\}; \quad B = \{b, k, n, v\};$$
$$C = \{c, l, m, u\}; \quad D = \{d, h, p, t\}; \quad E = \{e, j, o, s\}.$$

Construct the multiplication table for the quotient group S_4/Y.

Homomorphism

Let us examine again the group S of symmetries of a square, and the quotient group S/B, whose members are cosets of the normal subgroup B in S. For each member of S, let us form an ordered pair by matching that member of S with the member of S/B that contains it. The set of all these ordered pairs is shown one under the other in the list below:

$i \rightarrow I$ (also known as Q)
$p \rightarrow P$ (also known as R)
$q \rightarrow Q$ (also known as I)
$r \rightarrow R$ (also known as P)
$s \rightarrow S$ (also known as T)
$t \rightarrow T$ (also known as S)
$u \rightarrow U$ (also known as V)
$v \rightarrow V$ (also known as U)

We can use this list of ordered pairs as a dictionary for translating statements about members of S into another language. For example, this dictionary translates the statement $ps = u$ into the statement $PS = U$. Because of the way in which multiplication of cosets was defined, this dictionary translates every multiplication fact of S into a multiplication fact of S/B. In this respect, it is like the kind of dictionary that we have called an isomorphism from one group to another. However, it is not an isomorphism, because

each member of S/B appears in the second column twice. (In an isomorphism from one group to another, each member of the first group appears once and only once in the first column, and each member of the second group appears once and only once in the second column.) It is an example of another kind of dictionary called a *homomorphism*, which is defined as follows: A two-column dictionary that matches each member of one group with a member of another group is called a homomorphism if it has the following properties: 1) Every member of the first group appears once and only once in the first column of the dictionary; 2) every member of the second group appears at least once in the second column of the dictionary; 3) the dictionary translates every multiplication fact of the first group into a multiplication fact of the second group. Notice that each member of the second group may appear once or more than once in the second column of a homomorphism. In the special case where each member of the second group appears only once in the second column the homomorphism is an isomorphism.

If there is a homomorphism that matches the members of one group with the members of another, we say that the first group is *homomorphic* to the second, and that the second group is a *homomorphic image* of the first group.

The homomorphism relating the group S to the quotient group S/B is an example of a whole family of homomorphisms obtained in exactly the same way: If S is any group, and G is a normal subgroup of G, then the set of ordered pairs that matches each member of S with the member of S/G that contains it is a homomorphism.

In the example of a homomorphism that we just saw, we started with the quotient group S/B and showed that it is a homomorphic image of S. This procedure is also reversible. If we start with a group T that is a homomorphic image of S, we can show that there is a normal subgroup G of S such that T has the same structure as the quotient group S/G. To show how this is done, we first examine a specific example.

Consider the group R of symmetries of a rectangle, and the symmetric group S_2. Their multiplication tables appear on pages 86 and 144. To avoid confusion, let us use the capital letters I and A instead of the lower case letters i and a as names for the members of S_2. Let us examine the following set of ordered pairs matching each member of R with a member of S_2.

$$i \rightarrow I$$
$$a \rightarrow I$$
$$b \rightarrow A$$
$$c \rightarrow A$$

Each member of R appears once and only once in the first column. Each member of S_2 appears twice in the second column. Let us use this list of ordered pairs as a dictionary to translate every multiplication fact of R into another language. There are sixteen multiplication facts of R: $ii = i$, $ia = a$, $ib = b$, $ic = c$, $ai = a$, $aa = i$, $ab = c$, $ac = b$, $bi = b$, $ba = c$, $bb = i$, $bc = a$, $ci = c$, $ca = b$, $cb = a$, $cc = i$. It is easy to see that the translation of each of these multiplication facts of R is a multiplication fact of S_2. For example, the translation of the fact $ab = c$ is obtained by replacing a by I, b by A, and c by A. The translated statement is $IA = A$, which is a multiplication fact of S_2. (See exercise 22 below.) Consequently, this set of ordered pairs is a homomorphism, and S_2 is a homomorphic image of R.

Now let us rewrite the homomorphism by writing each member of S_2 only once, and by writing to the left of it the set of all members of R that it is matched with. Then the rewritten list looks like this:

$$\{i, a\} \rightarrow I$$
$$\{b, c\} \rightarrow A$$

Let $G = \{i, a\}$. It is easy to see that G is a normal subgroup of R, and that each set that appears in the left-hand column of the rewritten list is a coset of G. Moreover, each coset of G appears in the left-hand column once and only once. Then the rewritten

list is an isomorphism between the group S/G and the group S_2.

In general, it can be shown that if a group S is homomorphic to another group T, and if G is the set of all members of S that are matched with the identity element I of T, then G is a normal subgroup of S, the set of elements of S that are matched with any element A of T is a coset of G, and S/G is isomorphic to T. (See exercises 23 to 28 below.)

Exercises:

22. Verify that the dictionary on page 231 translates each multiplication fact of R into a multiplication fact of S_2.

For exercises 23 to 28, let G be the set of members of a group S that are matched with the identity element I of a group T by a homomorphism.

23. Let a and b be any two members of G, and let $ab = c$. Prove that c is a member of G.

24. Prove that i, the identity element of S, is a member of G.

25. Prove that if a is a member of G, then a^{-1} is also a member of G. (Exercises 23 to 25 show that G is a subgroup of S.)

26. Prove that every conjugate of a member of G is also a member of G. (This will show that G is a normal subgroup of S.)

27. Prove that the set of elements of S that are matched with any element A of T is a coset of G.

28. Prove that S/G is isomorphic to T.

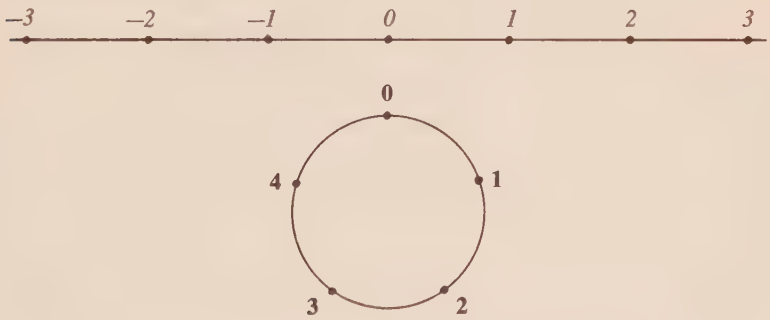
29. If G is a normal subgroup of a group S, under what conditions will the homomorphism that matches each member a of S with the coset aG turn out to be an isomorphism between S and S/G?

Integers and Clock-Face Numbers

The idea of a homomorphism gives us the means of understanding how the clock-face numbers we became acquainted with in

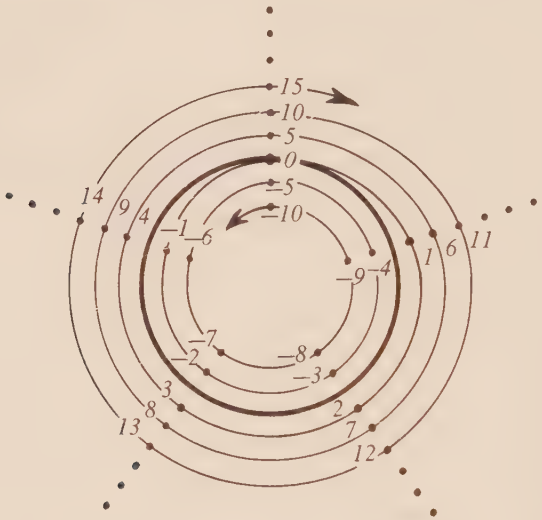
Chapter 6 are related to the integers we became acquainted with in Chapter 3. If we let Z stand for the additive group of integers, and let Z_n stand for the additive group of the n-point clock-face number system, then it is not hard to show that Z is homomorphic to Z_n.

For example, let us consider Z and Z_5. To avoid confusion between the members of the two groups, let us use italics for the members of Z, and boldface numerals for the members of Z_5. Then $Z = \{0, 1, 2, \ldots, -1, -2, \ldots\}$, and $Z_5 = \{\mathbf{0}, \mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}\}$. The members of Z may be pictured as points on a straight line that divide the line into equal segments each one unit long. As we know, the members of Z_5 may be pictured as points on a circle that divide the circle into five equal arcs, as shown below. Let us use a circle whose circumference is five units long, so that the arc from each member of Z_5 to its nearest neighbor in either direction is one unit long.



To set up a homomorphism between Z and Z_5, think of the line containing the members of Z as a string. Place the string so that the zero point of Z is on the zero point of Z_5. Then wind the string around the circle in both directions. In this way each point of Z is made to fall on a point of Z_5. The set of ordered pairs in which each member of Z is matched with the member of Z_5 that it falls on is a homomorphism. Winding the positive half of Z in one direction, we see that *1* falls on **1**, *2* falls on **2**, *3* falls on **3**, *4* falls on **4**, *5* falls on **0**, *6* falls on **1**, and so on. Winding the

negative half of Z in the other direction we see that -1 falls on **4**, -2 falls on **3**, -3 falls on **2**, -4 falls on **1**, -5 falls on **0**, -6 falls on **4**, and so on. Each turn of the string will fit tightly on the circle. However, to show the matching of points in the diagram below, we have distorted the picture a bit by separating each turn of the string from the next, and by winding the negative side of Z inside the circle while the positive side of Z is wound around the outside of the circle.



The set of members of Z that fall on 0 on the circle is $G_5 = \{0, 5, 10, \ldots, -5, -10, \ldots\}$. It is the set of all multiples of 5. As we learned in Chapter 11, exercise 11, this set is a subgroup of Z. It is a normal subgroup, since Z is a commutative group. The set of members of Z that fall on each member of Z_5 is a coset of G_5. In the diagram they lie on a line like a spoke of a wheel. The five spokes are the members of Z/G_5. The set of ordered pairs that matches each member of Z/G_5 with the member of Z_5 that it lies on is an isomorphism.

For any positive integer n, a similar diagram, in which Z is wound around a circle whose circumference is n units long, sets

up a homomorphism between Z and Z_n. The set of members of Z that fall on 0 on the circle is the set of all multiples of n. Let us call this set G_n. It is a normal subgroup of Z. The same diagram also sets up an isomorphism between Z/G_n and Z_n.

Exercises:

30. Draw a diagram showing the homomorphism between Z and Z_3 obtained by winding Z around a circle whose circumference is three units long.

31. Draw a diagram showing the homomorphism between Z and Z_4 obtained by winding Z around a circle whose circumference is four units long.

Answers to Exercises

Chapter 2

1. *9.* **2.** $\frac{2}{1}, \frac{4}{2}, \frac{6}{3}, \frac{8}{4}$, and so on. Any fraction whose numerator is double its denominator.

3. $\frac{3}{2}, \frac{6}{4}, \frac{9}{6}, \frac{12}{8}$, and so on. **4.** $\frac{2}{21}$. **5.** $\frac{3}{2}$.

6. Yes; Yes; Yes. **7.** *3*; No; No; No.

8. $8 - (5 - 2) = 8 - 3 = 5$; $(8 - 5) - 2 = 3 - 2 = 1$; No.

10. Yes; Yes; Yes; Yes. **11.** Yes; Yes; Yes; No.

12. *1.* **13.** $\frac{6}{5}$. **14.** *2.* **15.** $\frac{1}{5}$. **16.** No; No.

17. $\frac{5}{3}$. **18.** $\frac{1}{2} \times \frac{1}{2}; \frac{1}{4}$. **19.** $\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{16}{81}$.

20. $\frac{1}{9}$. **21.** *1, 3, 9, 27, 81, . . .* and $\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}, \ldots$; Yes; Yes; Yes.

22. 2^{10}. **23.** 4^9.

24. Closure, Associative Property, Identity Element.

25. Closure, Associative Property.

Chapter 3

1. $(9 + 16) + 43 = 25 + 43 = 68$.

2. $9 + (16 + 43) = 9 + 59 = 68$.

3. When we write the sum of three or more counting numbers, we may write it either without parentheses, or with parentheses around any two or more numbers that are written next to each other in the sum.

4. Yes, because if neither a nor b is 0, each is a counting number. Then their sum is a counting number, and every counting number is a whole number.

5. Yes, because if $a = 0$ and b is not 0, $a + b = 0 + b = b$, which is a counting number.

6. Yes, because if a is not 0 and $b = 0$, $a + b = a + 0 = a$, which is a counting number.

7. Yes, because if $a = 0$ and $b = 0$, $a + b = 0 + 0 = 0$.

8. The set of all whole numbers is closed with respect to addition.

9. If neither a nor b nor c is 0, each is a counting number. Since addition of counting numbers has the associative property, $a + (b + c) = (a + b) + c$.

10. If $a = 0$, $a + (b + c) = 0 + (b + c) = b + c$ by equation (6). Moreover, $(a + b) + c = (0 + b) + c = b + c$, since $0 + b = b$ by equation (6). Then $a + (b + c)$ and $(a + b) + c$ in this case are both equal to $b + c$, so they are equal to each other.

11. If $b = 0$, $a + (b + c) = a + (0 + c) = a + c$, since $0 + c = c$ by equation (6). Moreover, $(a + b) + c = (a + 0) + c = a + c$, since $a + 0 = a$ by equation (6). Then $a + (b + c)$ and $(a + b) + c$ in this case are both equal to $a + c$, so they are equal to each other.

12. If $c = 0$, $a + (b + c) = a + (b + 0) = a + b$, since $b + 0 = b$ by equation (6). Moreover, $(a + b) + c = (a + b) + 0 = a + b$ by equation (6). Then $a + (b + c)$ and $(a + b) + c$ in this case are both equal to $a + b$, so they are equal to each other.

13. Addition of whole numbers has the associative property.

14. 4. **15.** 3. **16.** 1. **17.** -2. **18.** -5.

19. -1. **20.** At 4. **21.** At -5. **22.** 2; -2.

23. -2. **24.** 4. **25.** 0. **26.** -6; 8.

27. $3 + ((-2) + 6) = 3 + 4 = 7$.

28. $(3 + (-2)) + 6 = 1 + 6 = 7$.

30. 5. **31.** 1. **32.** 1. **33.** $\frac{1}{9}$. **34.** $\frac{1}{32}$.

35. a^0, a^1, a^2, a^3, . . . , and a^{-1}, a^{-2}, a^{-3}, ...

36. *1*; *1*. **37.** a^{-7}; a^3; a^{-1}. **38.** a^3; a^4; a^5; a^{-3}.

39. a^{n+1}.

Chapter 4

1. Closure, Associative Property, Identity Element; No.

2. Yes; No; No, since $1 \div a$ need not be equal to a; No.

3. All four properties; Yes.

Chapter 5

1. *180°*; *270°*; *120°*; *1080°*. **2.** One sixth. **3.** *4*.

4. a) *20°* clockwise; b) *60°* clockwise; c) *280°* clockwise; d) *320°* clockwise; e) *0°* clockwise.

5. A *90°* turn. **6.** A *140°* turn.

7. A *40°* turn; a *50°* turn; a *60°* turn.

8. A *0°* turn; a *0°* turn; a *0°* turn. **9.** A *300°* turn.

10. A $(360 - x)°$ turn.

11. a)

	i	b
i	i	b
b	b	i

b) Yes.

c) Yes, because it has the associative property in the set of all turns. d) Yes, because it is an identity element in the set of all turns. e) Yes. The inverse of i is i, and the inverse of b is b. f) Yes.

12. a)

	i	p	q
i	i	p	q
p	p	q	i
q	q	i	p

b) Yes.

c) Yes, because it has the associative property in the set of all turns. d) Yes, because it is an identity element in the set of all turns. e) i, q and p respectively. Yes. f) Yes.

13. a)

	i
i	i

b) Yes.

c) $i(ii) = ii = i$. Also $(ii)i = ii = i$. So $i(ii) = (ii)i$.
d) Yes. e) Yes. f) Yes.

14. Infinite. **15.** Finite.

16. If n is any counting number, the group certainly includes among its members the n turns in the following list: a turn of $1°$, a turn of $\left(\frac{1}{2}\right)°$, . . . , a turn of $\left(\frac{1}{n}\right)°$. But it also includes among its members a turn of $\left(\frac{1}{n+1}\right)°$, which is not in this list. Therefore the group has more members than can be counted. That is, it is an infinite group.

17. One. It must contain an identity element i, and there is a group whose only member is i. (See exercise *13* on page *62*.)

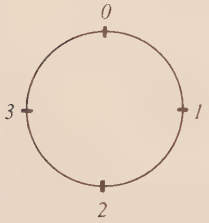
18. *3*.

Chapter 6

1.

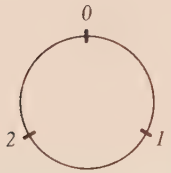
+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

2.



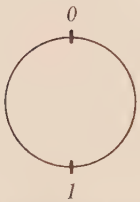
+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

3.



+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

4.



+	0	1
0	0	1
1	1	0

5.

×	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

6.

×	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

7.

×	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

8.

×	0	1
0	0	0
1	0	1

9. *0*, *3*, *2* and *1* respectively. **10.** *0*, *2* and *1* respectively.

11. *0* and *1* respectively.

12. a) Both are equal to *3*. b) Both are equal to *1*. c) Both are equal to *4*. d) Both are equal to *3*.

13. a)

×	1	2
1	1	2
2	2	1

b) *1* and *2* respectively.
c) Yes.

14. a)

×	1	2	3	4	5
1	1	2	3	4	5
2	2	4	0	2	4
3	3	0	3	0	3
4	4	2	0	4	2
5	5	4	3	2	1

b) *2*, *3* and *4*.
c) No.

15. a)

×	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

b) *1*, *4*, *5*, *2*, *3* and *6* respectively.
c) Yes.

16. If *3* has a multiplicative inverse *b*, then $3 \times b = 1$. Then $2 \times (3 \times b) = 2 \times 1$. This is impossible, because $2 \times (3 \times b) = (2 \times 3) \times b = 0 \times b = 0$, while $2 \times 1 = 2$, which is not zero.

17. If *2* has a multiplicative inverse *b*, then $2 \times b = 1$. Then $2 \times (2 \times b) = 2 \times 1$. This is impossible, because $2 \times (2 \times b) = (2 \times 2) \times b = 0 \times b = 0$, while $2 \times 1 = 2$, which is not zero.

18. a)

×	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7
2	2	4	6	0	2	4	6
3	3	6	1	4	7	2	5
4	4	0	4	0	4	0	4
5	5	2	7	4	1	6	3
6	6	4	2	0	6	4	2
7	7	6	5	4	3	2	1

b) *2*, *4* and *6* are zero divisors.
c) Proof that *4* has no multiplicative inverse: If it has a multiplicative inverse *b*, then $4 \times b = 1$. Then $2 \times (4 \times b) = 2 \times 1$. This is impossible, because $2 \times (4 \times b) = (2 \times 4) \times b = 0 \times b = 0$, while $2 \times 1 = 2$, which is not zero.

19. If *b* has a multiplicative inverse *c*, then $b \times c = 1$. Then $a \times (b \times c) = a \times 1$. This is impossible, because $a \times (b \times c) = (a \times b) \times c = 0 \times c = 0$, while $a \times 1 = a$, which is not zero.

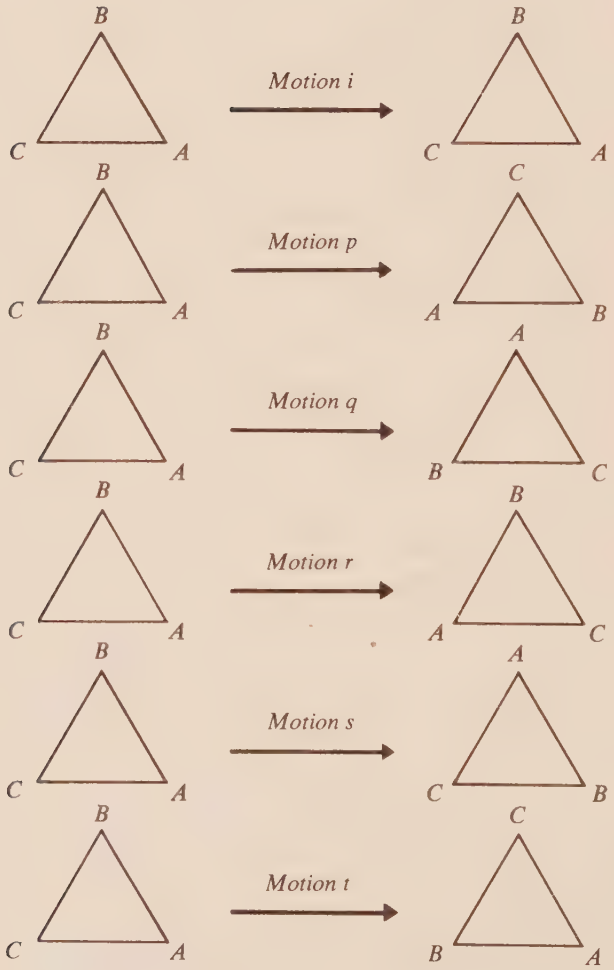
20. *9*, *10* and *12*.

21. No. If *n* is composite, there are smaller counting numbers *a* and *b* such that $a \times b = n$. Then *a* and *b* are zero divisors in the *n*-point

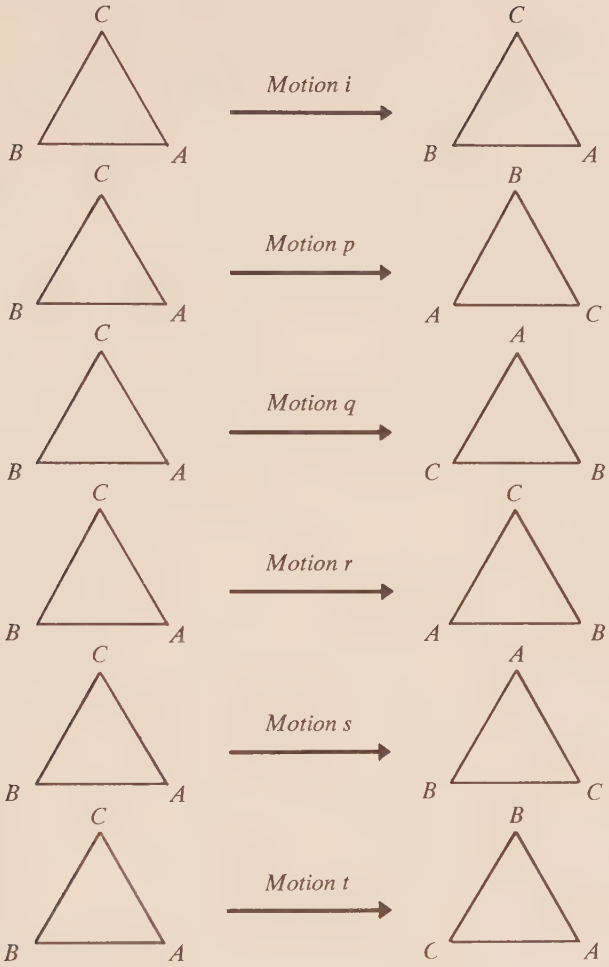
clock-face number system. Then a and b are non-zero numbers that do not have a multiplicative inverse in that system.

Chapter 7

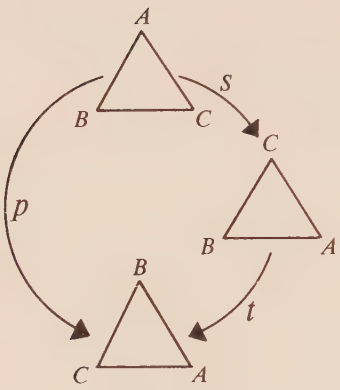
1.



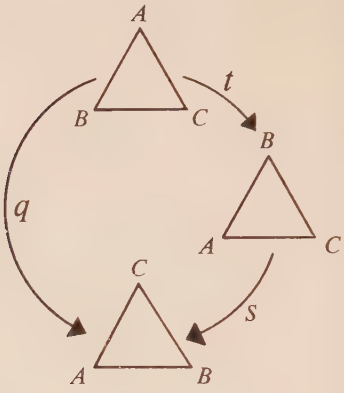
2.


C
B
A
Motion i
C
B
A
C
B
A
Motion p
B
A
C
C
B
A
Motion q
A
C
B
C
B
A
Motion r
C
A
B
C
B
A
Motion s
A
B
C
C
B
A
Motion t
B
C
A

3.



$st=p$



$ts=q$

4. $r^2 = i$; $s^2 = i$; $t^2 = i$. **5.** See page *81*.

6. $p(st) = pp = q$; $(ps)t = rt = q$.

7. i, q, p, r, s, and t respectively.

8.

	i	p	q
i	i	p	q
p	p	q	i
q	q	i	p

a) Yes.
b) Yes.
c) Yes.

9.

	r	s	t
r	i	p	q
s	q	i	p
t	p	q	i

a) No.
b) No.
c) No.

10.

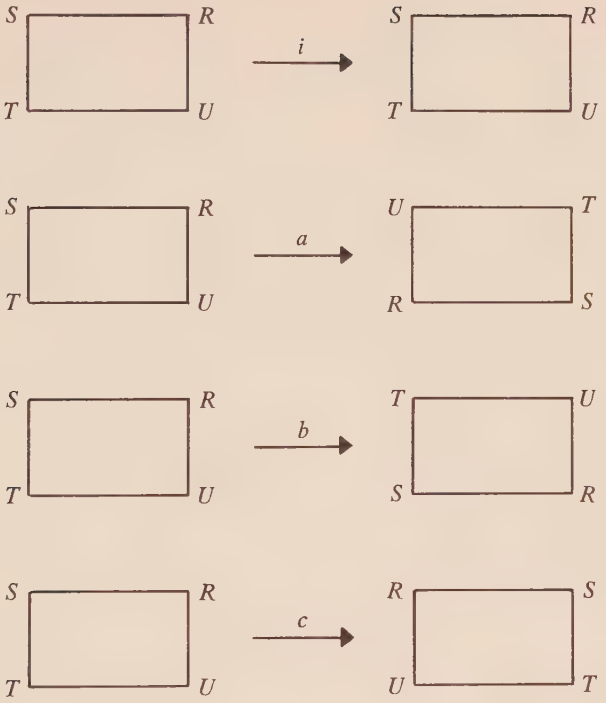
	i	r
i	i	r
r	r	i

Yes.

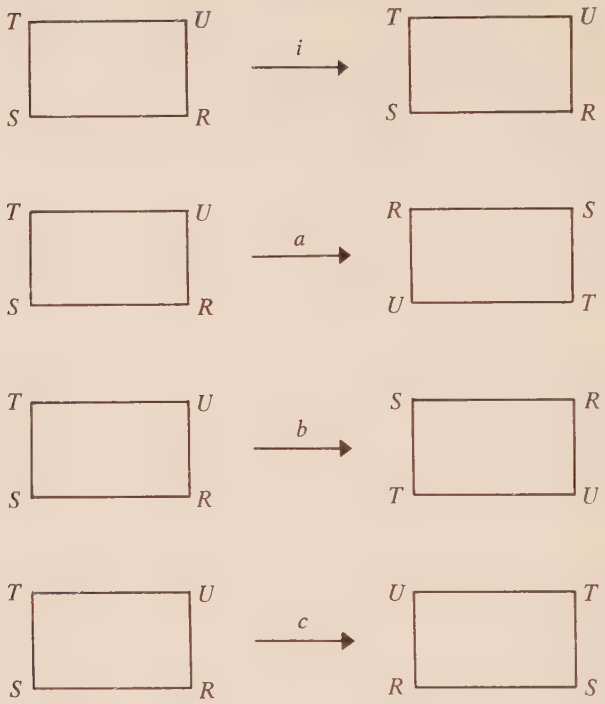
11. In pq and ir. **12.** Yes. **13.** Yes.

14. Yes. **15.** Yes.

16.



17.



18. See page *86*.

19. $a(bc) = aa = i$; $(ab)c = cc = i$.

20. a) i. b) i, a, b and c respectively.

21. a) $ab = ba = c$. b) $ac = ca = b$. c) $bc = cb = a$. d) Yes.

22. See page *89*. **23.** $s(tu) = sp = v$; $(st)u = qu = v$.

24. a) i. b) i, r, q, p, s, t, u and v respectively.

25. $pu = t$; $up = s$; no.

26.

	i	p	q	r
i	i	p	q	r
p	p	q	r	i
q	q	r	i	p
r	r	i	p	q

a) Yes.
b) Yes.
c) Yes.

27.

	i	q	s	t
i	i	q	s	t
q	q	i	t	s
s	s	t	i	q
t	t	s	q	i

a) Yes.
b) Yes.
c) Yes.

28.

	s	t	u	v
s	i	q	r	p
t	q	i	p	r
u	p	r	i	q
v	r	p	q	i

No.

29. a) M, T, V, W, Y. b)

	i	t
i	i	t
t	t	i

30. a) i, q, s, t. b) See answer to exercise *27*. c) I.

31. a) i, q. b)

	i	q
i	i	q
q	q	i

c) N.

32. a) i, u. b)

	i	u
i	i	u
u	u	i

c) No others.

33. a) All the symmetries of the square. b) See page *89*.
c) No others.

34. a) i. b)

	i
i	i

c) No others.

35. a)

	i	r	a	l
i	i	r	a	l
r	r	a	l	i
a	a	l	i	r
l	l	i	r	a

b) It is closed with respect to multiplication. Multiplication of commands has the associative property because *i*, *r*, *a* and *l* are like clockwise rotations of *0°*, *90°*, *180°* and *270°*. *i* is an identity element. The set contains the inverse of each of its members.

Chapter 8

1. a) $r(st) = rp = s$. b) $(iq)t = qt = r$.
c) $(pq)(rs) = ip = p$.

2. a) $(ab)c = cc = i$. b) $(ca)b = bb = i$.
c) $(aa)(bb) = ii = i$.

3. a) $(pu)v = tv = r$. b) $(pq)(rs) = rv = t$.
c) $(pu)(qt) = ts = q$.

4. $risvut = (ri)(sv)(ut) = rpr = (rp)r = ir = r$.

5. $tqrps = (tq)(rp)s = sss = (ss)s = is = s$.

6. a) $qir = (qi)r = qr = p$. b) $tsi = t(si) = ts = q$.
c) $(uu)p = ip = p$.

7. a) $pp^{-1}p^{-1} = (pp^{-1})p^{-1} = ip^{-1} = p^{-1} = r$.
b) $stq^{-1} = (st)q^{-1} = qq^{-1} = i$. c) $uv^{-1}v = u(v^{-1}v) = ui = u$.

8. $r(px) = rr$, so $(rp)x = rr$. But $rp = i$, and $rr = q$, so $ix = q$. Then $x = q$.

9. a) $(b^{-1}a^{-1})(ab) = b^{-1}a^{-1}ab = b^{-1}(a^{-1}a)b = b^{-1}ib = b^{-1}(ib) = b^{-1}b = i$. b) $(ab)(b^{-1}a^{-1}) = abb^{-1}a^{-1} = a(bb^{-1})a^{-1} = aia^{-1} = (ai)a^{-1} = aa^{-1} = i$.

10. $(abc)(c^{-1}b^{-1}a^{-1}) = abcc^{-1}b^{-1}a^{-1} = ab(cc^{-1})b^{-1}a^{-1} = abib^{-1}a^{-1} = a(bi)b^{-1}a^{-1} = abb^{-1}a^{-1} = a(bb^{-1})a^{-1} = aia^{-1} = (ai)a^{-1} = aa^{-1} = i$. Also, $(c^{-1}b^{-1}a^{-1})(abc) = c^{-1}b^{-1}a^{-1}abc = c^{-1}b^{-1}(a^{-1}a)bc = c^{-1}b^{-1}ibc = c^{-1}b^{-1}(ib)c = c^{-1}b^{-1}bc = c^{-1}(b^{-1}b)c = c^{-1}ic = c^{-1}(ic) = c^{-1}c = i$.

11. $a^2(a^{-1})^2 = (aa)(a^{-1}a^{-1}) = aaa^{-1}a^{-1} = a(aa^{-1})a^{-1} = aia^{-1} = (ai)a^{-1} = aa^{-1} = i$. Also, $(a^{-1})^2a^2 = (a^{-1}a^{-1})(aa) = a^{-1}a^{-1}aa = a^{-1}(a^{-1}a)a = a^{-1}ia = a^{-1}(ia) = a^{-1}a = i$.

12. $aa^2 = a(aa) = aaa = a^3$; $aa^3 = a(aaa) = aaaa = a^4$; $aa^4 = a(aaaa) = aaaaa = a^5$.

13. $a^3a^2 = (aaa)(aa) = aaaaa = a^5$.

14. $a^3a^{-2} = (aaa)(a^{-1}a^{-1}) = aaaa^{-1}a^{-1} = aa(aa^{-1})a^{-1} = aaia^{-1} = a(ai)a^{-1} = aaa^{-1} = a(aa^{-1}) = ai = a$.

15. $a^{-3}a^{-2} = (a^{-1}a^{-1}a^{-1})(a^{-1}a^{-1}) = a^{-1}a^{-1}a^{-1}a^{-1}a^{-1} = a^{-5}$.

16. $(a^3)^2 = a^3a^3 = (aaa)(aaa) = aaaaaa = a^6$.

17. $(a^{-3})^2 = a^{-3}a^{-3} = (a^{-1}a^{-1}a^{-1})(a^{-1}a^{-1}a^{-1}) = a^{-1}a^{-1}a^{-1}a^{-1}a^{-1}a^{-1} = a^{-6}$.

18. $(a^3)^{-2}$ = the inverse of $(a^3)^2$ = the inverse of $a^6 = a^{-6}$.

19. $(a^{-3})^{-2}$ = the inverse of $(a^{-3})^2$ = the inverse of $a^{-6} = a^6$.

20. $r^1 = r$; $r^2 = q$; $r^3 = rr^2 = rq = p$; $r^4 = rr^3 = rp = i$. The members of the subgroup are r, q, p and i.

21. $v^1 = v$; $v^2 = i$. The members of the subgroup are v and i.

22. $p^1 = p$; $p^2 = q$; $p^3 = pp^2 = pq = i$. The members of the subgroup are p, q and i.

23. $s^1 = s$; $s^2 = i$. The members of the subgroup are s and i.

24. $2^1 = 2$; $2^2 = 4$; $2^3 = 2(2^2) = 2(4) = 1$. The members of the subgroup are 2, 4 and 1.

25. $3^1 = 3$; $3^2 = 2$; $3^3 = 3(3^2) = 3(2) = 6$; $3^4 = 3(3^3) = 3(6) = 4$; $3^5 = 3(3^4) = 3(4) = 5$; $3^6 = 3(3^5) = 3(5) = 1$. The members of the subgroup are 3, 2, 6, 4, 5 and 1. That is, the subgroup generated by 3 is the whole group A.

26. $a^m = a^n$. Multiply both sides on the right by a^{-m}. Then $a^ma^{-m} = a^na^{-m}$. But $a^ma^{-m} = i$, and $a^na^{-m} = a^{n+(-m)} = a^{n-m}$. So $a^{n-m} = i$.

27. Suppose all members are powers of a. If x and y are any two members, $x = a^m$, and $y = a^n$, where m and n are integers. $xy = a^ma^n = a^{m+n}$. $yx = a^na^m = a^{n+m}$. But $m + n = n + m$, because addition of integers has the commutative property. Therefore $xy = yx$.

28. *1* and *3*. No.

29. If $ac = bc$, $(ac)c^{-1} = (bc)c^{-1}$, so $a(cc^{-1}) = b(cc^{-1})$. Then $ai = bi$, or $a = b$.

30. Since $b = bi$, we have $bb = bi$. Then, by the cancellation rule, $b = i$.

31. $a(a^{-1}b) = (aa^{-1})b = ib = b$.

32. If $ax = b$, $a^{-1}(ax) = a^{-1}b$. Then $(a^{-1}a)x = a^{-1}b$, $ix = a^{-1}b$, and $x = a^{-1}b$.

33. $(ba^{-1})a = b(a^{-1}a) = bi = b$.

34. If $xa = b$, $(xa)a^{-1} = ba^{-1}$. Then $x(aa^{-1}) = ba^{-1}$, $xi = ba^{-1}$, and $x = ba^{-1}$.

35. $x = p$. **36.** $x = r$. **37.** $x = q$. **38.** $x = p$.

39. a) $x = 4$. b) $x = 2$. c) $x = 3$.

40. $abc = a(bc) = (bc)a = bca$.

41. $(ab)^3 = (ab)(ab)(ab) = ababab = aaabbb = (aaa)(bbb) = a^3b^3$.

42. $(ab)^{-1} = (ba)^{-1} = a^{-1}b^{-1}$.

43. a) Let x and y be any two members of the group. Since $xx = i$, the inverse of x is x. That is, $x^{-1} = x$. Similarly, since $yy = i$, $y^{-1} = y$. Since xy is in the group, its square is also equal to i. So xy is also equal to its own inverse. That is, $xy = (xy)^{-1} = y^{-1}x^{-1} = yx$. Since $xy = yx$, and x and y are any two members of the group, the group is commutative. b) The group of symmetries of a rectangle.

44. a) $p + q$. b) $p + (-p) = 0$. c) $0 + p = p$.
d) $p + p = 2p$. e) $p + (q + r) = (p + q) + r$.
f) $p + q = q + p$. g) $2(p + q) = 2p + 2q$.

Chapter 9

4.

	i	a	b	c
i	i			
a		i		
b			i	
c				i

5.

	i	p	q	r	s	t	u	v
i	i							
p				i				
q			i					
r		i						
s					i			
t						i		
u							i	
v								i

10. *2* is in the same column with *1*. *2* is in the same row with *1*. *1* is diagonally opposite *1*. The table shows that $2 \times 2 = 1$.

11.

$\times$	1	2	3	4
1	1		3	
2	2		1	
3				
4				

$3 \times 2 = 1$

$\times$	1	2	3	4
1		2	3	
2		4	1	
3				
4				

$3 \times 4 = 2$

$\times$	1	2	3	4
1			3	4
2			1	3
3				
4				

$3 \times 3 = 4$

$\times$	1	2	3	4
1				
2	2		1	
3	3		4	
4				

$4 \times 2 = 3$

$\times$	1	2	3	4
1				
2		4	1	
3		1	4	
4				

$4 \times 4 = 1$

$\times$	1	2	3	4
1				
2			1	3
3			4	2
4				

$4 \times 3 = 2$

$\times$	1	2	3	4
1				
2	2		1	
3				
4	4		2	

$2 \times 2 = 4$

$\times$	1	2	3	4
1				
2		4	1	
3				
4		3	2	

$2 \times 4 = 3$

$\times$	1	2	3	4
1				
2			1	3
3				
4			2	1

$2 \times 3 = 1$

12. It shows the identity element pattern and the rectangle pattern. It does not show the closure pattern, the inverse pattern, and the once and only once pattern.

13.

	i	a	b	c
i	i	a	b	c
a	a	b	c	i
b	b	c	i	a
c	c	i	a	b

Chapter 10

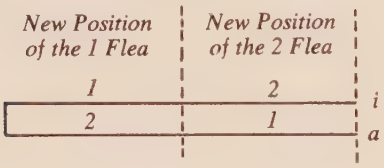
1. $-3 \rightarrow 2$. **2.** $6 \rightarrow 4$; $10 \rightarrow 8$; $x \rightarrow x - 2$.

3. $7 \rightarrow 2$; $8 \rightarrow 3$; $9 \rightarrow 4$; $x + 5 \rightarrow x$.

4. A jump of 3 units to the right; $12 \rightarrow 15$; $x \rightarrow x + 3$.

5. $3 \rightarrow 6$; $-2 \rightarrow -4$; $x \rightarrow 2x$.

6.



i

$1 \rightarrow 1$

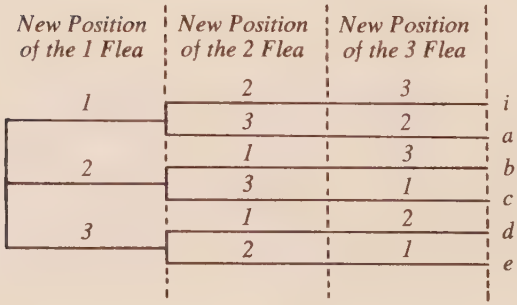
$2 \rightarrow 2$

a

$1 \rightarrow 2$

$2 \rightarrow 1$

7.



i	a	b
$1 \to 1$	$1 \to 1$	$1 \to 2$
$2 \to 2$	$2 \to 3$	$2 \to 1$
$3 \to 3$	$3 \to 2$	$3 \to 3$

c	d	e
$1 \to 2$	$1 \to 3$	$1 \to 3$
$2 \to 3$	$2 \to 1$	$2 \to 2$
$3 \to 1$	$3 \to 2$	$3 \to 1$

8. $1 \to 2$
$2 \to 3$
$3 \to 1$
$4 \to 4$
$ek = h.$

9. $5 \times 4 \times 3 \times 2 \times 1 = 120$; $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$; $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$.

10.

u q	$uq = l$
$1 \to 4 \to 2$	$1 \to 2$
$2 \to 2 \to 4$	$2 \to 4$
$3 \to 1 \to 3$	$3 \to 3$
$4 \to 3 \to 1$	$4 \to 1$

i a	$ia = a$
$1 \to 1 \to 1$	$1 \to 1$
$2 \to 2 \to 2$	$2 \to 2$
$3 \to 3 \to 4$	$3 \to 4$
$4 \to 4 \to 3$	$4 \to 3$

i b	$ib = b$
$1 \to 1 \to 1$	$1 \to 1$
$2 \to 2 \to 3$	$2 \to 3$
$3 \to 3 \to 2$	$3 \to 2$
$4 \to 4 \to 4$	$4 \to 4$

a a	$aa = i$
$1 \to 1 \to 1$	$1 \to 1$
$2 \to 2 \to 2$	$2 \to 2$
$3 \to 4 \to 3$	$3 \to 3$
$4 \to 3 \to 4$	$4 \to 4$

a	b	$ab = c$
$1 \to 1$	$\to 1$	$1 \to 1$
$2 \to 2$	$\to 3$	$2 \to 3$
$3 \to 4$	$\to 4$	$3 \to 4$
$4 \to 3$	$\to 2$	$4 \to 2$

11. See the table for S_4 on page *145*.

12. See the table for S_3 on page *144*.

13. See the table for S_2 on page *144*. **14.** S_1 and S_2.

15. a) $1 \to 3$, $2 \to 4$, $3 \to 1$, $4 \to 2$ b) q.

16. a, i;
d, c, i;
j, q, s, i.

17. $n^2 = k^3k^3 = k^6 = k^4k^2 = ix = x$.
$n^3 = k^3k^3k^3 = k^9 = k^4k^4k = iik = k$.
$n^4 = k^3k^3k^3k^3 = k^{12} = k^4k^4k^4 = iii = i$.

Chapter 11

1. Use $\begin{cases} \circ \to + \\ i \to 0 \\ b \to 1 \end{cases}$

2. Use $\begin{cases} \circ \to \circ \\ i \to i \\ b \to a \end{cases}$

3. Use $\begin{cases} \circ \to + \\ i \to 0 \\ p \to 1 \\ q \to 2 \end{cases}$

4. There are several dictionaries that may be used.
Example: Use $\begin{cases} \circ \to \circ \\ i \to i \\ p \to c \\ q \to d \\ r \to a \\ s \to b \\ t \to e \end{cases}$

o	i	c	d	a	b	e
i	i	c	d	a	b	e
c	c	d	i	e	a	b
d	d	i	c	b	e	a
a	a	b	e	i	c	d
b	b	e	a	d	i	c
e	e	a	b	c	d	i

Translated Table

○	i	a	b	c	d	e
i	i	a	b	c	d	e
c	c	e	a	d	i	b
d	d	b	e	i	c	a
a	a	i	c	b	e	d
b	b	d	i	e	a	c
e	e	c	d	a	b	i

Translated Table with Columns Rearranged

○	i	a	b	c	d	e
i	i	a	b	c	d	e
a	a	i	c	b	e	d
b	b	d	i	e	a	c
c	c	e	a	d	i	b
d	d	b	e	i	c	a
e	e	c	d	a	b	i

Translated Table with Rows and Columns Rearranged

5. Use $\begin{cases} \circ \to \circ \\ i \to i \\ r \to a \\ a \to b \\ l \to c \end{cases}$

6. Use $\begin{cases} \circ \to \circ \\ i \to i \\ q \to a \\ s \to b \\ t \to c \end{cases}$

7. Use $\begin{cases} \circ \to \circ \\ i \to i \\ a \to a \\ b \to b \\ c \to c \\ d \to d \\ e \to e \end{cases}$

8.

○	i	j	q	s
i	i	j	q	s
j	j	q	s	i
q	q	s	i	j
s	s	i	j	q

Use $\begin{cases} \circ \to + \\ i \to 0 \\ j \to 1 \\ q \to 2 \\ s \to 3 \end{cases}$

9.

○	i	m	h
i	i	m	h
m	m	h	i
h	h	i	m

Use $\begin{cases} \circ \to + \\ i \to 0 \\ m \to 1 \\ h \to 2 \end{cases}$

10. If $3a$ and $3b$ are any two members of the set, then $3a + 3b = 3(a + b)$ = three times an integer. Therefore the set is closed with respect to addition. It has the associative property, since addition of integers is associative. It contains the identity element 0. It contains the additive inverse of each of its members. Therefore it is a group with respect to addition. To show that it is isomorphic to the group of all integers, use the dictionary at the top of page 256:

$$
\begin{aligned}
+ &\rightarrow + \\
0 &\rightarrow 3(0) \\
1 &\rightarrow 3(1) \\
2 &\rightarrow 3(2) \\
&\cdots \\
-1 &\rightarrow 3(-1) \\
-2 &\rightarrow 3(-2) \\
&\cdots
\end{aligned}
$$

This dictionary obviously has properties 1 and 2. To show it has property 3, let a and b be any two integers, and let c be their sum. Then $a + b = c$ is an addition fact. The translated statement is $3a + 3b = 3c$. Since $c = a + b$, this statement says $3a + 3b = 3(a + b)$. This is an addition fact, since multiplication of integers is distributive with respect to addition.

11. The proof is the same as in exercise 10, with *5* used instead of *3*.

12. The proof is the same as in exercise 10, with k used instead of *3*.

13. $a^0a^n = a^{0+n} = a^n$; $a^na^0 = a^{n+0} = a^n$; the inverse of a^2 is a^3.

14.

$\circ$	a^0	a^1	a^2	a^3	a^4
a^0	a^0	a^1	a^2	a^3	a^4
a^1	a^1	a^2	a^3	a^4	a^0
a^2	a^2	a^3	a^4	a^0	a^1
a^3	a^3	a^4	a^0	a^1	a^2
a^4	a^4	a^0	a^1	a^2	a^3

15. Use $\begin{cases} \circ \rightarrow \circ \\ i \rightarrow a^0 \\ d \rightarrow a^1 \\ c \rightarrow a^2 \end{cases}$

16. Use $\begin{cases} + \rightarrow \circ \\ 0 \rightarrow a^0 \\ 1 \rightarrow a^1 \\ 2 \rightarrow a^2 \\ 3 \rightarrow a^3 \\ 4 \rightarrow a^4 \end{cases}$

17. The additive group of the n-point clock-face number system is isomorphic to C_n.

18. Number the elements i, a, b, c as *1*, *2*, *3*, *4* respectively. Match each element with a transformation as shown on page 257.

Match i with $\begin{cases} i \to ii = i \\ a \to ia = a \\ b \to ib = b \\ c \to ic = c \end{cases}$ or $\begin{cases} i \to i \\ a \to a \\ b \to b \\ c \to c \end{cases}$ or $\begin{cases} 1 \to 1 \\ 2 \to 2 \\ 3 \to 3 \\ 4 \to 4 \end{cases}$ which is called i.

Match a with $\begin{cases} i \to ai = a \\ a \to aa = i \\ b \to ab = c \\ c \to ac = b \end{cases}$ or $\begin{cases} i \to a \\ a \to i \\ b \to c \\ c \to b \end{cases}$ or $\begin{cases} 1 \to 2 \\ 2 \to 1 \\ 3 \to 4 \\ 4 \to 3 \end{cases}$ which is called g.

Match b with $\begin{cases} i \to bi = b \\ a \to ba = c \\ b \to bb = i \\ c \to bc = a \end{cases}$ or $\begin{cases} i \to b \\ a \to c \\ b \to i \\ c \to a \end{cases}$ or $\begin{cases} 1 \to 3 \\ 2 \to 4 \\ 3 \to 1 \\ 4 \to 2 \end{cases}$ which is called q.

Match c with $\begin{cases} i \to ci = c \\ a \to ca = b \\ b \to cb = a \\ c \to cc = i \end{cases}$ or $\begin{cases} i \to c \\ a \to b \\ b \to a \\ c \to i \end{cases}$ or $\begin{cases} 1 \to 4 \\ 2 \to 3 \\ 3 \to 2 \\ 4 \to 1 \end{cases}$ which is called x.

The dictionary

$$\begin{aligned} \circ &\to \circ \\ i &\to i \\ a &\to g \\ b &\to q \\ c &\to x \end{aligned}$$

changes the multiplication table for the symmetries of a rectangle into the following, which is really the multiplication table for the subgroup of S_4 whose elements are i, g, q and x:

	i	g	q	x
i	i	g	q	x
g	g	i	x	q
q	q	x	i	g
x	x	q	g	i

19. a)

○	i	c	b	a
i	i	c	b	a
c	c	b	a	i
b	b	a	i	c
a	a	i	c	b

b)

○	i	a	b	c
i	i	a	b	c
c	c	i	a	b
b	b	c	i	a
a	a	b	c	i

c)

○	i	a	b	c
i	i	a	b	c
a	a	b	c	i
b	b	c	i	a
c	c	i	a	b

d) Yes.

20. a)

○	i	b	c	a
i	i	b	c	a
b	b	c	a	i
c	c	a	i	b
a	a	i	b	c

b)

○	i	a	b	c
i	i	a	b	c
b	b	i	c	a
c	c	b	a	i
a	a	c	i	b

c)

○	i	a	b	c
i	i	a	b	c
a	a	c	i	b
b	b	i	c	a
c	c	b	a	i

d) No.

21. a)

○	i	a	c	b
i	i	a	c	b
a	a	i	b	c
c	c	b	i	a
b	b	c	a	i

b)

○	i	a	b	c
i	i	a	b	c
a	a	i	c	b
c	c	b	a	i
b	b	c	i	a

c)

○	i	a	b	c
i	i	a	b	c
a	a	i	c	b
b	b	c	i	a
c	c	b	a	i

d) Yes.

22. a)

○	i	b	a	c
i	i	b	a	c
b	b	i	c	a
a	a	c	i	b
c	c	a	b	i

b)

○	i	a	b	c
i	i	a	b	c
b	b	c	i	a
a	a	i	c	b
c	c	b	a	i

c)

○	i	a	b	c
i	i	a	b	c
a	a	i	c	b
b	b	c	i	a
c	c	b	a	i

d) Yes.

23. a)

○	i	b	c	a
i	i	b	c	a
b	b	i	a	c
c	c	a	i	b
a	a	c	b	i

b)

○	i	a	b	c
i	i	a	b	c
b	b	c	i	a
c	c	b	a	i
a	a	i	c	b

c)

○	i	a	b	c
i	i	a	b	c
a	a	i	c	b
b	b	c	i	a
c	c	b	a	i

d) Yes.

24. Yes. The identity transformation, which matches each element with itself, is an automorphism.

25. The elements a, b and c all have order 2.
There are six automorphisms as follows:

I:	P:	Q:	R:	S:	T:
$i \to i$	$i \to i$	$i \to i$	$i \to i$	$i \to i$	$i \to i$
$a \to a$	$a \to a$	$a \to b$	$a \to b$	$a \to c$	$a \to c$
$b \to b$	$b \to c$	$b \to a$	$b \to c$	$b \to b$	$b \to a$
$c \to c$	$c \to b$	$c \to c$	$c \to a$	$c \to a$	$c \to b$

26. The elements p and q are both of order *3*. There are two automorphisms as follows:

I:	A:
$i \to i$	$i \to i$
$p \to p$	$p \to q$
$q \to q$	$q \to p$

27. The elements p and q are of order *3*; the elements r, s and t are of order 2. There are six automorphisms as follows:

I:	A:	B:	C:	D:	E:
$i \to i$	$i \to i$	$i \to i$	$i \to i$	$i \to i$	$i \to i$
$p \to p$	$p \to q$	$p \to q$	$p \to p$	$p \to p$	$p \to q$
$q \to q$	$q \to p$	$q \to p$	$q \to q$	$q \to q$	$q \to p$
$r \to r$	$r \to s$	$r \to r$	$r \to t$	$r \to s$	$r \to t$
$s \to s$	$s \to r$	$s \to t$	$s \to r$	$s \to t$	$s \to s$
$t \to t$	$t \to t$	$t \to s$	$t \to s$	$t \to r$	$t \to r$

28.

	I	P	Q	R	S	T
I	I	P	Q	R	S	T
P	P	I	R	Q	T	S
Q	Q	T	I	S	R	P
R	R	S	P	T	Q	I
S	S	R	T	P	I	Q
T	T	Q	S	I	P	R

29.

	I	A
I	I	A
A	A	I

30.

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I	C	B	E	D
B	B	D	I	E	A	C
C	C	E	A	D	I	B
D	D	B	E	I	C	A
E	E	C	D	A	B	I

31. The tables are the same. The groups they represent are isomorphic.

32. The tables are the same. The groups they represent are isomorphic.

33. Use the transformation: $\begin{cases} a \to b \\ b \to c \\ c \to a. \end{cases}$

Table I, translated by means of this transformation, becomes Table IIB1 when the rows and columns are rearranged.

34. The tables are the same. The groups they represent are isomorphic.

35. The tables are the same. The groups they represent are isomorphic.

36. The cyclic groups C_1, C_2, C_3, and C_4 are commutative, and the group of symmetries of a rectangle is commutative. Therefore any group that is isomorphic to one of these groups is also commutative.

Chapter 12

2. The subgroups of B are $I = \{i\}$, $G = \{i, b\}$, and $B = \{i, a, b, c\}$.

3. The subgroups of R are $I = \{i\}$, $A = \{i, a\}$, $B = \{i, b\}$, $C = \{i, c\}$, and $R = \{i, a, b, c\}$.

4. The subgroups of T are $I = \{i\}$, $A = \{i, p, q\}$, $B = \{i, r\}$, $C = \{i, s\}$, $D = \{i, t\}$, and $T = \{i, p, q, r, s, t\}$.

5. $psp^{-1} = psq = rq = t$. **6.** $pqp^{-1} = pqq = iq = q$.

7. $bab^{-1} = bab = cb = a$.

8. $aba^{-1} = (ab)a^{-1} = (ba)a^{-1} = b(aa^{-1}) = bi = b$.

9. By exercise 8, for each element b in G, $aba^{-1} = b$. So the inner automorphism matches each element b with itself.

14. The group is commutative. So, by exercise 9, the only inner automorphism it has is the identity automorphism I.

15. Only I. **16.** All of them. **17.** See answers 8 and 9 above.

18. If the inner automorphism determined by a is the identity automorphism, then for each element b in G, $aba^{-1} = b$. Multiplying on

the right by a, we get $aba^{-1}a = ba$, or $ab = ba$. Then a commutes with every element of G.

19. i and q. **20.** Only i. **21.** All the elements in the group.

22. All the elements in the group.

23. S contains i, so aSa^{-1} contains $aia^{-1} = i$.

24. Each member of aSa^{-1} has the form aba^{-1}, where b is a member of S. Since S is a subgroup, it also contains b^{-1}. Then aSa^{-1} contains $ab^{-1}a^{-1}$, which is the inverse of aba^{-1}.

25. Any two members of aSa^{-1} have the form aba^{-1} and aca^{-1}, where b and c are members of S. Since S is a subgroup, it also contains bc. Then aSa^{-1} contains $abca^{-1}$. But the product $(aba^{-1})(aca^{-1}) = ab(a^{-1}a)ca^{-1} = abica^{-1} = abca^{-1}$.

26. Use a dictionary which matches each element b of S with its conjugate aba^{-1}. If $bc = d$ is a multiplication fact in S, this dictionary translates it into the statement $(aba^{-1})(aca^{-1}) = ada^{-1}$. By exercise 25, the statement $(aba^{-1})(aca^{-1}) = abca^{-1}$ is true. Replacing bc by d, we see that the statement $(aba^{-1})(aca^{-1}) = ada^{-1}$ is also true. Since the dictionary translates each multiplication fact into a multiplication fact, it is an isomorphism between s and aSa^{-1}. So S is isomorphic to aSa^{-1}.

27. $iCi^{-1} = iCi = \{iii, isi\} = \{i, s\}$.
$pCp^{-1} = pCq = \{piq, psq\} = \{i, t\}$.
$qCq^{-1} = qCp = \{qip, qsp\} = \{i, r\}$.
$rCr^{-1} = rCr = \{rir, rsr\} = \{i, t\}$.
$sCs^{-1} = sCs = \{sis, sss\} = \{i, s\}$.
$tCt^{-1} = tCt = \{tit, tst\} = \{i, r\}$.

28. $iAi^{-1} = iAi = \{iii, ipi, iqi\} = \{i, p, q\}$.
$pAp^{-1} = pAq = \{piq, ppq, pqq\} = \{i, p, q\}$.
$qAq^{-1} = qAp = \{qip, qpp, qqp\} = \{i, p, q\}$.
$rAr^{-1} = rAr = \{rir, rpr, rqr\} = \{i, q, p\}$.
$sAs^{-1} = sAs = \{sis, sps, sqs\} = \{i, q, p\}$.
$tAt^{-1} = tAt = \{tit, tpt, tqt\} = \{i, q, p\}$.

The only conjugate of A is A itself.

29. The only conjugate of A is A.

30. The only conjugate of G is G.

31. The only conjugate of B is B.

32. The conjugates of F are $F = \{i, v\}$ and $E = \{i, u\}$.

33. The only conjugate of H is H.

34. The conjugates of C are $C = \{i, s\}$ and $D = \{i, t\}$.

35. The conjugates of D are $D = \{i, t\}$ and $C = \{i, s\}$.

36. The conjugates of E are $E = \{i, u\}$ and $F = \{i, v\}$.

37. I, A and T are normal subgroups of T.

38. I, A, B, G, H and S are normal subgroups of S.

39. In a commutative group, every inner automorphism is the identity automorphism. So every conjugate of each subgroup is the subgroup itself. Then each subgroup is a normal subgroup.

40. *1*, *3*, *5* and *15*. **41.** *1* and *5*.

42. The only divisors of p are *1* and p. Then a subgroup contains either only *1* element or all p elements. If it contains only *1* element, the subgroup is I. If it contains all p elements, the subgroup is G.

43. p is greater than *1*. Then there is an element a in the group that is different from i. By exercise 42, the cyclic subgroup generated by a is either I or G. It cannot be I, because it contains a, which is different from i. So it must be G. Then G is cyclic.

44. By exercise 43, the group is cyclic. But every cyclic group is commutative. (See exercise 27 in Chapter VIII.)

45. By exercise 36 in Chapter XI, every group of order *1*, *2*, *3* or *4* is commutative. By exercise 44 above, every group of order *5* is commutative.

Chapter 13

1. $aGa^{-1} = G$. Then, for every member g in G, there is a member h in G for which $aga^{-1} = h$. Multiplying on the right by a, we get $ag = ha$. But ha is a member of Ga. Thus every member ag of aG is also a member of Ga. $aGa^{-1} = G$ also means that for every member h in G, there is a member g in G for which $aga^{-1} = h$. Then the same argument shows

that $ag = ha$. But ag is a member of aG. Thus every member ha of Ga is also a member of aG.

2. The left cosets of A are $\{i, p, q, r\}$ and $\{s, t, u, v\}$. The right cosets of A are $\{i, p, q, r\}$ and $\{s, t, u, v\}$.

3. The left cosets of B are $\{i, q\}$, $\{p, r\}$, $\{s, t\}$, and $\{u, v\}$. The right cosets of B are $\{i, q\}$, $\{p, r\}$, $\{s, t\}$, and $\{u, v\}$.

4. The left cosets of H are $\{i, q, u, v\}$ and $\{p, r, s, t\}$. The right cosets of H are $\{i, q, u, v\}$ and $\{p, r, s, t\}$.

5. The left cosets of A are $\{i, p, q\}$ and $\{r, s, t\}$. The right cosets of A are $\{i, p, q\}$ and $\{r, s, t\}$.

6. $uB = \{u, v\}$, $rB = \{p, r\}$, and $tB = \{s, t\}$. $up = s$, $ur = t$, $vp = t$, $vr = s$.

7. $uH = \{i, q, u, v\}$, $sH = \{p, r, s, t\}$, and $pH = \{p, r, s, t\}$. $ip = p$, $ir = r$, $is = s$, $it = t$; $qp = r$, $qr = p$, $qs = t$, $qt = s$; $up = s$, $ur = t$, $us = p$, $ut = r$; $vp = t$, $vr = s$, $vs = r$, $vt = p$.

8. $pA = \{i, p, q\}$, $rA = \{r, s, t\}$, and $tA = \{r, s, t\}$. Moreover, $ir = r$, $is = s$, $it = t$; $pr = t$, $ps = r$, $pt = s$; $qr = s$, $qs = t$, $qt = r$.

9. The left cosets of A are $\{i, a\}$ and $\{b, c\}$. The right cosets of A are $\{i, a\}$ and $\{b, c\}$. $aA = \{i, a\}$, $bA = \{b, c\}$, and $cA = \{b, c\}$. $ib = b$, $ic = c$, $ab = c$, $ac = b$.

10. The left cosets of A are $\{0, 3\}$, $\{1, 4\}$, and $\{2, 5\}$. The right cosets of A are $\{0, 3\}$, $\{1, 4\}$, and $\{2, 5\}$. $2 + A = \{2, 5\}$, $3 + A = \{0, 3\}$, and $5 + A = \{2, 5\}$. $2 + 0 = 2$, $2 + 3 = 5$, $5 + 0 = 5$, and $5 + 3 = 2$.

11. a) $0 + E = \{0, 2, 4, \ldots, -2, -4, \ldots\}$.
$1 + E = \{1, 3, 5, \ldots, -1, -3, \ldots\}$.
$2 + E = \{2, 4, 6, \ldots, 0, -2, \ldots\}$
$= \{0, 2, 4, \ldots, -2, -4, \ldots\} = 0 + E$.
$3 + E = \{3, 5, 7, \ldots, 1, -1, \ldots\}$
$= \{1, 3, 5, \ldots, -1, -3, \ldots\} = 1 + E$.

b) Same as in a). c) Two. d) Two.
f) $1 + E = 3 + E =$ the set of all odd integers. $2 + E =$ the set of all even integers. An odd integer plus an even integer equals an odd integer.

12. a) $0 + T = \{0, 3, 6, \ldots, -3, -6, \ldots\}$.
$1 + T = \{1, 4, 7, \ldots, -2, -5, \ldots\}$.
$2 + T = \{2, 5, 8, \ldots, -1, -4, \ldots\}$.
$3 + T = \{3, 6, 9, \ldots, 0, -3, \ldots\}$
$= \{0, 3, 6, \ldots, -3, -6, \ldots\} = 0 + T$.
$4 + T = \{4, 7, 10, \ldots, 1, -2, \ldots\}$
$= \{1, 4, 7, \ldots, -2, -5, \ldots\} = 1 + T$.

b) Same as in a). c) Three. d) Three.
f) $(3a + 1) + (3b + 2) = 3a + 3b + 3 = 3(a + b + 1)$ = a multiple of 3 = a member of $3 + T$.

13.

	I	P	S	U
I	I	P	S	U
P	P	I	U	S
S	S	U	I	P
U	U	S	P	I

Note that $pp = q$
Therefore $PP = Q = I$
Note that $pu = t$.
Therefore $PU = T = S$. Etc.

14. By the definition of multiplication of cosets, $(AB)C$ is the coset that contains the product $(ab)c$, and $A(BC)$ is the coset that contains the product $a(bc)$. But, since S is a group, $(ab)c = a(bc)$. Therefore these two cosets have an element in common. Then the cosets must be the same. That is, $(AB)C = A(BC)$.

15. Let a be a member of the coset A. Since $ia = a$, and $ai = a$, then $IA = A$, and $AI = A$, by the definition of coset multiplication.

16. Let a be any member of coset A. Since S is a group, a has an inverse b with the property that $ab = i$ and $ba = i$. Then $AB = I$ and $BA = I$, by the definition of coset multiplication.

17. n/m.

18.

	I	S
I	I	S
S	S	I

19.

	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

20. Same as in 19.

21. a)

	i	g	q	x
i	i	g	q	x
g	g	i	x	q
q	q	x	i	g
x	x	q	g	i

Y is a subgroup of S_4, since it is closed with respect to multiplication.

b) The left cosets of Y are $\{i, g, q, x\}$, $\{a, f, r, w\}$, $\{b, k, n, v\}$, $\{c, l, m, u\}$, $\{d, h, p, t\}$, and $\{e, j, o, s\}$. The right cosets of Y are the same six sets. Y is a normal subgroup because every left coset of Y is a right coset of Y, and vice versa.

c)

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I	C	B	E	D
B	B	D	I	E	A	C
C	C	E	A	D	I	B
D	D	B	E	I	C	A
E	E	C	D	A	B	I

23. Since a and b are in G, each is matched with I by the homomorphism. Suppose that the homomorphism matches c with X. The homomorphism translates the multiplication fact $ab = c$ into the multiplication fact $II = X$. But $II = I$. Therefore $X = I$. Then c is a member of G.

24. Suppose that the homomorphism matches i with X. Let A be any member of T. Let b be any member of S that is matched with A. $ib = bi = b$. The homomorphism translates these multiplication facts into the multiplication facts $XA = AX = A$. Since this is true for every member A of T, X must be the identity element of T. That is, $X = I$. Then i is a member of G.

25. Suppose that the homomorphism matches a^{-1} with X. The homomorphism translates the multiplication fact $aa^{-1} = i$ into the multiplication fact $IX = I$. But $IX = X$. Therefore $X = I$, and a^{-1} is a member of G.

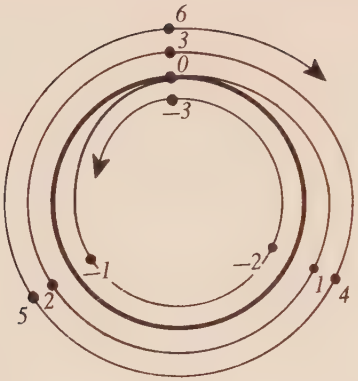
26. Let g be any member of G, and let a be any member of S. Let $aga^{-1} = h$. Let X, Y and Z be the members of T that the homomorphism matches with h, a and a^{-1} respectively. The homomorphism translates the multiplication fact $aa^{-1} = i$ into the multiplication fact $YZ = I$. It translates the multiplication fact $aga^{-1} = h$ into the multiplication fact $YIZ = X$. But $YIZ = (YI)Z = YZ = I$. Therefore $X = I$, and h is a member of G.

27. Let a be any member of S that is matched with A by the homomorphism, let g be any member of G, and let $ag = c$. Let X be the member of T that is matched with c. The homomorphism translates the multiplication fact $ag = c$ into the multiplication fact $AI = X$. But $AI = A$. Therefore $X = A$. Thus every member of the coset aG is matched with A. We now show that no other members of S are matched with A by showing that if any element b is matched with A, then b is a member of the coset aG. Let B be the member of T that a^{-1} is matched with. Let $a^{-1}b = h$, and let X be the member of T that is matched with h. The homomorphism translates the multiplication facts $a^{-1}a = i$ and $a^{-1}b = h$ into the multiplication facts $BA = I$ and $BA = X$. Then $X = I$, and h is a member of G. Multiplying the equation $a^{-1}b = h$ by a on the left, we get $b = ah$. That is, b is a member of the coset aG.

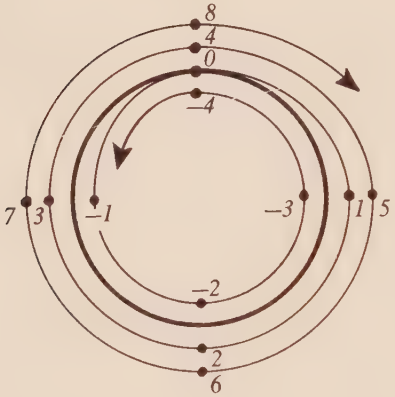
28. Construct a dictionary that matches each member aG of S/G with the member of T that a was matched with by the homomorphism. Each member of S/G appears once and only once in the left-hand column. Each member of T appears once and only once in the right-hand column. (See example on page 231.) Let aG and bG be any two members of S/G. Let $ab = c$. $(aG)(bG) = cG$. Suppose that the dictionary matches aG, bG, and cG with X, Y and Z respectively. Then the original homomorphism matches a, b and c with X, Y and Z respectively. The homomorphism translates the multiplication fact $ab = c$ into the multiplication fact $XY = Z$. The dictionary relating S/G to T translates the multiplication fact $(aG)(bG) = cG$ into the statement $XY = Z$. Thus it translates every multiplication fact of S/G into a multiplication fact of T. Then the dictionary is an isomorphism between S/G and T.

29. If and only if G and its cosets each contain only one member. This happens when and only when G is the trivial subgroup $\{i\}$.

30.



31.



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From the point of view of the general reader, there is a serious gap in the available literature on groups. On the one hand, there are many textbooks on the theory of groups. But these are too abstract, too technical, and too difficult for the general reader. On the other hand, there are many books for the general reader that mention groups, but do not give enough detail to give the reader a real understanding of what groups are. This book is intended to bridge that gap by developing a substantial part of the elements of group theory through simple, concrete examples; and by requiring of the reader that he know no more than sixth grade arithmetic.

It explains what a group is and acquaints the reader with certain important groups. As it shows, the members of a group may or may not be numbers, but they behave like numbers in many ways.

Because groups have so many uses, it is important that young people become acquainted with them early. This book shows how the idea of a group may be introduced through familiar examples that can be understood by school children above the sixth grade. It is likely that some day in the near future groups will be a part of the school mathematics curriculum as early as the seventh grade, and experiences from which the idea of a group may be developed will be included as early as the second grade. To prepare for that day, school teachers have to learn now what they may have to teach later about groups. This book, written for the general reader, will also help achieve that purpose for the teacher.

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